



موضوع

پروژه- زمینه داده ها و خروجی ها

اعضای گروه

فریده مطیعان ۴۰۳۲۰۵۴۶۳

نام درس

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بهار ۱۴۰۵

این فایل شامل مثال خروجی تولید شده و یکی داده های تصادفی تولید شده است.

مثال خروجی کد برای داده های تولید شده

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↻ Iteration 1

Master Problem Objective Value : -6,400,000.00

Decision Variables:

x[1]: 0.0

x[2]: 0.0

x[3]: 0.0

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-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, 1.0, -0.0]$$

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [0.0, 0.0, 1.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (0.0, 0.0, 0.0)-----

scenario1

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

➤ w = [9.785496],theta =-100000.0

➤ e = [9.785496]

➤ E = [[0.00951779 0.01007766 0.01151733]]

✓ Added optimality cut to master problem.

optimality_cut : Size=1, Index={1}, Active=True

Key : Lower : Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

=====

scenario2

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

➤ w = [60.25830153],theta =-100000.0

➤ e = [60.25830153]

➤ E = [[0.05304883 0.05616935 0.06419354]]

✓ Added optimality cut to master problem.

optimality_cut : Size=2, Index={1, 2}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

=====

scenario3

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ w = [179.09259324],theta =-100000.0

➤ e = [179.09259324]

➤ E = [[0.16560308 0.17534444 0.20039365]]

✓ Added optimality cut to master problem.

optimality_cut : Size=3, Index={1, 2, 3}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

=====

scenario4

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

➤ w = [316.22021003],theta =-100000.0

➤ e = [316.22021003]

➤ E = [[0.30008169 0.31773355 0.36312406]]

✓ Added optimality cut to master problem.

optimality_cut : Size=4, Index={1, 2, 3, 4}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

=====

scenario5

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

➤ w = [271.85361107],theta =-100000.0

➤ e = [271.85361107]

➤ E = [[0.26887293 0.28468899 0.32535884]]

✓ Added optimality cut to master problem.

optimality_cut : Size=5, Index={1, 2, 3, 4, 5}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

=====

scenario6

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

➤ w = [193.68136379],theta =-100000.0

➤ e = [193.68136379]

➤ E = [[0.17427172 0.184523 0.21088343]]

✓ Added optimality cut to master problem.

optimality_cut : Size=6, Index={1, 2, 3, 4, 5, 6}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

=====

scenario7

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [94.92932352],theta =-100000.0

➤ e = [94.92932352]

➤ E = [[0.09025744 0.09556671 0.10921909]]

✓ Added optimality cut to master problem.

optimality_cut : Size=7, Index={1, 2, 3, 4, 5, 6, 7}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

=====

scenario8

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

► w = [18.39287286],theta =-100000.0

► e = [18.39287286]

► E = [[0.01987744 0.0210467 0.02405337]]

✓ Added optimality cut to master problem.

optimality_cut : Size=8, Index={1, 2, 3, 4, 5, 6, 7, 8}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

=====

scenario9

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{9: [436.5264598399147, 385.17040574110115, 66.02921241276017, -0.0]}

► w = [107.84771361], theta = -100000.0

► e = [107.84771361]

► E = [[0.25955863 0.27482679 0.31408776]]

✓ Added optimality cut to master problem.

optimality_cut : Size=9, Index={1, 2, 3, 4, 5, 6, 7, 8, 9}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

=====

scenario10

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{10: [436.2208913180268, 384.90078645708235, 65.98299196407125, -0.0]}

► w = [431.08888083],theta =-100000.0

► e = [431.08888083]

► E = [[1.0723182 1.13539574 1.29759513]]

✓ Added optimality cut to master problem.

optimality_cut : Size=10, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

=====

scenario11

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{11: [446.4982555174795, 393.9690489860112, 67.53755125474477, -0.0]}

➤ w = [1636.28478346],theta =-100000.0

➤ e = [1636.28478346]

➤ E = [[4.04437004 4.28227416 4.89402761]]

✓ Added optimality cut to master problem.

optimality_cut : Size=11, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

=====

scenario12

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{12: [467.5104634221321, 412.5092324312929, 70.71586841679306, -0.0]}

► w = [2945.31591956], theta = -100000.0

► e = [2945.31591956]

► E = [[7.30669766 7.7365034 8.84171817]]

✓ Added optimality cut to master problem.

optimality_cut : Size=12, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

=====

scenario13

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{13: [501.2553686719416, 442.2841488281836, 75.82013979911719, -0.0]}

➤ w = [2621.27072206],theta =-100000.0

➤ e = [2621.27072206]

➤ E = [[6.32456455 6.69659776 7.65325458]]

✓ Added optimality cut to master problem.

optimality_cut : Size=13, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

=====

scenario14

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{14: [549.5312732287364, 484.880535201826, 83.12237746317017, -0.0]}

► w = [1945.98721461], theta = -100000.0

► e = [1945.98721461]

► E = [[4.68579821 4.9614334 5.6702096]]

✓ Added optimality cut to master problem.

optimality_cut : Size=14, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] +
5.670209603125645*x[3] + theta[14] : +Inf : True

=====

scenario15

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{15: [614.7056822336644, 542.3873666767626, 92.98069143030214, -0.0]}

► w = [683.40808201],theta =-100000.0

► e = [683.40808201]

► E = [[1.69538901 1.79511777 2.05156317]]

✓ Added optimality cut to master problem.

optimality_cut : Size=15, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

=====

scenario16

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{16: [700.6169483826413, 618.1914250435068, 105.97567286460118, -0.0]}

► w = [201.94253218], theta = -100000.0

► e = [201.94253218]

► E = [[0.50661612 0.53641706 0.61304807]]

✓ Added optimality cut to master problem.

optimality_cut : Size=16, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

=====

scenario17

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{17: [667.5127975715615, 588.9818802102011, 100.96832232174877, -0.0]}

➤ w = [384.80149507], theta = -100000.0

➤ e = [384.80149507]

➤ E = [[1.50901281 1.59777826 1.8260323]]

✓ Added optimality cut to master problem.

optimality_cut : Size=17, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

=====

scenario18

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{18: [651.7194447810183, 575.0465689244278, 98.5794118156162, -0.0]}

► w = [2254.18255018],theta =-100000.0

► e = [2254.18255018]

► E = [[8.87680987 9.39897516 10.74168589]]

✓ Added optimality cut to master problem.

optimality_cut : Size=18, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

=====

scenario19

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{19: [651.7911339199442, 575.1098240470095, 98.59025555091591, -0.0]}

► w = [6494.90694624],theta =-100000.0

► e = [6494.90694624]

► E = [[25.38325615 26.87638887 30.71587299]]

✓ Added optimality cut to master problem.

optimality_cut : Size=19, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

=====

scenario20

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{20: [667.2646554392036, 588.7629312698856, 100.93078821769465, -0.0]}

► w = [12144.21672899], theta = -100000.0

► e = [12144.21672899]

► E = [[47.32904865 50.11311033 57.27212609]]

✓ Added optimality cut to master problem.

optimality_cut : Size=20, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

=====

scenario21

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{21: [699.3400674261661, 617.064765376029, 105.78253120731924, -0.0]}

► w = [13073.54549577], theta = -100000.0

► e = [13073.54549577]

► E = [[50.60162475 53.57819092 61.23221819]]

✓ Added optimality cut to master problem.

optimality_cut : Size=21, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

=====

scenario22

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{22: [749.8953609004036, 661.6723772650621, 113.42955038829635, -0.0]}

► w = [7534.24280246], theta = -100000.0

► e = [7534.24280246]

► E = [[29.03804808 30.74616856 35.13847835]]

✓ Added optimality cut to master problem.

optimality_cut : Size=22, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

=====

scenario23

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{23: [821.0004390209799, 724.4121520773353, 124.18494035611461, -0.0]}

► w = [2704.47203442], theta = -100000.0

► e = [2704.47203442]

► E = [[10.67628971 11.30430675 12.91920772]]

✓ Added optimality cut to master problem.

optimality_cut : Size=23, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

=====

scenario24

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{24: [918.7980113171591, 810.7041276327875, 138.97785045133497, -0.0]}

➤ w = [529.66003005], theta = -100000.0

➤ e = [529.66003005]

➤ E = [[2.04910332 2.16963881 2.47958722]]

✓ Added optimality cut to master problem.

optimality_cut : Size=24, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] +
2.479587216612538*x[3] + theta[24] : +Inf : True

=====

scenario25

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{25: [852.5618626813051, 752.2604670717399, 128.95893721229825, -0.0]}

➤ w = [1018.05916544],theta =-100000.0

➤ e = [1018.05916544]

➤ E = [[5.43501793 5.75472487 6.57682842]]

✓ Added optimality cut to master problem.

optimality_cut : Size=25, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

=====

scenario26

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{26: [811.1955611040082, 715.760789209419, 122.70184957875753, -0.0]}

➤ w = [5043.72769463],theta =-100000.0

➤ e = [5043.72769463]

➤ E = [[26.99997502 28.58820884 32.67223867]]

✓ Added optimality cut to master problem.

optimality_cut : Size=26, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

=====

scenario27

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{27: [791.8323230604556, 698.6755791709902, 119.77295642931259, -0.0]}

► $w = [13726.64621211], \theta = -100000.0$

► $e = [13726.64621211]$

► $E = [[73.33117573 \ 77.6447743 \ 88.73688491]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=27, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

=====

scenario28

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{28: [792.0619544341431, 698.8781950889498, 119.80769058667708, -0.0]}

► w = [25243.48040661], theta = -100000.0

► e = [25243.48040661]

► E = [[134.96426799 142.90334258 163.31810581]]

✓ Added optimality cut to master problem.

optimality_cut : Size=28, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] +
2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] +
6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] +
32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] +
88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

=====

scenario29

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{29: [811.0872825796512, 715.6652493349864, 122.68547131456907, -0.0]}

➤ w = [25716.23795944],theta =-100000.0

➤ e = [25716.23795944]

➤ E = [[137.2576648 145.33164508 166.09330866]]

✓ Added optimality cut to master problem.

optimality_cut : Size=29, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

=====

scenario30

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{30: [849.71936984892, 749.7523851608117, 128.52898031328198, -0.0]}

► w = [15080.0196402], theta = -100000.0

► e = [15080.0196402]

► E = [[80.71688227 85.46493417 97.67421048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=30, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] +
88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] +
166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
97.67421047609123*x[3] + theta[30] : +Inf : True

=====

scenario31

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{31: [910.3468468876405, 803.2472178420356, 137.69952305863467, -0.0]}

➤ w = [5472.79104423],theta =-100000.0

➤ e = [5472.79104423]

➤ E = [[29.26027732 30.9814701 35.4073944]]

✓ Added optimality cut to master problem.

optimality_cut : Size=31, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

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scenario32

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{32: [1000.4984951349236, 882.7927898249325, 151.33590682713128, -0.0]}

► w = [947.53092775], theta = -100000.0

► e = [947.53092775]

► E = [[4.98485869 5.27808567 6.03209791]]

✓ Added optimality cut to master problem.

optimality_cut : Size=32, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

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scenario33

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{33: [905.7312776757436, 799.1746567727149, 137.0013697324654, -0.0]}

➤ w = [1044.25488485], theta = -100000.0

➤ e = [1044.25488485]

➤ E = [[7.03245993 7.44613405 8.50986748]]

✓ Added optimality cut to master problem.

optimality_cut : Size=33, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

=====

scenario34

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{34: [839.6853529076284, 740.8988408008486, 127.01122985157404, -0.0]}

► w = [5428.31884027],theta =-100000.0

► e = [5428.31884027]

► E = [[37.06746907 39.24790843 44.85475249]]

✓ Added optimality cut to master problem.

optimality_cut : Size=34, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487x[1] + 0.1753444411069076x[2] + 0.20039364697932285x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066x[1] + 0.31773354960289335x[2] + 0.363124056689021x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609x[1] + 0.28468898513900553x[2] + 0.3253588401588636x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484x[1] + 0.18452300215960046x[2] + 0.2108834310395433x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525x[1] + 0.09556670612297144x[2] + 0.10921909271196732x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873x[1] + 0.021046701662377272x[2] + 0.024053373328431164x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133x[1] + 0.27482678790439047x[2] + 0.3140877576050175x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732x[1] + 1.1353957359225595x[2] + 1.2975951267686392x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941x[1] + 4.282274155786878x[2] + 4.894027606613574x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133x[1] + 7.7365034018484655x[2] + 8.8417181735411x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895x[1] + 6.696597760358598x[2] + 7.653254583266969x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111x[1] + 4.96143340273494x[2] + 5.670209603125645x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

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scenario35

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{35: [799.8590766192195, 705.7580087816643, 120.98708721971389, -0.0]}

► w = [14162.21070955], theta = -100000.0

► e = [14162.21070955]

► E = [[96.20844949 101.86777005 116.42030863]]

✓ Added optimality cut to master problem.

optimality_cut : Size=35, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

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scenario36

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{36: [780.6064686449948, 688.7704135102896, 118.07492803033537, -0.0]}

► w = [25360.52662545], theta = -100000.0

► e = [25360.52662545]

► E = [[172.45158105 182.5957917 208.6809048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=36, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] +
88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] +
166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] +
116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] +
208.68090480369352*x[3] + theta[36] : +Inf : True

=====

scenario37

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{37: [780.3098381869098, 688.5086807531557, 118.03005955768384, -0.0]}

► w = [24611.89030799], theta = -100000.0

► e = [24611.89030799]

► E = [[167.7643133 177.63280231 203.00891693]]

✓ Added optimality cut to master problem.

optimality_cut : Size=37, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

=====

scenario38

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{38: [799.021668106632, 705.0191189176163, 120.86042038587709, -0.0]}

- $w = [14476.39257511], \theta = -100000.0$
- $e = [14476.39257511]$
- $E = [[98.40431256 \ 104.19280153 \ 119.07748746]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=38, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38}, Active=True

Key : Lower	: Body	: Upper : Active
1 :	$9.785496 : 0.009517792788000001 * x[1] + 0.010077662951999999 * x[2] + 0.011517329088000003 * x[3] + \theta[1] :$	$+\text{Inf} : \text{True}$
2 :	$60.25830152831999 : 0.05304882616689596 * x[1] + 0.056169345353184 * x[2] + 0.06419353754649597 * x[3] + \theta[2] :$	$+\text{Inf} : \text{True}$
3 :	$179.09259323679183 : 0.16560308326763487 * x[1] + 0.1753444411069076 * x[2] + 0.20039364697932285 * x[3] + \theta[3] :$	$+\text{Inf} : \text{True}$
4 :	$316.22021002633863 : 0.300081685736066 * x[1] + 0.31773354960289335 * x[2] + 0.363124056689021 * x[3] + \theta[4] :$	$+\text{Inf} : \text{True}$
5 :	$271.85361107332903 : 0.2688729304090609 * x[1] + 0.28468898513900553 * x[2] + 0.3253588401588636 * x[3] + \theta[5] :$	$+\text{Inf} : \text{True}$
6 :	$193.68136379025395 : 0.17427172426184484 * x[1] + 0.18452300215960046 * x[2] + 0.2108834310395433 * x[3] + \theta[6] :$	$+\text{Inf} : \text{True}$
7 :	$94.92932352217971 : 0.09025744467169525 * x[1] + 0.09556670612297144 * x[2] + 0.10921909271196732 * x[3] + \theta[7] :$	$+\text{Inf} : \text{True}$
8 :	$18.392872863500727 : 0.019877440458911873 * x[1] + 0.021046701662377272 * x[2] + 0.024053373328431164 * x[3] + \theta[8] :$	$+\text{Inf} : \text{True}$
9 :	$107.84771360750833 : 0.2595586330208133 * x[1] + 0.27482678790439047 * x[2] + 0.3140877576050175 * x[3] + \theta[9] :$	$+\text{Inf} : \text{True}$
10 :	$431.0888808319322 : 1.0723181950379732 * x[1] + 1.1353957359225595 * x[2] + 1.2975951267686392 * x[3] + \theta[10] :$	$+\text{Inf} : \text{True}$
11 :	$1636.2847834552335 : 4.044370036020941 * x[1] + 4.282274155786878 * x[2] + 4.894027606613574 * x[3] + \theta[11] :$	$+\text{Inf} : \text{True}$
12 :	$2945.3159195594317 : 7.30669765730133 * x[1] + 7.7365034018484655 * x[2] + 8.8417181735411 * x[3] + \theta[12] :$	$+\text{Inf} : \text{True}$

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

=====

scenario39

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{39: [837.1669625420426, 738.6767316547433, 126.63029685509886, -0.0]}

- $w = [5687.81083374], \theta = -100000.0$
- $e = [5687.81083374]$
- $E = [[38.68747361 \ 40.96320735 \ 46.81509412]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=39, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

=====

scenario40

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{40: [899.301494501913, 793.5013186781583, 136.0287974876843, -0.0]}

► w = [888.72147692], theta = -100000.0

► e = [888.72147692]

► E = [[5.96722514 6.31823838 7.22084386]]

✓ Added optimality cut to master problem.

optimality_cut : Size=40, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

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scenario41

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{41: [790.9176783845425, 697.8685397510667, 119.6346068144686, -0.0]}

► w = [618.77677191], theta = -100000.0

► e = [618.77677191]

► E = [[5.13374779 5.43573295 6.21226623]]

✓ Added optimality cut to master problem.

optimality_cut : Size=41, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] +
88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] +
166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] +
116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] +
208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] +
203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] +
119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] +
46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] +
7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] +
6.212266228054911*x[3] + theta[41] : +Inf : True

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scenario42

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{42: [714.9342170221395, 630.8243091371817, 108.14131013780259, -0.0]}

► w = [2531.70822734], theta = -100000.0

► e = [2531.70822734]

► E = [[21.02149676 22.25805539 25.43777759]]

✓ Added optimality cut to master problem.

optimality_cut : Size=42, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] +
6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] +
25.43777758804692*x[3] + theta[42] : +Inf : True

=====

scenario43

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{43: [663.6591349773116, 585.581589685863, 100.38541537471939, -0.0]}

► w = [7132.38376237],theta =-100000.0

► e = [7132.38376237]

► E = [[58.72855736 62.18317838 71.06648957]]

✓ Added optimality cut to master problem.

optimality_cut : Size=43, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + \text{theta}[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + \text{theta}[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

=====

scenario44

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{44: [631.9362283253961, 557.5907896988788, 95.58699251980781, -0.0]}

➤ w = [12125.74103998], theta = -100000.0

➤ e = [12125.74103998]

➤ E = [[100.08716573 105.97464607 121.11388122]]

✓ Added optimality cut to master problem.

optimality_cut : Size=44, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

=====

scenario45

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{45: [616.7128845850373, 544.1584275750328, 93.28430187000563, -0.0]}

► w = [10868.6576601], theta = -100000.0

► e = [10868.6576601]

► E = [[89.75725662 95.03709525 108.61382314]]

✓ Added optimality cut to master problem.

optimality_cut : Size=45, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

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scenario46

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{46: [616.4291966581281, 543.9081146983483, 93.24139109114543, -0.0]}

► w = [6497.8889436],theta =-100000.0

► e = [6497.8889436]

► E = [[53.59482007 56.74745655 64.85423606]]

✓ Added optimality cut to master problem.

optimality_cut : Size=46, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

=====

scenario47

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{47: [630.9276113635274, 556.7008335560535, 95.43442860960917, -0.0]}

► w = [2286.18475647], theta = -100000.0

► e = [2286.18475647]

► E = [[18.85792156 19.96721107 22.81966979]]

✓ Added optimality cut to master problem.

optimality_cut : Size=47, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] +
64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

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scenario48

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{48: [660.688466791545, 582.9604118748925, 99.93607060712444, -0.0]}

► w = [326.45783065],theta =-100000.0

► e = [326.45783065]

► E = [[2.68668965 2.84473022 3.25112025]]

✓ Added optimality cut to master problem.

optimality_cut : Size=48, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + \text{theta}[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + \text{theta}[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

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scenario49

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{49: [565.8400304989508, 499.27061514613297, 85.58924831076564, -0.0]}

► w = [139.79577224], theta = -100000.0

► e = [139.79577224]

► E = [[1.35597905 1.43574252 1.6408486]]

✓ Added optimality cut to master problem.

optimality_cut : Size=49, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

=====

scenario50

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{50: [498.4145324646958, 439.77752864531976, 75.3904334820548, -0.0]}

➤ w = [656.73444278], theta = -100000.0

➤ e = [656.73444278]

➤ E = [[6.35538339 6.72922947 7.69054796]]

✓ Added optimality cut to master problem.

optimality_cut : Size=50, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] +
7.690547963331018*x[3] + theta[50] : +Inf : True

=====

scenario51

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{51: [450.9854055553554, 397.9282990194311, 68.21627983190247, -0.0]}

► w = [1689.86884317],theta =-100000.0

► e = [1689.86884317]

► E = [[16.3614912 17.32393186 19.79877926]]

✓ Added optimality cut to master problem.

optimality_cut : Size=51, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

=====

scenario52

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{52: [418.84818555548077, 369.57192843130645, 63.355187731081095, -0.0]}

► w = [2604.25018901], theta = -100000.0

► e = [2604.25018901]

► E = [[25.13873407 26.61748313 30.41998072]]

✓ Added optimality cut to master problem.

optimality_cut : Size=52, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

=====

scenario53

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{53: [398.9068234411843, 351.97660891869197, 60.33884724320433, -0.0]}

► w = [2365.28281193],theta =-100000.0

► e = [2365.28281193]

► E = [[22.88799703 24.23434979 27.69639976]]

✓ Added optimality cut to master problem.

optimality_cut : Size=53, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

=====

scenario54

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{54: [389.0737702433591, 343.3003855088462, 58.85149465865934, -0.0]}

- $w = [1457.88230379], \theta = -100000.0$
- $e = [1457.88230379]$
- $E = [[14.15785952 \ 14.99067479 \ 17.13219976]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=54, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

=====

scenario55

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{55: [389.9530769641091, 344.07624438009617, 58.98449903658791, -0.0]}

► w = [465.64985073],theta =-100000.0

► e = [465.64985073]

► E = [[4.47284953 4.73595833 5.4125238]]

✓ Added optimality cut to master problem.

optimality_cut : Size=55, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

=====

scenario56

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{56: [399.31974987278693, 352.3409557701061, 60.40130670344676, -0.0]}

► w = [82.21288968],theta =-100000.0

► e = [82.21288968]

► E = [[0.78965481 0.83610509 0.95554867]]

✓ Added optimality cut to master problem.

optimality_cut : Size=56, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] +
1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] +
7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] +
19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] +
30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] +
27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] +
17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] +
5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] +
0.9555486720485278*x[3] + theta[56] : +Inf : True

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scenario57

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{57: [333.14048772636994, 293.94748917032643, 50.39099814348452, -0.0]}

- $w = [13.71754949], \theta = -100000.0$
- $e = [13.71754949]$
- $E = [[0.14736469 \ 0.15603321 \ 0.17832366]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=57, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 12.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

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scenario58

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{58: [283.5258748892817, 250.16988960818972, 42.88626678997537, -0.0]}

► w = [46.69837939], theta = -100000.0

► e = [46.69837939]

► E = [[0.52908764 0.56021044 0.6402405]]

✓ Added optimality cut to master problem.

optimality_cut : Size=58, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

=====

scenario59

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{59: [249.0151053977583, 219.71921064508086, 37.66615039629957, -0.0]}

► w = [164.05701061], theta = -100000.0

► e = [164.05701061]

► E = [[1.83872754 1.94688798 2.22501484]]

✓ Added optimality cut to master problem.

optimality_cut : Size=59, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] +
2.2250148362102085*x[3] + theta[59] : +Inf : True

=====

scenario60

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{60: [225.23416283227237, 198.73602602847564, 34.06903303345296, -0.0]}

► w = [268.95608856], theta = -100000.0

► e = [268.95608856]

► E = [[2.99694888 3.17323999 3.62655999]]

✓ Added optimality cut to master problem.

optimality_cut : Size=60, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

=====

scenario61

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{61: [209.11865848162327, 184.5164633661382, 31.6313937199094, -0.0]}

➤ w = [232.49074384], theta = -100000.0

➤ e = [232.49074384]

➤ E = [[2.57989166 2.73164999 3.12188571]]

✓ Added optimality cut to master problem.

optimality_cut : Size=61, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

=====

scenario62

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{62: [199.2294371220273, 175.7906798135535, 30.135545110894885, -0.0]}

► w = [131.25704093], theta = -100000.0

► e = [131.25704093]

► E = [[1.46314099 1.5492081 1.77052355]]

✓ Added optimality cut to master problem.

optimality_cut : Size=62, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

=====

scenario63

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{63: [194.80853591228953, 171.88988462849076, 29.466837364884128, -0.0]}

► w = [48.1291677], theta = -100000.0

► e = [48.1291677]

► E = [[0.53141821 0.5626781 0.64306069]]

✓ Added optimality cut to master problem.

optimality_cut : Size=63, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63}, Active=True

Key : Lower : Body : Upper : Active

1 : $9.785496 : 0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : $60.25830152831999 : 0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + \text{theta}[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + \text{theta}[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

=====

scenario64

0.0

w[1]:0.0

w[2]:0.0

w[3]:0.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{64: [195.54491217803798, 172.53962839238645, 29.57822201012339, -0.0]}

► w = [8.05184932], theta = -100000.0

► e = [8.05184932]

► E = [[0.08685616 0.09196535 0.10510325]]

✓ Added optimality cut to master problem.

optimality_cut : Size=64, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

=====

[Thetatheta[1] from previous iteration: -100,000.00]

[Thetatheta[2] from previous iteration: -100,000.00]

[Thetatheta[3] from previous iteration: -100,000.00]

[Thetatheta[4] from previous iteration: -100,000.00]

[Thetatheta[5] from previous iteration: -100,000.00]

[Thetatheta[6] from previous iteration: -100,000.00]

[Thetatheta[7] from previous iteration: -100,000.00]

[Thetatheta[8] from previous iteration: -100,000.00]

[Thetatheta[9] from previous iteration: -100,000.00]

[Thetatheta[10] from previous iteration: -100,000.00]

[Thetatheta[11] from previous iteration: -100,000.00]

[Thetatheta[12] from previous iteration: -100,000.00]

[Thetatheta[13] from previous iteration: -100,000.00]

[Thetatheta[14] from previous iteration: -100,000.00]

[Thetatheta[15] from previous iteration: -100,000.00]

[Thetatheta[16] from previous iteration: -100,000.00]

[Thetatheta[17] from previous iteration: -100,000.00]

[Thetatheta[18] from previous iteration: -100,000.00]

[Thetatheta[19] from previous iteration: -100,000.00]

[Thetatheta[20] from previous iteration: -100,000.00]

[Thetatheta[21] from previous iteration: -100,000.00]

[Thetatheta[22] from previous iteration: -100,000.00]

[Thetatheta[23] from previous iteration: -100,000.00]

[Thetatheta[24] from previous iteration: -100,000.00]

[Thetatheta[25] from previous iteration: -100,000.00]

[Thetatheta[26] from previous iteration: -100,000.00]

[Thetatheta[27] from previous iteration: -100,000.00]

[Thetatheta[28] from previous iteration: -100,000.00]

[Thetatheta[29] from previous iteration: -100,000.00]

[Thetatheta[30] from previous iteration: -100,000.00]

[Thetatheta[31] from previous iteration: -100,000.00]

[Thetatheta[32] from previous iteration: -100,000.00]

[Thetatheta[33] from previous iteration: -100,000.00]

[Thetatheta[34] from previous iteration: -100,000.00]

[Thetatheta[35] from previous iteration: -100,000.00]

[Thetatheta[36] from previous iteration: -100,000.00]

[Thetatheta[37] from previous iteration: -100,000.00]

[Thetatheta[38] from previous iteration: -100,000.00]

[Thetatheta[39] from previous iteration: -100,000.00]

[Thetatheta[40] from previous iteration: -100,000.00]

[Thetatheta[41] from previous iteration: -100,000.00]

[Thetatheta[42] from previous iteration: -100,000.00]

[Thetatheta[43] from previous iteration: -100,000.00]

[Thetatheta[44] from previous iteration: -100,000.00]

[Thetatheta[45] from previous iteration: -100,000.00]

[Thetatheta[46] from previous iteration: -100,000.00]

[Thetatheta[47] from previous iteration: -100,000.00]

[Thetatheta[48] from previous iteration: -100,000.00]

[Thetatheta[49] from previous iteration: -100,000.00]

[Thetatheta[50] from previous iteration: -100,000.00]

[Thetatheta[51] from previous iteration: -100,000.00]

[Thetatheta[52] from previous iteration: -100,000.00]

[Thetatheta[53] from previous iteration: -100,000.00]

[Thetatheta[54] from previous iteration: -100,000.00]

[Thetatheta[55] from previous iteration: -100,000.00]

[Thetatheta[56] from previous iteration: -100,000.00]

[Thetatheta[57] from previous iteration: -100,000.00]

[Thetatheta[58] from previous iteration: -100,000.00]

[Thetatheta[59] from previous iteration: -100,000.00]

[Thetatheta[60] from previous iteration: -100,000.00]

[Thetatheta[61] from previous iteration: -100,000.00]

[Thetatheta[62] from previous iteration: -100,000.00]

[Thetatheta[63] from previous iteration: -100,000.00]

[Thetatheta[64] from previous iteration: -100,000.00]

=====

↻ Iteration 2

Master Problem Objective Value : -647,738.91

Decision Variables:

x[1]: 0.0

x[2]: 0.0

x[3]: 500.0

=====

-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (0.0, 0.0, 500.0)-----

scenario1

0.0

w[1]:0.0

w[2]:0.0

w[3]:1602.0000000000002

w[4]:0.0

y[1]:200.0

y[2]:240.0

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

➤ w = [4.02683146],theta =4.026831455999998

➤ e = [9.785496]

➤ E = [[0.00951779 0.01007766 0.01151733]]

scenario2

0.0

w[1]:0.0

w[2]:0.0

w[3]:1449.9999999999993

w[4]:0.0

y[1]:200.0

y[2]:240.0

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

➤ w = [28.16153276],theta =28.161532755072002

➤ e = [60.25830153]

➤ E = [[0.05304883 0.05616935 0.06419354]]

scenario3

0.0

w[1]:0.0

w[2]:0.0

w[3]:1522.9999999999998

w[4]:0.0

y[1]:200.0

y[2]:240.0

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ w = [78.89576975],theta =78.8957697471304

➤ e = [179.09259324]

➤ E = [[0.16560308 0.17534444 0.20039365]]

scenario4

0.0

w[1]:0.0

w[2]:0.0

w[3]:1563.0000000000007

w[4]:0.0

y[1]:200.0

y[2]:240.0

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

➤ w = [134.65818168],theta =134.65818168182815

➤ e = [316.22021003]

➤ E = [[0.30008169 0.31773355 0.36312406]]

scenario5

0.0

w[1]:0.0

w[2]:0.0

w[3]:1629.00000000000014

w[4]:0.0

y[1]:200.0

y[2]:240.0

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

➤ w = [109.17419099],theta =109.17419099389721

➤ e = [271.85361107]

➤ E = [[0.26887293 0.28468899 0.32535884]]

scenario6

0.0

w[1]:0.0

w[2]:0.0

w[3]:1481.9999999999993

w[4]:0.0

y[1]:200.0

y[2]:240.0

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

➤ w = [88.23964827],theta =88.2396482704823

➤ e = [193.68136379]

➤ E = [[0.17427172 0.184523 0.21088343]]

scenario7

0.0

w[1]:0.0

w[2]:0.0

w[3]:1565.9999999999989

w[4]:0.0

y[1]:200.0

y[2]:240.0

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [40.31977717],theta =40.319777166196054

➤ e = [94.92932352]

➤ E = [[0.09025744 0.09556671 0.10921909]]

scenario8

0.0

w[1]:0.0

w[2]:0.0

w[3]:1780.0000000000011

w[4]:0.0

y[1]:200.0

y[2]:240.0

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

➤ w = [6.3661862],theta =6.366186199285146

➤ e = [18.39287286]

➤ E = [[0.01987744 0.0210467 0.02405337]]

scenario9

0.0

w[1]:0.0

w[2]:0.0

w[3]:3963.9999999999995

w[4]:0.0

y[1]:200.0

y[2]:240.0

{9: [436.5264598399147, 385.17040574110115, 66.02921241276017, -0.0]}

➤ $w = [-49.1961652], \theta = -49.19616519500042$

➤ $e = [107.84771361]$

➤ $E = [[0.25955863 \ 0.27482679 \ 0.31408776]]$

scenario10

0.0

$w[1]:0.0$

$w[2]:0.0$

$w[3]:4096.999999999999$

$w[4]:0.0$

$y[1]:200.0$

$y[2]:240.0$

{10: [436.2208913180268, 384.90078645708235, 65.98299196407125, -0.0]}

➤ $w = [-217.70868255], \theta = -217.70868255238736$

➤ $e = [431.08888083]$

➤ $E = [[1.0723182 \ 1.13539574 \ 1.29759513]]$

scenario11

0.0

$w[1]:0.0$

$w[2]:0.0$

$w[3]:4070.9999999999995$

$w[4]:0.0$

$y[1]:200.0$

y[2]:240.0

{11: [446.4982555174795, 393.9690489860112, 67.53755125474477, -0.0]}

➤ w = [-810.72901985],theta =-810.7290198515534

➤ e = [1636.28478346]

➤ E = [[4.04437004 4.28227416 4.89402761]]

scenario12

0.0

w[1]:0.0

w[2]:0.0

w[3]:4085.9999999999995

w[4]:0.0

y[1]:200.0

y[2]:240.0

{12: [467.5104634221321, 412.5092324312929, 70.71586841679306, -0.0]}

➤ w = [-1475.54316721],theta =-1475.543167211118

➤ e = [2945.31591956]

➤ E = [[7.30669766 7.7365034 8.84171817]]

scenario13

0.0

w[1]:0.0

w[2]:0.0

w[3]:3974.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{13: [501.2553686719416, 442.2841488281836, 75.82013979911719, -0.0]}

➤ w = [-1205.35656958],theta =-1205.3565695784491

➤ e = [2621.27072206]

➤ E = [[6.32456455 6.69659776 7.65325458]]

scenario14

0.0

w[1]:0.0

w[2]:0.0

w[3]:3965.9999999999999

w[4]:0.0

y[1]:200.0

y[2]:240.0

{14: [549.5312732287364, 484.880535201826, 83.12237746317017, -0.0]}

➤ w = [-889.11758695],theta =-889.1175869528272

➤ e = [1945.98721461]

➤ E = [[4.68579821 4.9614334 5.6702096]]

scenario15

0.0

w[1]:0.0

w[2]:0.0

w[3]:4085.9999999999995

w[4]:0.0

y[1]:200.0

y[2]:240.0

{15: [614.7056822336644, 542.3873666767626, 92.98069143030214, -0.0]}

➤ w = [-342.37350198],theta =-342.3735019846581

➤ e = [683.40808201]

➤ E = [[1.69538901 1.79511777 2.05156317]]

scenario16

0.0

w[1]:0.0

w[2]:0.0

w[3]:4132.0

w[4]:0.0

y[1]:200.0

y[2]:240.0

{16: [700.6169483826413, 618.1914250435068, 105.97567286460118, -0.0]}

➤ w = [-104.58150401],theta =-104.5815040126935

➤ e = [201.94253218]

➤ E = [[0.50661612 0.53641706 0.61304807]]

scenario17

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:459.0

y[1]:200.0

y[2]:240.0

{17: [667.5127975715615, 588.9818802102011, 28.046756200485788, 72.92156612126298]}

► w = [-481.35525797], theta = -528.2146563559807

► e = [-227.73966035]

► E = [[1.50901281 1.59777826 0.5072312]]

✓ Added optimality cut to master problem.

optimality_cut : Size=65, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

=====

scenario18

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:486.0000000000009

y[1]:200.0

y[2]:240.0

{18: [651.7194447810183, 575.0465689244278, 27.383169948782292, 71.1962418668339]}

➤ w = [-2826.00885832],theta =-3116.6603961193705

➤ e = [-1334.1080399]

➤ E = [[8.87680987 9.39897516 2.98380164]]

✓ Added optimality cut to master problem.

optimality_cut : Size=66, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] +
2.983801636835073*x[3] + theta[18] : +Inf : True

=====

scenario19

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:436.9999999999991

y[1]:200.0

y[2]:240.0

{19: [651.7911339199442, 575.1098240470095, 27.386182097476656, 71.20407345343926]}

► w = [-8110.01798991],theta =-8863.029548308585

► e = [-3843.9245192]

► E = [[25.38325615 26.87638887 8.53218694]]

✓ Added optimality cut to master problem.

optimality_cut : Size=67, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

=====

scenario20

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:419.00000000000009

y[1]:200.0

y[2]:240.0

{20: [667.2646554392036, 588.7629312698856, 28.036330060470753, 72.89445815722391]}

► w = [-15141.85553179], theta = -16491.84631797318

► e = [-7187.3935743]

► E = [[47.32904865 50.11311033 15.90892391]]

✓ Added optimality cut to master problem.

optimality_cut : Size=68, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

=====

scenario21

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:375.0

y[1]:200.0

y[2]:240.0

{21: [699.3400674261661, 617.064765376029, 29.384036446477584, 76.39849476084166]}

► $w = [-16241.87922561], \theta = -17542.563598911573$

► $e = [-7737.40447709]$

► $E = [[50.60162475 \ 53.57819092 \ 17.0089495 \]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=69, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

=====

scenario22

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:348.0

y[1]:200.0

y[2]:240.0

{22: [749.8953609004036, 661.6723772650621, 31.50820844119345, 81.9213419471029]}

➤ w = [-9339.3858739], theta = -10034.996372645512

➤ e = [-4459.0416586]

➤ E = [[29.03804808 30.74616856 9.76068843]]

✓ Added optimality cut to master problem.

optimality_cut : Size=70, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] +
8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] +
15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] +
17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] +
9.760688430613165*x[3] + theta[22] : +Inf : True

=====

scenario23

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:502.0

y[1]:200.0

y[2]:240.0

{23: [821.0004390209799, 724.4121520773353, 34.49581676558741, 89.6891235905272]}

➤ w = [-3394.9403028],theta =-3755.1318231416058

➤ e = [-1600.60589792]

➤ E = [[10.67628971 11.30430675 3.58866881]]

✓ Added optimality cut to master problem.

optimality_cut : Size=71, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] +
17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] +
9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] +
3.5886688097575896*x[3] + theta[23] : +Inf : True

=====

scenario24

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:372.0

y[1]:200.0

y[2]:240.0

{24: [918.7980113171591, 810.7041276327875, 38.60495845870419, 100.37289199263078]}

► w = [-657.8593761],theta =-710.1335782528479

► e = [-313.47226268]

► E = [[2.04910332 2.16963881 0.68877423]]

✓ Added optimality cut to master problem.

optimality_cut : Size=72, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

=====

scenario25

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2793.0

y[1]:200.0

y[2]:240.0

{25: [852.5618626813051, 752.2604670717399, 35.8219270034162, 93.13701020888206]}

► w = [-1515.97320421],theta =-2270.3550457953534

► e = [-602.5248122]

► E = [[5.43501793 5.75472487 1.82689678]]

✓ Added optimality cut to master problem.

optimality_cut : Size=73, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

=====

scenario26

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2817.0

y[1]:200.0

y[2]:240.0

{26: [811.1955611040082, 715.760789209419, 34.083847105210445, 88.61800247354708]}

➤ w = [-7522.87425637],theta =-11292.39164218313

➤ e = [-2985.06332947]

➤ E = [[26.99997502 28.58820884 9.07562185]]

✓ Added optimality cut to master problem.

optimality_cut : Size=74, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

=====

scenario27

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2799.0

y[1]:200.0

y[2]:240.0

{27: [791.8323230604556, 698.6755791709902, 33.270265674809075, 86.5026907545035]}

➤ w = [-20448.50082149],theta =-30641.796244353332

➤ e = [-8123.93347247]

➤ E = [[73.33117573 77.6447743 24.6491347]]

✓ Added optimality cut to master problem.

optimality_cut : Size=75, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

=====

scenario28

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2806.0000000000002

y[1]:200.0

y[2]:240.0

{28: [792.0619544341431, 698.8781950889498, 33.27991405185477, 86.52777653482232]}

► $w = [-37623.08926724], \theta = -56415.57249729308$

► $e = [-14940.01901616]$

► $E = [[134.96426799 \ 142.90334258 \ 45.3661405 \]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=76, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

=====

scenario29

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2791.0

y[1]:200.0

y[2]:240.0

{29: [811.0872825796512, 715.6652493349864, 34.07929758738032, 88.60617372718875]}

► w = [-38288.3293945], theta = -57330.416371693835

► e = [-15219.81430252]

► E = [[137.2576648 145.33164508 46.13703018]]

✓ Added optimality cut to master problem.

optimality_cut : Size=77, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] +
46.1370301839617*x[3] + theta[29] : +Inf : True

=====

scenario30

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2815.999999999998

y[1]:200.0

y[2]:240.0

{30: [849.71936984892, 749.7523851608117, 35.70249453146724, 92.82648578181474]}

► w = [-22490.7721491],theta =-33757.08559784449

► e = [-8924.90958298]

► E = [[80.71688227 85.46493417 27.13172513]]

✓ Added optimality cut to master problem.

optimality_cut : Size=78, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

=====

scenario31

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2806.0000000000002

y[1]:200.0

y[2]:240.0

{31: [910.3468468876405, 803.2472178420356, 38.24986751628743, 99.44965554234724]}

► $w = [-8156.69244817], \theta = -12230.906156562916$

► $e = [-3238.99878128]$

► $E = [[29.26027732 \ 30.9814701 \ 9.83538733]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=79, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

=====

scenario32

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:2665.0

y[1]:200.0

y[2]:240.0

{32: [1000.4984951349236, 882.7927898249325, 42.03775189642538, 109.2981549307059]}

► w = [-1398.57498672],theta =-2068.5180273658852

► e = [-560.7836103]

► E = [[4.98485869 5.27808567 1.67558275]]

✓ Added optimality cut to master problem.

optimality_cut : Size=80, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80}, Active=True

Key	: Lower	: Body	: Upper	: Active
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1 :	9.785496	: 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1]	: +Inf	: True
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2 :	60.25830152831999	: 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2]	: +Inf	: True
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3 :	179.09259323679183	: 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3]	: +Inf	: True
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4 :	316.22021002633863	: 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4]	: +Inf	: True
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5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] +
1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] +
9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] +
24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] +
46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
1.6755827528396188*x[3] + theta[32] : +Inf : True

=====

scenario33

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5092.0

y[1]:200.0

y[2]:240.0

{33: [905.7312776757436, 799.1746567727149, 38.05593603679596, 98.94543369566944]}

► w = [-1799.95444029],theta =-3210.6788557533364

► e = [-618.02840124]

► E = [[7.03245993 7.44613405 2.36385208]]

✓ Added optimality cut to master problem.

optimality_cut : Size=81, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
2.3638520781127887*x[3] + theta[33] : +Inf : True

=====

scenario34

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5247.0

y[1]:200.0

y[2]:240.0

{34: [839.6853529076284, 740.8988408008486, 35.2808971809928, 91.73033267058125]}

► w = [-9442.50523164],theta =-16999.057403340703

► e = [-3212.6784973]

► E = [[37.06746907 39.24790843 12.45965347]]

✓ Added optimality cut to master problem.

optimality_cut : Size=82, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

=====

scenario35

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5189.0

y[1]:200.0

y[2]:240.0

{35: [799.8590766192195, 705.7580087816643, 33.60752422769831, 87.37956299201556]}

► w = [-24551.20385149], theta = -44047.94360320722

► e = [-8381.71654239]

► E = [[96.20844949 101.86777005 32.33897462]]

✓ Added optimality cut to master problem.

optimality_cut : Size=83, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] +
12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] +
32.33897461819961*x[3] + theta[35] : +Inf : True

=====

scenario36

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5200.0

y[1]:200.0

y[2]:240.0

{36: [780.6064686449948, 688.7704135102896, 32.79859111953762, 85.27633691079775]}

➤ w = [-43992.75026864],theta =-78979.9257763979

➤ e = [-15009.29126812]

➤ E = [[172.45158105 182.5957917 57.966918]]

✓ Added optimality cut to master problem.

optimality_cut : Size=84, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] +
57.966918001025995*x[3] + theta[36] : +Inf : True

=====

scenario37

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5227.0

y[1]:200.0

y[2]:240.0

{37: [780.3098381869098, 688.5086807531557, 32.7861276549122, 85.24393190277164]}

► w = [-42761.90370144],theta =-76892.56815691586

► e = [-14566.22079452]

► E = [[167.7643133 177.63280231 56.39136581]]

✓ Added optimality cut to master problem.

optimality_cut : Size=85, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

=====

scenario38

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5196.0

y[1]:200.0

y[2]:240.0

{38: [799.021668106632, 705.0191189176163, 33.572338996076994, 87.2880813898001]}

► w = [-25106.2008374], theta = -45062.351157063924

► e = [-8567.6609118]

► E = [[98.40431256 104.19280153 33.07707985]]

✓ Added optimality cut to master problem.

optimality_cut : Size=86, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] +
24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] +
45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] +
46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] +
12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] +
32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] +
57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] +
56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] +
33.07707985120687*x[3] + theta[38] : +Inf : True

=====

scenario39

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5203.0

y[1]:200.0

y[2]:240.0

{39: [837.1669625420426, 738.6767316547433, 35.17508245974971, 91.45521439534915]}

► w = [-9868.35179654], theta = -17719.73622477507

► e = [-3366.2553914]

► E = [[38.68747361 40.96320735 13.00419281]]

✓ Added optimality cut to master problem.

optimality_cut : Size=87, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] +
12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] +
32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] +
57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] +
56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] +
33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] +
13.004192810287009*x[3] + theta[39] : +Inf : True

=====

scenario40

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:5059.0

y[1]:200.0

y[2]:240.0

{40: [899.301494501913, 793.5013186781583, 37.78577707991233, 98.24302040777198]}

► w = [-1528.8729979],theta =-2721.7004544795846

► e = [-525.97801695]

► E = [[5.96722514 6.31823838 2.00578996]]

✓ Added optimality cut to master problem.

optimality_cut : Size=88, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

=====

scenario41

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7665.0

y[1]:200.0

y[2]:240.0

{41: [790.9176783845425, 697.8685397510667, 33.231835226241294, 86.4027715882273]}

► w = [-1229.02957809], theta = -2487.356342114843

► e = [-366.21482419]

► E = [[5.13374779 5.43573295 1.72562951]]

✓ Added optimality cut to master problem.

optimality_cut : Size=89, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

=====

scenario42

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7676.0

y[1]:200.0

y[2]:240.0

{42: [714.9342170221395, 630.8243091371817, 30.03925281605629, 78.1020573217463]}

► w = [-5031.38259547],theta =-10187.180566686236

► e = [-1498.35793046]

► E = [[21.02149676 22.25805539 7.06604933]]

✓ Added optimality cut to master problem.

optimality_cut : Size=90, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.13219975857664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

=====

scenario43

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7562.000000000002

y[1]:200.0

y[2]:240.0

{43: [663.6591349773116, 585.581589685863, 27.884837604088734, 72.50057777063066]}

► w = [-14091.55249052], theta = -28400.861024067934

► e = [-4221.20671651]

► E = [[58.72855736 62.18317838 19.74069155]]

✓ Added optimality cut to master problem.

optimality_cut : Size=91, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] +
27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] +
9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] +
1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] +
2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] +
12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] +
32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] +
57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] +
56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] +
33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] +
13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] +
2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] +
19.740691548023207*x[3] + theta[43] : +Inf : True

=====

scenario44

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7594.999999999998

y[1]:200.0

y[2]:240.0

{44: [631.9362283253961, 557.5907896988788, 26.551942366613293, 69.03505015319452]}

► w = [-23997.83137454], theta = -48431.19957011133

► e = [-7176.45898285]

► E = [[100.08716573 105.97464607 33.64274478]]

✓ Added optimality cut to master problem.

optimality_cut : Size=92, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25] : +\text{Inf} : \text{True}$

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26] : +\text{Inf} : \text{True}$

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27] : +\text{Inf} : \text{True}$

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28] : +\text{Inf} : \text{True}$

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29] : +\text{Inf} : \text{True}$

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30] : +\text{Inf} : \text{True}$

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31] : +\text{Inf} : \text{True}$

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32] : +\text{Inf} : \text{True}$

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33] : +\text{Inf} : \text{True}$

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34] : +\text{Inf} : \text{True}$

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35] : +\text{Inf} : \text{True}$

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36] : +\text{Inf} : \text{True}$

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37] : +\text{Inf} : \text{True}$

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38] : +\text{Inf} : \text{True}$

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39] : +\text{Inf} : \text{True}$

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40] : +\text{Inf} : \text{True}$

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41] : +\text{Inf} : \text{True}$

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

=====

scenario45

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7602.0

y[1]:200.0

y[2]:240.0

{45: [616.7128845850373, 544.1584275750328, 25.912306075001577, 67.37199579500405]}

► w = [-21517.7240736],theta =-43438.25390863429

► e = [-6432.47086006]

► E = [[89.75725662 95.03709525 30.17050643]]

✓ Added optimality cut to master problem.

optimality_cut : Size=93, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

=====

scenario46

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7585.0

y[1]:200.0

y[2]:240.0

{46: [616.4291966581281, 543.9081146983483, 25.900386414207077, 67.34100467693835]}

➤ w = [-12853.22216037],theta =-25929.22908451793

➤ e = [-3845.68937478]

➤ E = [[53.59482007 56.74745655 18.01506557]]

✓ Added optimality cut to master problem.

optimality_cut : Size=94, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

=====

scenario47

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7586.0

y[1]:200.0

y[2]:240.0

{47: [630.9276113635274, 556.7008335560535, 26.509563502669227, 68.92486510693995]}

► $w = [-4522.44670295], \theta = -9123.650137923129$

► $e = [-1353.04812118]$

► $E = [[18.85792156 \ 19.96721107 \ 6.33879716]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=95, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

=====

scenario48

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:7555.0

y[1]:200.0

y[2]:240.0

{48: [660.688466791545, 582.9604118748925, 27.760019613090133, 72.1760509940343]}

► w = [-644.75421553],theta =-1299.102293845546

► e = [-193.20973651]

► E = [[2.68668965 2.84473022 0.90308896]]

✓ Added optimality cut to master problem.

optimality_cut : Size=96, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

=====

scenario49

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9976.0

y[1]:200.0

y[2]:240.0

{49: [565.8400304989508, 499.27061514613297, 23.774791197434915, 61.814457113330725]}

➤ w = [-310.63191187],theta =-680.6285263667578

➤ e = [-82.73627337]

➤ E = [[1.35597905 1.43574252 0.45579128]]

✓ Added optimality cut to master problem.

optimality_cut : Size=97, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

=====

scenario50

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9939.0

y[1]:200.0

y[2]:240.0

{50: [498.4145324646958, 439.77752864531976, 20.941787078348572, 54.448646403706235]}

► w = [-1456.81122975], theta = -3188.539538888498

► e = [-388.67956817]

► E = [[6.35538339 6.72922947 2.13626332]]

✓ Added optimality cut to master problem.

optimality_cut : Size=98, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

=====

scenario51

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9947.0

y[1]:200.0

y[2]:240.0

{51: [450.9854055553554, 397.9282990194311, 18.94896661997292, 49.26731321192955]}

➤ w = [-3749.95691147], theta = -8209.520788592888

➤ e = [-1000.1264582]

➤ E = [[16.3614912 17.32393186 5.49966091]]

✓ Added optimality cut to master problem.

optimality_cut : Size=99, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

=====

scenario52

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9899.0

y[1]:200.0

y[2]:240.0

{52: [418.84818555548077, 369.57192843130645, 17.598663258633653, 45.75652447244744]}

► w = [-5766.28825014],theta =-12605.740170007915

► e = [-1541.29092819]

► E = [[25.13873407 26.61748313 8.44999464]]

✓ Added optimality cut to master problem.

optimality_cut : Size=100, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

=====

scenario53

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9937.999999999998

y[1]:200.0

y[2]:240.0

{53: [398.9068234411843, 351.97660891869197, 16.760790900890107, 43.57805634231422]}

► w = [-5246.58344549],theta =-11482.917070081932

► e = [-1399.86125604]

► E = [[22.88799703 24.23434979 7.69344438]]

✓ Added optimality cut to master problem.

optimality_cut : Size=101, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.13219975857664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

=====

scenario54

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9995.000000000002

y[1]:200.0

y[2]:240.0

{54: [389.0737702433591, 343.3003855088462, 16.347637405183168, 42.50385725347617]}

➤ w = [-3242.30049094],theta =-7108.217575499598

➤ e = [-862.82830225]

➤ E = [[14.15785952 14.99067479 4.75894438]]

✓ Added optimality cut to master problem.

optimality_cut : Size=102, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] +
4.7589443773854665*x[3] + theta[54] : +Inf : True

=====

scenario55

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9821.0

y[1]:200.0

y[2]:240.0

{55: [389.9530769641091, 344.07624438009617, 16.384583065718882, 42.599915970869034]}

► w = [-1027.32810435],theta =-2240.6120511200556

► e = [-275.58868717]

► E = [[4.47284953 4.73595833 1.50347883]]

✓ Added optimality cut to master problem.

optimality_cut : Size=103, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

=====

scenario56

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:9820.0

y[1]:200.0

y[2]:240.0

{56: [399.31974987278693, 352.3409557701061, 16.77814075095745, 43.6231659524893]}

► w = [-181.37170152],theta =-395.56144634457246

► e = [-48.65660818]

► E = [[0.78965481 0.83610509 0.26543019]]

✓ Added optimality cut to master problem.

optimality_cut : Size=104, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : $0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + \text{theta}[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \text{theta}[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

=====

scenario57

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:11694.0

y[1]:200.0

y[2]:240.0

{57: [333.14048772636994, 293.94748917032643, 13.997499484301272, 36.39349865918325]}

► w = [-32.88572529],theta =-75.44428262046628

► e = [-8.1185497]

► E = [[0.14736469 0.15603321 0.04953435]]

✓ Added optimality cut to master problem.

optimality_cut : Size=105, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

=====

scenario58

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:12661.0

y[1]:200.0

y[2]:240.0

{58: [283.5258748892817, 250.16988960818972, 11.912851886104283, 30.973414903871088]}

► w = [-116.56010799],theta =-273.42187043356347

► e = [-27.63781638]

► E = [[0.52908764 0.56021044 0.17784458]]

✓ Added optimality cut to master problem.

optimality_cut : Size=106, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

=====

scenario59

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:12460.0

y[1]:200.0

y[2]:240.0

{59: [249.0151053977583, 219.71921064508086, 10.46281955452767, 27.2033308417719]}

► w = [-406.12480383],theta =-948.4504074901106

► e = [-97.09496547]

► E = [[1.83872754 1.94688798 0.61805968]]

✓ Added optimality cut to master problem.

optimality_cut : Size=107, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \text{theta}[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

=====

scenario60

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:12353.0

y[1]:200.0

y[2]:240.0

{60: [225.23416283227237, 198.73602602847564, 9.463620287070277, 24.605412746382683]}

► w = [-662.8669803],theta =-1544.3239049040533

► e = [-159.17809323]

► E = [[2.99694888 3.17323999 1.00737777]]

✓ Added optimality cut to master problem.

optimality_cut : Size=108, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

=====

scenario61

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:12277.0

y[1]:200.0

y[2]:240.0

{61: [209.11865848162327, 184.5164633661382, 8.786498255530399, 22.844895464379]}

➤ w = [-571.19179995], theta = -1328.4521103093832

➤ e = [-137.59656268]

➤ E = [[2.57989166 2.73164999 0.86719047]]

✓ Added optimality cut to master problem.

optimality_cut : Size=109, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

=====

scenario62

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:12360.0

y[1]:200.0

y[2]:240.0

{62: [199.2294371220273, 175.7906798135535, 8.370984753026367, 21.76456035786852]}

► $w = [-323.58878661], \theta = -754.0047322501949$

► $e = [-77.68273851]$

► $E = \begin{bmatrix} 1.46314099 & 1.5492081 & 0.4918121 \end{bmatrix}$

✓ Added optimality cut to master problem.

optimality_cut : Size=110, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

=====

scenario63

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:12186.0

y[1]:200.0

y[2]:240.0

{63: [194.80853591228953, 171.88988462849076, 8.185232601356711, 21.28160476352742]}

► w = [-117.79859351], theta = -273.40117489469225

► e = [-28.48460945]

► E = [[0.53141821 0.5626781 0.17862797]]

✓ Added optimality cut to master problem.

optimality_cut : Size=111, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

=====

scenario64

0.0

w[1]:0.0

w[2]:0.0

w[3]:6000.0

w[4]:11767.0

y[1]:200.0

y[2]:240.0

{64: [195.54491217803798, 172.53962839238645, 8.21617278058984, 21.36204922953355]}

➤ w = [-19.36305439],theta =-44.499777720408204

➤ e = [-4.76538021]

➤ E = [[0.08685616 0.09196535 0.02919535]]

✓ Added optimality cut to master problem.

optimality_cut : Size=112, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

=====

[Thetatheta[1] from previous iteration: 4.03]

[Thetatheta[2] from previous iteration: 28.16]

[Thetatheta[3] from previous iteration: 78.90]

[Thetatheta[4] from previous iteration: 134.66]

[Thetatheta[5] from previous iteration: 109.17]

[Thetatheta[6] from previous iteration: 88.24]

[Thetatheta[7] from previous iteration: 40.32]

[Thetatheta[8] from previous iteration: 6.37]

[Thetatheta[9] from previous iteration: -49.20]

[Thetatheta[10] from previous iteration: -217.71]

[Thetatheta[11] from previous iteration: -810.73]

[Thetatheta[12] from previous iteration: -1,475.54]

[Thetatheta[13] from previous iteration: -1,205.36]

[Thetatheta[14] from previous iteration: -889.12]

[Thetatheta[15] from previous iteration: -342.37]

[Thetatheta[16] from previous iteration: -104.58]

[Thetatheta[17] from previous iteration: -528.21]

[Thetatheta[18] from previous iteration: -3,116.66]

[Thetatheta[19] from previous iteration: -8,863.03]

[Thetatheta[20] from previous iteration: -16,491.85]

[Thetatheta[21] from previous iteration: -17,542.56]

[Thetatheta[22] from previous iteration: -10,035.00]

[Thetatheta[23] from previous iteration: -3,755.13]

[Thetatheta[24] from previous iteration: -710.13]

[Thetatheta[25] from previous iteration: -2,270.36]

[Thetatheta[26] from previous iteration: -11,292.39]

[Thetatheta[27] from previous iteration: -30,641.80]

[Thetatheta[28] from previous iteration: -56,415.57]

[Thetatheta[29] from previous iteration: -57,330.42]

[Thetatheta[30] from previous iteration: -33,757.09]

[Thetatheta[31] from previous iteration: -12,230.91]

[Thetatheta[32] from previous iteration: -2,068.52]

[Thetatheta[33] from previous iteration: -3,210.68]

[Thetatheta[34] from previous iteration: -16,999.06]

[Thetatheta[35] from previous iteration: -44,047.94]

[Thetatheta[36] from previous iteration: -78,979.93]

[Thetatheta[37] from previous iteration: -76,892.57]

[Thetatheta[38] from previous iteration: -45,062.35]

[Thetatheta[39] from previous iteration: -17,719.74]

[Thetatheta[40] from previous iteration: -2,721.70]

[Thetatheta[41] from previous iteration: -2,487.36]

[Thetatheta[42] from previous iteration: -10,187.18]

[Thetatheta[43] from previous iteration: -28,400.86]

[Thetatheta[44] from previous iteration: -48,431.20]

[Thetatheta[45] from previous iteration: -43,438.25]

[Thetatheta[46] from previous iteration: -25,929.23]

[Thetatheta[47] from previous iteration: -9,123.65]

[Thetatheta[48] from previous iteration: -1,299.10]

[Thetatheta[49] from previous iteration: -680.63]

[Thetatheta[50] from previous iteration: -3,188.54]

[Thetatheta[51] from previous iteration: -8,209.52]

[Thetatheta[52] from previous iteration: -12,605.74]

[Thetatheta[53] from previous iteration: -11,482.92]

[Thetatheta[54] from previous iteration: -7,108.22]

[Thetatheta[55] from previous iteration: -2,240.61]

[Thetatheta[56] from previous iteration: -395.56]

[Thetatheta[57] from previous iteration: -75.44]

[Thetatheta[58] from previous iteration: -273.42]

[Thetatheta[59] from previous iteration: -948.45]

[Thetatheta[60] from previous iteration: -1,544.32]

[Thetatheta[61] from previous iteration: -1,328.45]

[Thetatheta[62] from previous iteration: -754.00]

[Thetatheta[63] from previous iteration: -273.40]

[Theta[64] from previous iteration: -44.50]

=====

↻ Iteration 3

Master Problem Objective Value : -578,176.16

Decision Variables:

x[1]: 0.0

x[2]: 279.3306362633321

x[3]: 220.6693637366679

=====

-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, -0.0, -0.0, -0.0]$$

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [0.0, -0.0, -0.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (0.0, 279.3306362633321, 220.6693637366679)-----

scenario1

0.0

w[1]:0.0

w[2]:0.0

w[3]:707.024641412284

w[4]:0.0

y[1]:200.0

y[2]:105.75369621184262

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

➤ w = [4.42897431],theta =4.428974313775652

➤ e = [9.785496]

➤ E = [[0.00951779 0.01007766 0.01151733]]

scenario2

0.0

w[1]:0.0

w[2]:0.0

w[3]:639.9411548363365

w[4]:0.0

y[1]:200.0

y[2]:118.49117322545052

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

➤ w = [30.40293547],theta =30.402935465929104

➤ e = [60.25830153]

➤ E = [[0.05304883 0.05616935 0.06419354]]

scenario3

0.0

w[1]:0.0

w[2]:0.0

w[3]:672.1588819418903

w[4]:0.0

y[1]:200.0

y[2]:112.37383229128353

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ w = [85.89278036],theta =85.89278036136335

➤ e = [179.09259324]

➤ E = [[0.16560308 0.17534444 0.20039365]]

scenario4

0.0

w[1]:0.0

w[2]:0.0

w[3]:689.8124310408241

w[4]:0.0

y[1]:200.0

y[2]:109.02186465612354

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

➤ w = [147.33714091],theta =147.33714090651148

➤ e = [316.22021003]

➤ E = [[0.30008169 0.31773355 0.36312406]]

scenario5

0.0

w[1]:0.0

w[2]:0.0

w[3]:718.9407870540646

w[4]:0.0

y[1]:200.0

y[2]:103.49111805810955

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

► w = [120.53452747],theta =120.53452747333165

► e = [271.85361107]

► E = [[0.26887293 0.28468899 0.32535884]]

scenario6

0.0

w[1]:0.0

w[2]:0.0

w[3]:654.0639941154833

w[4]:0.0

y[1]:200.0

y[2]:115.80959911732258

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

► w = [95.60292364],theta =95.60292364169105

► e = [193.68136379]

► E = [[0.17427172 0.184523 0.21088343]]

scenario7

0.0

w[1]:0.0

w[2]:0.0

w[3]:691.1364472232434

w[4]:0.0

y[1]:200.0

y[2]:108.77046708348664

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [44.133307],theta =44.13330699861325

➤ e = [94.92932352]

➤ E = [[0.09025744 0.09556671 0.10921909]]

scenario8

0.0

w[1]:0.0

w[2]:0.0

w[3]:785.5829349025382

w[4]:0.0

y[1]:200.0

y[2]:90.83744023538057

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

➤ w = [7.20604171],theta =7.206041708798912

➤ e = [18.39287286]

➤ E = [[0.01987744 0.0210467 0.02405337]]

scenario9

0.0

w[1]:0.0

w[2]:92.17999264435457

w[3]:1749.4667157043027

w[4]:0.0

y[1]:200.0

y[2]:0.0

{9: [436.5264598399147, 275.1217183865008, 66.02921241276017, -0.0]}

► w = [-32.14280123],theta =-38.22937354820887

► e = [92.00070263]

► E = [[0.25955863 0.19630485 0.31408776]]

✓ Added optimality cut to master problem.

optimality_cut : Size=113, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

=====

scenario10

0.0

w[1]:0.0

w[2]:103.32528503126144

w[3]:1808.1647664582565

w[4]:0.0

y[1]:200.0

y[2]:0.0

{10: [436.2208913180268, 274.92913318363026, 65.98299196407125, -0.0]}

➤ w = [-145.1305777],theta =-172.40142350582704

➤ e = [367.74520855]

➤ E = [[1.0723182 0.81099695 1.29759513]]

✓ Added optimality cut to master problem.

optimality_cut : Size=114, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

=====

scenario11

0.0

w[1]:0.0

w[2]:101.14650606840752

w[3]:1796.68995954395

w[4]:0.0

y[1]:200.0

y[2]:0.0

{11: [446.4982555174795, 281.40646356143657, 67.53755125474477, -0.0]}

► w = [-538.51826035],theta =-639.8475391958434

► e = [1395.85110099]

► E = [[4.04437004 3.05876725 4.89402761]]

✓ Added optimality cut to master problem.

optimality_cut : Size=115, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

=====

scenario12

0.0

w[1]:0.0

w[2]:102.4034939315925

w[3]:1803.3100404560498

w[4]:0.0

y[1]:200.0

y[2]:0.0

{12: [467.5104634221321, 294.6494517366378, 70.71586841679306, -0.0]}

► $w = [-982.16324577], \theta = -1166.8228218265815$

► $e = [2512.53480485]$

► $E = [[7.30669766 \ 5.52607386 \ 8.84171817]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=116, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

=====

scenario13

0.0

w[1]:0.0

w[2]:93.01798455314452

w[3]:1753.8801029790366

w[4]:0.0

y[1]:200.0

y[2]:0.0

{13: [501.2553686719416, 315.9172491629883, 75.82013979911719, -0.0]}

► w = [-788.85220267], theta = -938.1330105497959

► e = [2236.10441188]

► E = [[6.32456455 4.78328411 7.65325458]]

✓ Added optimality cut to master problem.

optimality_cut : Size=117, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] +
1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] +
0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] +
0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] +
0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] +
0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] +
1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] +
0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] +
0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] +
0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] +
0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

=====

scenario14

0.0

w[1]:0.0

w[2]:92.34759102611253

w[3]:1750.3493931592493

w[4]:0.0

y[1]:200.0

y[2]:0.0

{14: [549.5312732287364, 346.34323942987567, 83.12237746317017, -0.0]}

► w = [-581.10984436],theta =-691.1346799293847

► e = [1660.04623614]

► E = [[4.68579821 3.543881 5.6702096]]

✓ Added optimality cut to master problem.

optimality_cut : Size=118, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

=====

scenario15

0.0

w[1]:0.0

w[2]:102.4034939315925

w[3]:1803.3100404560498

w[4]:0.0

y[1]:200.0

y[2]:0.0

{15: [614.7056822336644, 387.4195476262589, 92.98069143030214, -0.0]}

► w = [-227.89348184], theta = -270.7404463533592

► e = [582.98893527]

► E = [[1.69538901 1.28222698 2.05156317]]

✓ Added optimality cut to master problem.

optimality_cut : Size=119, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

=====

scenario16

0.0

w[1]:0.0

w[2]:106.25825671202648

w[3]:1823.6116219198234

w[4]:0.0

y[1]:200.0

y[2]:0.0

{16: [700.6169483826413, 441.56530360250486, 105.97567286460118, -0.0]}

► w = [-70.03852687],theta =-83.17611549770467

► e = [172.26934378]

► E = [[0.50661612 0.38315505 0.61304807]]

✓ Added optimality cut to master problem.

optimality_cut : Size=120, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

=====

scenario17

0.0

w[1]:0.0

w[2]:301.25897388745864

w[3]:2850.6068407502758

w[4]:0.0

y[1]:200.0

y[2]:0.0

{17: [667.5127975715615, 420.70134300728654, 100.96832232174877, -0.0]}

➤ w = [-393.48187998], theta = -464.45631073180465

➤ e = [328.25923457]

➤ E = [[1.50901281 1.14127019 1.8260323]]

✓ Added optimality cut to master problem.

optimality_cut : Size=121, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

=====

scenario18

0.0

w[1]:0.0

w[2]:303.5215520411916

w[3]:2862.5229863920563

w[4]:0.0

y[1]:200.0

y[2]:0.0

{18: [651.7194447810183, 410.7475492317342, 98.5794118156162, -0.0]}

➤ w = [-2322.70648672], theta = -2741.60015175405

➤ e = [1922.95572648]

➤ E = [[8.87680987 6.71355368 10.74168589]]

✓ Added optimality cut to master problem.

optimality_cut : Size=122, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

=====

scenario19

0.0

w[1]:0.0

w[2]:299.4153916881206

w[3]:2840.8973887458624

w[4]:0.0

y[1]:200.0

y[2]:0.0

{19: [651.7911339199442, 410.7927314621497, 98.59025555091591, -0.0]}

➤ w = [-6599.92659251], theta = -7790.544005130229

➤ e = [5540.5532725]

➤ E = [[25.38325615 19.19742062 30.71587299]]

✓ Added optimality cut to master problem.

optimality_cut : Size=123, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] +
5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] +
2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

=====

scenario20

0.0

w[1]:0.0

w[2]:297.90700625229863

w[3]:2832.9532916513426

w[4]:0.0

y[1]:200.0

y[2]:0.0

{20: [667.2646554392036, 420.5449509070611, 100.93078821769465, -0.0]}

➤ w = [-12277.10537042], theta = -14492.11389022666

➤ e = [10359.7603933]

➤ E = [[47.32904865 35.79507881 57.27212609]]

✓ Added optimality cut to master problem.

optimality_cut : Size=124, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124},

Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

=====

scenario21

0.0

w[1]:0.0

w[2]:294.2198418536227

w[3]:2813.5343876425154

w[4]:0.0

y[1]:200.0

y[2]:0.0

{21: [699.3400674261661, 440.7605466971635, 105.78253120731924, -0.0]}

► w = [-13049.56144064],theta =-15404.559290580552

► e = [11152.53472904]

► E = [[50.60162475 38.27013637 61.23221819]]

✓ Added optimality cut to master problem.

optimality_cut : Size=125, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \theta[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \theta[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \theta[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \theta[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \theta[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

=====

scenario22

0.0

w[1]:0.0

w[2]:291.9572636998896

w[3]:2801.6182420007353

w[4]:0.0

y[1]:200.0

y[2]:0.0

{22: [749.8953609004036, 472.6231266179014, 113.42955038829635, -0.0]}

► w = [-7461.34871635],theta =-8808.08968328442

► e = [6427.17039067]

► E = [[29.03804808 21.96154897 35.13847835]]

✓ Added optimality cut to master problem.

optimality_cut : Size=126, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

=====

scenario23

0.0

w[1]:0.0

w[2]:304.8623390952556

w[3]:2869.584406031629

w[4]:0.0

y[1]:200.0

y[2]:0.0

{23: [821.0004390209799, 517.4372514838108, 124.18494035611461, -0.0]}

► w = [-2799.24969062],theta =-3304.0405092560236

► e = [2307.08022528]

► E = [[10.67628971 8.07450482 12.91920772]]

✓ Added optimality cut to master problem.

optimality_cut : Size=127, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

=====

scenario24

0.0

w[1]:0.0

w[2]:293.96844428098564

w[3]:2812.2103714600958

w[4]:0.0

y[1]:200.0

y[2]:0.0

{24: [918.7980113171591, 579.0743768805622, 138.97785045133497, -0.0]}

► w = [-528.22692142],theta =-623.5554938919973

► e = [451.8324338]

► E = [[2.04910332 1.54974201 2.47958722]]

✓ Added optimality cut to master problem.

optimality_cut : Size=128, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

=====

scenario25

0.0

w[1]:0.0

w[2]:496.84628539904384

w[3]:3880.6914306730414

w[4]:0.0

y[1]:200.0

y[2]:0.0

{25: [852.5618626813051, 537.3289050512425, 128.95893721229825, -0.0]}

► $w = [-1731.03128752], \theta = -2040.7163373151186$

► $e = [868.46679827]$

► $E = [[5.43501793 \ 4.11051776 \ 6.57682842]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=129, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

=====

scenario26

0.0

w[1]:0.0

w[2]:498.85746598013986

w[3]:3891.283560132402

w[4]:0.0

y[1]:200.0

y[2]:0.0

{26: [811.1955611040082, 511.2577065781562, 122.70184957875753, -0.0]}

► w = [-8611.12685732],theta =-10151.596990076767

► e = [4302.60852317]

► E = [[26.99997502 20.42014917 32.67223867]]

✓ Added optimality cut to master problem.

optimality_cut : Size=130, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

=====

scenario27

0.0

w[1]:0.0

w[2]:497.34908054431776

w[3]:3883.339463037881

w[4]:0.0

y[1]:200.0

y[2]:0.0

{27: [791.8323230604556, 499.0539851221356, 119.77295642931259, -0.0]}

➤ w = [-23363.67388459],theta =-27543.42992900891

➤ e = [11709.66962584]

➤ E = [[73.33117573 55.46055307 88.73688491]]

✓ Added optimality cut to master problem.

optimality_cut : Size=131, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

=====

scenario28

0.0

w[1]:0.0

w[2]:497.93567488047074

w[3]:3886.4288341301954

w[4]:0.0

y[1]:200.0

y[2]:0.0

{28: [792.0619544341431, 499.198710777821, 119.80769058667708, -0.0]}

► w = [-43017.41219462],theta =-50713.103696215796

► e = [21534.23430605]

► E = [[134.96426799 102.07381613 163.31810581]]

✓ Added optimality cut to master problem.

optimality_cut : Size=132, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

=====

scenario29

0.0

w[1]:0.0

w[2]:496.6786870172858

w[3]:3879.808753218095

w[4]:0.0

y[1]:200.0

y[2]:0.0

{29: [811.0872825796512, 511.1894638107043, 122.68547131456907, -0.0]}

► $w = [-43711.02279279], \theta = -51531.047673229885$

► $e = [21937.52544295]$

► $E = [[137.2576648 \ 103.80831791 \ 166.09330866]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=133, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

=====

scenario30

0.0

w[1]:0.0

w[2]:498.7736667892608

w[3]:3890.842221404928

w[4]:0.0

y[1]:200.0

y[2]:0.0

{30: [849.71936984892, 535.5374179720081, 128.52898031328198, -0.0]}

► w = [-25741.65045889],theta =-30346.660677993856

► e = [12864.1800196]

► E = [[80.71688227 61.04638155 97.67421048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=134, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

=====

scenario31

0.0

w[1]:0.0

w[2]:497.93567488047074

w[3]:3886.4288341301954

w[4]:0.0

y[1]:200.0

y[2]:0.0

{31: [910.3468468876405, 573.7480127443109, 137.69952305863467, -0.0]}

► w = [-9326.18261874],theta =-10994.609905735171

► e = [4668.62582957]

► E = [[29.26027732 22.1296215 35.4073944]]

✓ Added optimality cut to master problem.

optimality_cut : Size=135, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

=====

scenario32

0.0

w[1]:0.0

w[2]:486.11998896653176

w[3]:3824.2000735564543

w[4]:0.0

y[1]:200.0

y[2]:0.0

{32: [1000.4984951349236, 630.56627844638, 151.33590682713128, -0.0]}

► w = [-1575.89090642],theta =-1857.899308957735

► e = [808.30189346]

► E = [[4.98485869 3.77006119 6.03209791]]

✓ Added optimality cut to master problem.

optimality_cut : Size=136, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

=====

scenario33

0.0

w[1]:0.0

w[2]:689.5006252298639

w[3]:4895.329165134241

w[4]:0.0

y[1]:200.0

y[2]:0.0

{33: [905.7312776757436, 570.8390405519389, 137.0013697324654, -0.0]}

► w = [-2472.72037809],theta =-2913.5455184980965

► e = [890.81335075]

► E = [[7.03245993 5.31866718 8.50986748]]

✓ Added optimality cut to master problem.

optimality_cut : Size=137, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

=====

scenario34

0.0

w[1]:0.0

w[2]:702.4894998161088

w[3]:4963.736667892607

w[4]:0.0

y[1]:200.0

y[2]:0.0

{34: [839.6853529076284, 529.2134577148915, 127.01122985157404, -0.0]}

➤ w = [-13098.19796993], theta = -15432.894084379646

➤ e = [4630.6883168]

➤ E = [[37.06746907 28.0342203 44.85475249]]

✓ Added optimality cut to master problem.

optimality_cut : Size=138, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

=====

scenario35

0.0

w[1]:0.0

w[2]:697.629146745127

w[3]:4938.139021699154

w[4]:0.0

y[1]:200.0

y[2]:0.0

{35: [799.8590766192195, 504.1128634154743, 120.98708721971389, -0.0]}

► w = [-33934.01192251], theta = -39982.973742914764

► e = [12081.23280937]

► E = [[96.20844949 72.76269289 116.42030863]]

✓ Added optimality cut to master problem.

optimality_cut : Size=139, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

=====

scenario36

0.0

w[1]:0.0

w[2]:698.5509378447958

w[3]:4942.99374770136

w[4]:0.0

y[1]:200.0

y[2]:0.0

{36: [780.6064686449948, 491.9788667930638, 118.07492803033537, -0.0]}

➤ w = [-60847.25678698],theta =-71693.55453704495

➤ e = [21634.08189681]

➤ E = [[172.45158105 130.4255655 208.6809048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=140, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

=====

scenario37

0.0

w[1]:0.0

w[2]:700.8135159985289

w[3]:4954.9098933431405

w[4]:0.0

y[1]:200.0

y[2]:0.0

{37: [780.3098381869098, 491.7919148236825, 118.03005955768384, -0.0]}

➤ w = [-59244.03045684],theta =-69804.2419152741

➤ e = [20995.44928314]

➤ E = [[167.7643133 126.88057308 203.00891693]]

✓ Added optimality cut to master problem.

optimality_cut : Size=141, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

=====

scenario38

0.0

w[1]:0.0

w[2]:698.2157410812799

w[3]:4941.228392791467

w[4]:0.0

y[1]:200.0

y[2]:0.0

{38: [799.021668106632, 503.5850849411544, 120.86042038587709, -0.0]}

► w = [-34716.24817908],theta =-40904.60236480713

► e = [12349.24917632]

► E = [[98.40431256 74.42342967 119.07748746]]

✓ Added optimality cut to master problem.

optimality_cut : Size=142, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

=====

scenario39

0.0

w[1]:0.0

w[2]:698.8023354174329

w[3]:4944.317763883781

w[4]:0.0

y[1]:200.0

y[2]:0.0

{39: [837.1669625420426, 527.6262368962451, 126.63029685509886, -0.0]}

► w = [-13651.6624241],theta =-16085.124971470239

► e = [4852.0508745]

► E = [[38.68747361 29.25943382 46.81509412]]

✓ Added optimality cut to master problem.

optimality_cut : Size=143, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

=====

scenario40

0.0

w[1]:0.0

w[2]:686.7352519308569

w[3]:4880.76498712762

w[4]:0.0

y[1]:200.0

y[2]:0.0

{40: [899.301494501913, 566.7866561986845, 136.0287974876843, -0.0]}

► w = [-2095.91200861],theta =-2469.5750906603826

► e = [758.13383133]

► E = [[5.96722514 4.51302741 7.22084386]]

✓ Added optimality cut to master problem.

optimality_cut : Size=144, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

=====

scenario41

0.0

w[1]:0.0

w[2]:905.1159433615301

w[3]:6000.0

w[4]:30.893710923132858

y[1]:200.0

y[2]:0.0

{41: [790.9176783845425, 498.477528393619, 33.231835226241294, 86.4027715882273]}

➤ w = [-1922.47836473],theta =-2265.3751330779014

➤ e = [-457.13712537]

➤ E = [[5.13374779 3.88266639 1.72562951]]

✓ Added optimality cut to master problem.

optimality_cut : Size=145, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] +
5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] +
2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

=====

scenario42

0.0

w[1]:0.0

w[2]:906.0377344611991

w[3]:6000.0

w[4]:35.748436925339774

y[1]:200.0

y[2]:0.0

{42: [714.9342170221395, 450.58879224084404, 30.03925281605629, 78.1020573217463]}

► w = [-7870.59377137], theta = -9274.975314195763

► e = [-1870.36403734]

► E = [[21.02149676 15.89861099 7.06604933]]

✓ Added optimality cut to master problem.

optimality_cut : Size=146, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

=====

scenario43

0.0

w[1]:0.0

w[2]:896.484626700993

w[3]:5985.43582199338

w[4]:0.0

y[1]:200.0

y[2]:0.0

{43: [663.6591349773116, 418.2725640613307, 100.38541537471939, -0.0]}

➤ w = [-22004.74185456],theta =-25919.48005539356

➤ e = [6084.36002586]

➤ E = [[58.72855736 44.41655598 71.06648957]]

✓ Added optimality cut to master problem.

optimality_cut : Size=147, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,
147}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

=====

scenario44

0.0

w[1]:0.0

w[2]:899.25

w[3]:5999.999999999999

w[4]:0.0

y[1]:200.0

y[2]:0.0

{44: [631.9362283253961, 398.27913549919907, 95.58699251980781, -0.0]}

► w = [-37526.3845476], theta = -44202.34738241658

► e = [10343.99949942]

► E = [[100.08716573 75.69617576 121.11388122]]

✓ Added optimality cut to master problem.

optimality_cut : Size=148, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

=====

scenario45

0.0

w[1]:0.0

w[2]:899.8365943361532

w[3]:6000.0

w[4]:3.0893710923137405

y[1]:200.0

y[2]:0.0

{45: [616.7128845850373, 388.68459112502336, 25.912306075001577, 67.37199579500405]}

➤ w = [-33649.18476744],theta =-39636.94960041753

➤ e = [-8029.49810807]

➤ E = [[89.75725662 67.88363946 30.17050643]]

✓ Added optimality cut to master problem.

optimality_cut : Size=149, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,
147, 148, 149}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 3.7008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

=====

scenario46

0.0

w[1]:0.0

w[2]:898.41200809121

w[3]:5995.586612725267

w[4]:0.0

y[1]:200.0

y[2]:0.0

{46: [616.4291966581281, 388.50579621310584, 93.24139109114543, -0.0]}

► w = [-20090.60529606], theta = -23664.7572067731

► e = [5543.09709882]

► E = [[53.59482007 40.53389754 64.85423606]]

✓ Added optimality cut to master problem.

optimality_cut : Size=150, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

=====

scenario47

0.0

w[1]:0.0

w[2]:898.4958072820891

w[3]:5996.02795145274

w[4]:0.0

y[1]:200.0

y[2]:0.0

{47: [630.9276113635274, 397.6434525400381, 95.43442860960917, -0.0]}

► w = [-7069.2419961],theta =-8326.871027745514

► e = [1950.25556776]

► E = [[18.85792156 14.26229362 22.81966979]]

✓ Added optimality cut to master problem.

optimality_cut : Size=151, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

=====

scenario48

0.0

w[1]:0.0

w[2]:895.89803236484

w[3]:5982.346450901066

w[4]:0.0

y[1]:200.0

y[2]:0.0

{48: [660.688466791545, 416.4002941963517, 99.93607060712444, -0.0]}

➤ w = [-1006.52004984],theta =-1185.5851078806397

➤ e = [278.48851676]

➤ E = [[2.68668965 2.03195016 3.25112025]]

✓ Added optimality cut to master problem.

optimality_cut : Size=152, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

=====

scenario49

0.0

w[1]:0.0

w[2]:1098.775873482898

w[3]:6000.0

w[4]:1050.8275101140116

y[1]:200.0

y[2]:0.0

{49: [565.8400304989508, 356.62186796152344, 23.774791197434915, 61.814457113330725]}

➤ w = [-490.31891572],theta =-584.3623167982925

➤ e = [-103.27769296]

➤ E = [[1.35597905 1.02553037 0.45579128]]

✓ Added optimality cut to master problem.

optimality_cut : Size=153, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,
147, 148, 149, 150, 151, 152, 153}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

=====

scenario50

0.0

w[1]:0.0

w[2]:1095.675303420375

w[3]:6000.0

w[4]:1034.4979771974986

y[1]:200.0

y[2]:0.0

{50: [498.4145324646958, 314.1268061752283, 20.941787078348572, 54.448646403706235]}

► w = [-2299.21572621],theta =-2739.7673853016495

► e = [-485.17932303]

► E = [[6.35538339 4.80659248 2.13626332]]

✓ Added optimality cut to master problem.

optimality_cut : Size=154, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

=====

scenario51

0.0

w[1]:0.0

w[2]:1096.3456969474073

w[3]:6000.0

w[4]:1038.028687017285

y[1]:200.0

y[2]:0.0

{51: [450.9854055553554, 284.23449929959355, 18.94896661997292, 49.26731321192955]}

► w = [-5918.54389538],theta =-7052.838039017345

► e = [-1248.43371679]

► E = [[16.3614912 12.37423704 5.49966091]]

✓ Added optimality cut to master problem.

optimality_cut : Size=155, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] +
203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] +
119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] +
46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] +
7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] +
64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] +
0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] +
5.499660906534489*x[3] + theta[51] : +Inf : True

=====

scenario52

0.0

w[1]:0.0

w[2]:1092.3233357852152

w[3]:6000.0

w[4]:1016.8444280985659

y[1]:200.0

y[2]:0.0

{52: [418.84818555548077, 263.9799488795045, 17.598663258633653, 45.75652447244744]}

► w = [-9099.38155942],theta =-10841.024367791475

► e = [-1923.95626209]

► E = [[25.13873407 19.01248795 8.44999464]]

✓ Added optimality cut to master problem.

optimality_cut : Size=156, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

=====

scenario53

0.0

w[1]:0.0

w[2]:1095.591504229496

w[3]:6000.0

w[4]:1034.056638470025

y[1]:200.0

y[2]:0.0

{53: [398.9068234411843, 251.4118635133513, 16.760790900890107, 43.57805634231422]}

► w = [-8280.40359738],theta =-9866.965079331207

► e = [-1747.41301616]

► E = [[22.88799703 17.31024985 7.69344438]]

✓ Added optimality cut to master problem.

optimality_cut : Size=157, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

=====

scenario54

0.0

w[1]:0.0

w[2]:1100.3680581095994

w[3]:6000.0

w[4]:1059.2129459360067

y[1]:200.0

y[2]:0.0

{54: [389.0737702433591, 245.2145610777472, 16.347637405183168, 42.50385725347617]}

► w = [-5118.16863259],theta =-6100.336256823607

► e = [-1077.0477428]

► E = [[14.15785952 10.70762485 4.75894438]]

✓ Added optimality cut to master problem.

optimality_cut : Size=158, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,
147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

=====

scenario55

0.0

w[1]:0.0

w[2]:1085.7869988966531

w[3]:6000.0

w[4]:982.4200073556449

y[1]:200.0

y[2]:0.0

{55: [389.9530769641091, 245.7687459857829, 16.384583065718882, 42.599915970869034]}

➤ w = [-1620.70974749],theta =-1930.2586580772588

➤ e = [-344.01070605]

➤ E = [[4.47284953 3.38282738 1.50347883]]

✓ Added optimality cut to master problem.

optimality_cut : Size=159, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

=====

scenario56

0.0

w[1]:0.0

w[2]:1085.7031997057743

w[3]:6000.0

w[4]:981.9786686281723

y[1]:200.0

y[2]:0.0

{56: [399.31974987278693, 251.67211126436143, 16.77814075095745, 43.6231659524893]}

► w = [-286.13044152],theta =-340.7786848148997

► $e = [-60.73686952]$

► $E = [[0.78965481 \ 0.59721792 \ 0.26543019]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=160, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918x[1] + 55.46055307058254x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557x[1] + 102.07381612988239x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286x[1] + 103.80831791391374x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654x[1] + 61.046381547557026x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856x[1] + 22.12962150099164x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864x[1] + 3.770061193889139x[2] + 6.0320979102226255x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544x[1] + 5.3186671757537685x[2] + 8.509867481206035x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976x[1] + 28.0342203045103x[2] + 44.85475248721651x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438x[1] + 72.76269289094907x[2] + 116.42030862551856x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225x[1] + 130.42556550230836x[2] + 208.68090480369352x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295x[1] + 126.8805730811316x[2] + 203.00891692981068x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036x[1] + 74.42342966521537x[2] + 119.07748746434463x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824x[1] + 29.25943382314574x[2] + 46.81509411703319x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925x[1] + 4.5130274142489x[2] + 7.220843862798243x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267x[1] + 3.882666392534318x[2] + 1.7256295077930317x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775x[1] + 15.89861099252932x[2] + 7.066049330013037x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902x[1] + 44.41655598305218x[2] + 71.0664895728835x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

=====

scenario57

0.0

w[1]:0.0

w[2]:1242.7428834130194

w[3]:6000.0

w[4]:1809.047443913203

y[1]:200.0

y[2]:0.0

{57: [333.14048772636994, 209.9624922645188, 13.997499484301272, 36.39349865918325]}

► w = [-52.1969425],theta =-62.63411822424335

► e = [-10.13418963]

► E = [[0.14736469 0.11145229 0.04953435]]

✓ Added optimality cut to master problem.

optimality_cut : Size=161, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

=====

scenario58

0.0

w[1]:0.0

w[2]:1323.7767009930121

w[3]:6000.0

w[4]:2235.821993379919

y[1]:200.0

y[2]:0.0

{58: [283.5258748892817, 178.692778291564, 11.912851886104283, 30.973414903871088]}

► w = [-185.51871142],theta =-223.36660526645633

► e = [-34.49961906]

► E = [[0.52908764 0.40015031 0.17784458]]

✓ Added optimality cut to master problem.

optimality_cut : Size=162, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

=====

scenario59

0.0

w[1]:0.0

w[2]:1306.9330636263333

w[3]:6000.0

w[4]:2147.1129091577786

y[1]:200.0

y[2]:0.0

{59: [249.0151053977583, 156.94229331791482, 10.46281955452767, 27.2033308417719]}

► $w = [-646.03489352], \theta = -777.3072597374332$

► $e = [-121.20130172]$

► $E = [[1.83872754 \ 1.39063427 \ 0.61805968]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=163, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

=====

scenario60

0.0

w[1]:0.0

w[2]:1297.9665502022804

w[3]:6000.0

w[4]:2099.889665318132

y[1]:200.0

y[2]:0.0

{60: [225.23416283227237, 141.95430430605396, 9.463620287070277, 24.605412746382683]}

► w = [-1054.12640188],theta =-1267.858650712338

► e = [-198.69817155]

► E = [[2.99694888 2.26659999 1.00737777]]

✓ Added optimality cut to master problem.

optimality_cut : Size=164, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

=====

scenario61

0.0

w[1]:0.0

w[2]:1291.5978116954764

w[3]:6000.0

w[4]:2066.3479220301588

y[1]:200.0

y[2]:0.0

{61: [209.11865848162327, 131.7974738329558, 8.786498255530399, 22.844895464379]}

► w = [-908.14478893],theta =-1091.9924640203253

► e = [-171.7584679]

► E = [[2.57989166 1.95117857 0.86719047]]

✓ Added optimality cut to master problem.

optimality_cut : Size=165, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \text{theta}[31]$: +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] +
0.8671904745281772*x[3] + theta[61] : +Inf : True

=====

scenario62

0.0

w[1]:0.0

w[2]:1298.5531445384333

w[3]:6000.0

w[4]:2102.9790364104447

y[1]:200.0

y[2]:0.0

{62: [199.2294371220273, 125.56477129539532, 8.370984753026367, 21.76456035786852]}

► w = [-514.59826768],theta =-618.9518859890057

► e = [-96.96948738]

► E = [[1.46314099 1.10657722 0.4918121]]

✓ Added optimality cut to master problem.

optimality_cut : Size=166, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,
147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

=====

scenario63

0.0

w[1]:0.0

w[2]:1283.9720853254873

w[3]:6000.0

w[4]:2026.1860978300847

y[1]:200.0

y[2]:0.0

{63: [194.80853591228953, 122.7784890203505, 8.185232601356711, 21.28160476352742]}

► w = [-187.24096446],theta =-225.07556107441326

► e = [-35.55665042]

► E = [[0.53141821 0.40191293 0.17862797]]

✓ Added optimality cut to master problem.

optimality_cut : Size=167, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

=====

scenario64

0.0

w[1]:0.0

w[2]:1248.8602243471864

w[3]:6000.0

w[4]:1841.2651710187565

y[1]:200.0

y[2]:0.0

{64: [195.54491217803798, 123.24259170884743, 8.21617278058984, 21.36204922953355]}

➤ w = [-30.74012731],theta =-36.8966381428001

➤ e = [-5.94850909]

➤ E = [[0.08685616 0.06568953 0.02919535]]

✓ Added optimality cut to master problem.

optimality_cut : Size=168, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

=====

[Thetatheta[1] from previous iteration: 4.43]

[Thetatheta[2] from previous iteration: 30.40]

[Thetatheta[3] from previous iteration: 85.89]

[Thetatheta[4] from previous iteration: 147.34]

[Thetatheta[5] from previous iteration: 120.53]

[Thetatheta[6] from previous iteration: 95.60]

[Thetatheta[7] from previous iteration: 44.13]

[Thetatheta[8] from previous iteration: 7.21]

[Thetatheta[9] from previous iteration: -38.23]

[Thetatheta[10] from previous iteration: -172.40]

[Thetatheta[11] from previous iteration: -639.85]

[Thetatheta[12] from previous iteration: -1,166.82]

[Thetatheta[13] from previous iteration: -938.13]

[Thetatheta[14] from previous iteration: -691.13]

[Thetatheta[15] from previous iteration: -270.74]

[Thetatheta[16] from previous iteration: -83.18]

[Thetatheta[17] from previous iteration: -464.46]

[Thetatheta[18] from previous iteration: -2,741.60]

[Thetatheta[19] from previous iteration: -7,790.54]

[Thetatheta[20] from previous iteration: -14,492.11]

[Thetatheta[21] from previous iteration: -15,404.56]

[Thetatheta[22] from previous iteration: -8,808.09]

[Thetatheta[23] from previous iteration: -3,304.04]

[Thetatheta[24] from previous iteration: -623.56]

[Thetatheta[25] from previous iteration: -2,040.72]

[Thetatheta[26] from previous iteration: -10,151.60]

[Thetatheta[27] from previous iteration: -27,543.43]

[Thetatheta[28] from previous iteration: -50,713.10]

[Thetatheta[29] from previous iteration: -51,531.05]

[Thetatheta[30] from previous iteration: -30,346.66]

[Thetatheta[31] from previous iteration: -10,994.61]

[Thetatheta[32] from previous iteration: -1,857.90]

[Thetatheta[33] from previous iteration: -2,913.55]

[Thetatheta[34] from previous iteration: -15,432.89]

[Thetatheta[35] from previous iteration: -39,982.97]

[Thetatheta[36] from previous iteration: -71,693.55]

[Thetatheta[37] from previous iteration: -69,804.24]

[Thetatheta[38] from previous iteration: -40,904.60]

[Thetatheta[39] from previous iteration: -16,085.12]

[Thetatheta[40] from previous iteration: -2,469.58]

[Thetatheta[41] from previous iteration: -2,265.38]

[Thetatheta[42] from previous iteration: -9,274.98]

[Thetatheta[43] from previous iteration: -25,919.48]

[Thetatheta[44] from previous iteration: -44,202.35]

[Thetatheta[45] from previous iteration: -39,636.95]

[Thetatheta[46] from previous iteration: -23,664.76]

[Thetatheta[47] from previous iteration: -8,326.87]

[Thetatheta[48] from previous iteration: -1,185.59]

[Thetatheta[49] from previous iteration: -584.36]

[Thetatheta[50] from previous iteration: -2,739.77]

[Thetatheta[51] from previous iteration: -7,052.84]

[Thetatheta[52] from previous iteration: -10,841.02]

[Thetatheta[53] from previous iteration: -9,866.97]

[Thetatheta[54] from previous iteration: -6,100.34]

[Thetatheta[55] from previous iteration: -1,930.26]

[Thetatheta[56] from previous iteration: -340.78]

[Thetatheta[57] from previous iteration: -62.63]

[Thetatheta[58] from previous iteration: -223.37]

[Thetatheta[59] from previous iteration: -777.31]

[Thetatheta[60] from previous iteration: -1,267.86]

[Thetatheta[61] from previous iteration: -1,091.99]

[Thetatheta[62] from previous iteration: -618.95]

[Thetatheta[63] from previous iteration: -225.08]

[Thetatheta[64] from previous iteration: -36.90]

=====

↻ Iteration 4

Master Problem Objective Value : -572,621.81

Decision Variables:

x[1]: 229.0130062237069

x[2]: 50.317630039625215

x[3]: 220.6693637366679

=====

-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

$$\pi \text{ (Dual values)} = [-0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

$$\pi \text{ (Dual values)} = [-0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

$$\pi \text{ (Dual values)} = [-0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

$$\pi \text{ (Dual values)} = [-0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (229.0130062237069, 50.317630039625215,
220.6693637366679)-----

scenario1

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:707.024641412284

w[4]:0.0

y[1]:108.28029100740537

y[2]:215.81734700295613

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

➤ w = [4.55719186],theta =4.557191863128252

➤ e = [9.785496]

➤ E = [[0.00951779 0.01007766 0.01151733]]

scenario2

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:639.9411548363365

w[4]:0.0

y[1]:116.98278524390629

y[2]:218.11183093276304

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

➤ w = [31.11757495],theta =31.117574945759685

➤ e = [60.25830153]

➤ E = [[0.05304883 0.05616935 0.06419354]]

scenario3

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:672.1588819418903

w[4]:0.0

y[1]:112.80329788032361

y[2]:217.00987483489524

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ w = [88.123678],theta =88.12367800483605

➤ e = [179.09259324]

➤ E = [[0.16560308 0.17534444 0.20039365]]

scenario4

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:689.8124310408241

w[4]:0.0

y[1]:110.5131678180865

y[2]:216.40606327441972

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

➤ w = [151.37964732],theta =151.3796473161052

➤ e = [316.22021003]

➤ E = [[0.30008169 0.31773355 0.36312406]]

scenario5

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:718.9407870540646

w[4]:0.0

y[1]:106.73445321539529

y[2]:215.40977419963514

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

➤ w = [124.15660971],theta =124.15660971363496

➤ e = [271.85361107]

➤ E = [[0.26887293 0.28468899 0.32535884]]

scenario6

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:654.0639941154833

w[4]:0.0

y[1]:115.15068119411663

y[2]:217.62878168438263

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

➤ w = [97.95059961],theta =97.9505996106907

➤ e = [193.68136379]

➤ E = [[0.17427172 0.184523 0.21088343]]

scenario7

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:691.1364472232434

w[4]:0.0

y[1]:110.34140806341881

y[2]:216.3607774073841

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [45.34919692],theta =45.349196924397646

➤ e = [94.92932352]

► E = [[0.09025744 0.09556671 0.10921909]]

scenario8

229.0130062237069

w[1]:0.0

w[2]:0.0

w[3]:785.5829349025382

w[4]:0.0

y[1]:98.08921223045037

y[2]:213.13038555884012

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

► w = [7.47381773],theta =7.473817732065271

► e = [18.39287286]

► E = [[0.01987744 0.0210467 0.02405337]]

scenario9

229.0130062237069

w[1]:26.95188916769351

w[2]:0.0

w[3]:1749.4667157043027

w[4]:0.0

y[1]:0.0

y[2]:180.1622743568777

{9: [311.8046141713676, 385.17040574110115, 66.02921241276017, -0.0]}

► $w = [-32.71587388], \theta = -34.732767498831684$

► $e = [92.88109213]$

► $E = [[0.18539902 \ 0.27482679 \ 0.31408776]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=169, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.13219975857664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

=====

scenario10

229.0130062237069

w[1]:34.56657162463176

w[2]:0.0

w[3]:1808.1647664582565

w[4]:0.0

y[1]:0.0

y[2]:178.15460091829667

{10: [311.5863509414476, 384.90078645708235, 65.98299196407125, -0.0]}

➤ w = [-147.6161932], theta = -157.95584624264916

➤ e = [371.26430145]

➤ E = [[0.76594157 1.13539574 1.29759513]]

✓ Added optimality cut to master problem.

optimality_cut : Size=170, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

=====

scenario11

229.0130062237069

w[1]:33.077987084177664

w[2]:0.0

w[3]:1796.68995954395

w[4]:0.0

y[1]:0.0

y[2]:178.54707843260573

{11: [318.92732536962814, 393.9690489860112, 67.53755125474477, -0.0]}

➤ w = [-547.80827428],theta =-585.3644015352413

➤ e = [1409.20852779]

➤ E = [[2.88883574 4.28227416 4.89402761]]

✓ Added optimality cut to master problem.

optimality_cut : Size=171, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177x[1] + 2.0319501556193567x[2] + 3.2511202489909716x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855x[1] + 1.0255303732595935x[2] + 0.45579127700426414x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383x[1] + 4.806592477081885x[2] + 2.136263323147507x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401x[1] + 12.374237039702589x[2] + 5.499660906534489x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373x[1] + 19.012487948775647x[2] + 8.449994643900297x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014x[1] + 17.310249852519426x[2] + 7.6934443788975315x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756x[1] + 10.707624849117288x[2] + 4.7589443773854665x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648x[1] + 3.3828273773097317x[2] + 1.5034788343598828x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362x[1] + 0.5972179200303298x[2] + 0.2654301866801469x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975x[1] + 0.11145229014385186x[2] + 0.04953435117504534x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886x[1] + 0.4001503122838651x[2] + 0.17784458323727365x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476x[1] + 1.3906342726313798x[2] + 0.6180596767250586x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004x[1] + 2.2665999918282376x[2] + 1.0073777741458847x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325x[1] + 1.9511785676883961x[2] + 0.8671904745281772x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684x[1] + 1.1065772164720598x[2] + 0.4918120962098051x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346x[1] + 0.4019129282383369x[2] + 0.17862796810592776x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002x[1] + 0.06568953380673276x[2] + 0.029195348358547937x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

=====

scenario12

229.0130062237069

w[1]:33.936785857516554

w[2]:0.0

w[3]:1803.3100404560498

w[4]:0.0

y[1]:0.0

y[2]:178.3206490974274

{12: [333.9360453015228, 412.5092324312929, 70.71586841679306, -0.0]}

► w = [-999.03549389],theta =-1068.391716175623

► e = [2536.57820011]

► E = [[5.21906976 7.7365034 8.84171817]]

✓ Added optimality cut to master problem.

optimality_cut : Size=172, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,

147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

=====

scenario13

229.0130062237069

w[1]:27.524421683252825

w[2]:0.0

w[3]:1753.8801029790366

w[4]:0.0

y[1]:0.0

y[2]:180.01132146675883

{13: [358.03954905138676, 442.2841488281836, 75.82013979911719, -0.0]}

► w = [-802.8700228],theta =-852.9325669625373

► e = [2257.50254022]

► E = [[4.51754611 6.69659776 7.65325458]]

✓ Added optimality cut to master problem.

optimality_cut : Size=173, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125,
126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,

147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

=====

scenario14

229.0130062237069

w[1]:27.06639567080535

w[2]:0.0

w[3]:1750.3493931592493

w[4]:0.0

y[1]:0.0

y[2]:180.1320837788539

{14: [392.5223380205258, 484.880535201826, 83.12237746317017, -0.0]}

➤ w = [-591.46350937],theta =-628.0106366661048

➤ e = [1675.93184605]

➤ E = [[3.34699872 4.9614334 5.6702096]]

✓ Added optimality cut to master problem.

optimality_cut : Size=174, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177x[1] + 2.0319501556193567x[2] + 3.2511202489909716x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855x[1] + 1.0255303732595935x[2] + 0.45579127700426414x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383x[1] + 4.806592477081885x[2] + 2.136263323147507x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401x[1] + 12.374237039702589x[2] + 5.499660906534489x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373x[1] + 19.012487948775647x[2] + 8.449994643900297x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014x[1] + 17.310249852519426x[2] + 7.6934443788975315x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756x[1] + 10.707624849117288x[2] + 4.7589443773854665x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648x[1] + 3.3828273773097317x[2] + 1.5034788343598828x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362x[1] + 0.5972179200303298x[2] + 0.2654301866801469x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975x[1] + 0.11145229014385186x[2] + 0.04953435117504534x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886x[1] + 0.4001503122838651x[2] + 0.17784458323727365x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476x[1] + 1.3906342726313798x[2] + 0.6180596767250586x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004x[1] + 2.2665999918282376x[2] + 1.0073777741458847x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325x[1] + 1.9511785676883961x[2] + 0.8671904745281772x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684x[1] + 1.1065772164720598x[2] + 0.4918120962098051x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346x[1] + 0.4019129282383369x[2] + 0.17862796810592776x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002x[1] + 0.06568953380673276x[2] + 0.029195348358547937x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

=====

scenario15

229.0130062237069

w[1]:33.936785857516554

w[2]:0.0

w[3]:1803.3100404560498

w[4]:0.0

y[1]:0.0

y[2]:178.3206490974274

{15: [439.07548730976015, 542.3873666767626, 92.98069143030214, -0.0]}

► w = [-231.8083864], theta = -247.90126204834405

► e = [588.56777675]

► E = [[1.21099215 1.79511777 2.05156317]]

✓ Added optimality cut to master problem.

optimality_cut : Size=175, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

=====

scenario16

229.0130062237069

w[1]:36.5704354290892

w[2]:0.0

w[3]:1823.6116219198234

w[4]:0.0

y[1]:0.0

y[2]:177.6262658028806

{16: [500.4406774161722, 618.1914250435068, 105.97567286460118, -0.0]}

► $w = [-71.22693744], \theta = -76.35131081632137$

► $e = [173.91785425]$

► $E = [[0.36186865 \ 0.53641706 \ 0.61304807]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=176, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : $0.009517792788000001 \cdot x[1] + 0.010077662951999999 \cdot x[2] + 0.011517329088000003 \cdot x[3] + \theta[1]$: +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.13219975857664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

=====

scenario17

229.0130062237069

w[1]:169.79875179973067

w[2]:0.0

w[3]:2850.6068407502758

w[4]:0.0

y[1]:0.0

y[2]:142.49952827221824

{17: [476.7948554082581, 588.9818802102011, 100.96832232174877, -0.0]}

► w = [-398.79073014],theta =-444.1278660792616

► e = [331.40047126]

► E = [[1.07786629 1.59777826 1.8260323]]

✓ Added optimality cut to master problem.

optimality_cut : Size=177, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] +
1.8260323028532908*x[3] + theta[17] : +Inf : True

=====

scenario18

229.0130062237069

w[1]:171.34458959174077

w[2]:0.0

w[3]:2862.5229863920563

w[4]:0.0

y[1]:0.0

y[2]:142.09195546889725

{18: [465.5138891292987, 575.0465689244278, 98.5794118156162, -0.0]}

► w = [-2354.01286785],theta =-2622.0175097593574

► e = [1941.35721669]

► E = [[6.34057848 9.39897516 10.74168589]]

✓ Added optimality cut to master problem.

optimality_cut : Size=178, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

=====

scenario19

229.0130062237069

w[1]:168.5391802655003

w[2]:0.0

w[3]:2840.8973887458624

w[4]:0.0

y[1]:0.0

y[2]:142.83162463047975

{19: [465.5650956571029, 575.1098240470095, 98.59025555091591, -0.0]}

➤ w = [-6689.04670468],theta =-7448.597193434929

➤ e = [5593.57292105]

➤ E = [[18.13089725 26.87638887 30.71587299]]

✓ Added optimality cut to master problem.

optimality_cut : Size=179, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

=====

scenario20

229.0130062237069

w[1]:167.50862173749368

w[2]:0.0

w[3]:2832.9532916513426

w[4]:0.0

y[1]:0.0

y[2]:143.10333983269373

{20: [476.61761102800256, 588.7629312698856, 100.93078821769465, -0.0]}

► w = [-12442.99950898], theta = -13854.527554204684

► e = [10458.8968564]

► E = [[33.80646332 50.11311033 57.27212609]]

✓ Added optimality cut to master problem.

optimality_cut : Size=180, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

=====

scenario21

229.0130062237069

w[1]:164.98947866903285

w[2]:0.0

w[3]:2813.5343876425154

w[4]:0.0

y[1]:0.0

y[2]:143.76753254921675

{21: [499.52861959011864, 617.064765376029, 105.78253120731924, -0.0]}

➤ w = [-13226.19481348],theta =-14722.88692560115

➤ e = [11259.25754942]

➤ E = [[36.14401768 53.57819092 61.23221819]]

✓ Added optimality cut to master problem.

optimality_cut : Size=181, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

=====

scenario22

229.0130062237069

w[1]:163.44364087702286

w[2]:0.0

w[3]:2801.6182420007353

w[4]:0.0

y[1]:0.0

y[2]:144.17510535253774

{22: [535.6395435002883, 661.6723772650621, 113.42955038829635, -0.0]}

► w = [-7562.45035688], theta = -8416.907878230577

► e = [6488.67441355]

► E = [[20.74146292 30.74616856 35.13847835]]

✓ Added optimality cut to master problem.

optimality_cut : Size=182, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

=====

scenario23

229.0130062237069

w[1]:172.26064161663555

w[2]:0.0

w[3]:2869.584406031629

w[4]:0.0

y[1]:0.0

y[2]:141.85043084470703

{23: [586.4288850149857, 724.4121520773353, 124.18494035611461, -0.0]}

➤ w = [-2836.95686739],theta =-3160.216438574852

➤ e = [2329.15754801]

➤ E = [[7.62592122 11.30430675 12.91920772]]

✓ Added optimality cut to master problem.

optimality_cut : Size=183, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] +
12.919207715127316*x[3] + theta[23] : +Inf : True

=====

scenario24

229.0130062237069

w[1]:164.81771891436506

w[2]:0.0

w[3]:2812.2103714600958

w[4]:0.0

y[1]:0.0

y[2]:143.81281841625244

{24: [656.2842937979708, 810.7041276327875, 138.97785045133497, -0.0]}

► w = [-535.37762209],theta =-595.9512990399683

► e = [456.15618915]

► E = [[1.46364523 2.16963881 2.47958722]]

✓ Added optimality cut to master problem.

optimality_cut : Size=184, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

=====

scenario25

229.0130062237069

w[1]:303.42784093126363

w[2]:0.0

w[3]:3880.6914306730414

w[4]:0.0

y[1]:0.0

y[2]:107.26712371847265

{25: [608.9727590580751, 752.2604670717399, 128.95893721229825, -0.0]}

► w = [-1753.15531473], theta = -1967.49929051565

► e = [876.77748534]

► E = [[3.88215567 5.75472487 6.57682842]]

✓ Added optimality cut to master problem.

optimality_cut : Size=185, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

=====

scenario26

229.0130062237069

w[1]:304.80191896860595

w[2]:0.0

w[3]:3891.283560132402

w[4]:0.0

y[1]:0.0

y[2]:106.90483678218735

{26: [579.4254007885773, 715.760789209419, 122.70184957875753, -0.0]}

► $w = [-8721.14654404], \theta = -9787.870787361931$

► $e = [4343.78181048]$

► $E = [[19.28569644 \ 28.58820884 \ 32.67223867]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=186, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186},

Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.13219975857664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

=====

scenario27

229.0130062237069

w[1]:303.7713604405992

w[2]:0.0

w[3]:3883.339463037881

w[4]:0.0

y[1]:0.0

y[2]:107.17655198440133

{27: [565.594516471754, 698.6755791709902, 119.77295642931259, -0.0]}

➤ w = [-23662.25551153],theta =-26555.55975235708

➤ e = [11821.72388064]

➤ E = [[52.37941123 77.6447743 88.73688491]]

✓ Added optimality cut to master problem.

optimality_cut : Size=187, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

=====

scenario28

229.0130062237069

w[1]:304.1721332014908

w[2]:0.0

w[3]:3886.4288341301954

w[4]:0.0

y[1]:0.0

y[2]:107.07088496131811

{28: [565.7585388815307, 698.8781950889498, 119.80769058667708, -0.0]}

➤ w = [-43567.10844637],theta =-48894.95235821122

➤ e = [21740.30353386]

➤ E = [[96.40304857 142.90334258 163.31810581]]

✓ Added optimality cut to master problem.

optimality_cut : Size=188, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] +
6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] +
32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.37941123327986*x[1] + 77.6447742988156*x[2] +
88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

=====

scenario29

229.0130062237069

w[1]:303.31333442815185

w[2]:0.0

w[3]:3879.808753218095

w[4]:0.0

y[1]:0.0

y[2]:107.29731429649641

{29: [579.3480589854651, 715.6652493349864, 122.68547131456907, -0.0]}

➤ w = [-44269.7022364],theta =-49682.00117661387

➤ e = [22147.45391609]

➤ E = [[98.04118914 145.33164508 166.09330866]]

✓ Added optimality cut to master problem.

optimality_cut : Size=189, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

=====

scenario30

229.0130062237069

w[1]:304.74466571704994

w[2]:0.0

w[3]:3890.842221404928

w[4]:0.0

y[1]:0.0

y[2]:106.91993207119924

{30: [606.9424070349428, 749.7523851608117, 128.52898031328198, -0.0]}

➤ w = [-26070.54221246],theta =-29259.29503909387

➤ e = [12987.28222074]

➤ E = [[57.65491591 85.46493417 97.67421048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=190, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

=====

scenario31

229.0130062237069

w[1]:304.1721332014908

w[2]:0.0

w[3]:3886.4288341301954

w[4]:0.0

y[1]:0.0

y[2]:107.07088496131811

{31: [650.247747776886, 803.2472178420356, 137.69952305863467, -0.0]}

► w = [-9445.35686394], theta = -10600.434372115797

► e = [4713.30167483]

► E = [[20.90019808 30.9814701 35.4073944]]

✓ Added optimality cut to master problem.

optimality_cut : Size=191, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

=====

scenario32

229.0130062237069

w[1]:296.09942473210504

w[2]:0.0

w[3]:3824.2000735564543

w[4]:0.0

y[1]:0.0

y[2]:109.19932071199426

{32: [714.641782239231, 882.7927898249325, 151.33590682713128, -0.0]}

► w = [-1596.0698974],theta =-1790.7465163619659

► e = [816.03683981]

► E = [[3.56061335 5.27808567 6.03209791]]

✓ Added optimality cut to master problem.

optimality_cut : Size=192, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.13219975857664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

=====

scenario33

229.0130062237069

w[1]:435.0530662583393

w[2]:0.0

w[3]:4895.329165134241

w[4]:0.0

y[1]:0.0

y[2]:72.56305428014315

{33: [646.950912625531, 799.1746567727149, 137.0013697324654, -0.0]}

► w = [-2503.57583048],theta =-2818.8087661296627

► e = [899.33788042]

► E = [[5.02318567 7.44613405 8.50986748]]

✓ Added optimality cut to master problem.

optimality_cut : Size=193, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \theta[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \theta[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \theta[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

=====

scenario34

229.0130062237069

w[1]:443.92732024950783

w[2]:0.0

w[3]:4963.736667892607

w[4]:0.0

y[1]:0.0

y[2]:70.22328448330057

{34: [599.7752520768773, 740.8988408008486, 127.01122985157404, -0.0]}

► w = [-13261.45353619],theta =-14933.54511234059

► e = [4675.00112366]

► E = [[26.47676362 39.24790843 44.85475249]]

✓ Added optimality cut to master problem.

optimality_cut : Size=194, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

=====

scenario35

229.0130062237069

w[1]:440.60663165926417

w[2]:0.0

w[3]:4938.139021699154

w[4]:0.0

y[1]:0.0

y[2]:71.09881124599005

{35: [571.327911870871, 705.7580087816643, 120.98708721971389, -0.0]}

➤ w = [-34357.14481936],theta =-38686.915728701206

➤ e = [12196.84269272]

➤ E = [[68.72032106 101.86777005 116.42030863]]

✓ Added optimality cut to master problem.

optimality_cut : Size=195, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

=====

scenario36

229.0130062237069

w[1]:441.23641742637926

w[2]:0.0

w[3]:4942.99374770136

w[4]:0.0

y[1]:0.0

y[2]:70.93276306685928

{36: [557.5760490321393, 688.7704135102896, 118.07492803033537, -0.0]}

► w = [-61605.91695177],theta =-69370.39836028079

► e = [21841.10660396]

► E = [[123.17970075 182.5957917 208.6809048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=196, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] +
208.68090480369352*x[3] + theta[36] : +Inf : True

=====

scenario37

229.0130062237069

w[1]:442.7822552183893

w[2]:0.0

w[3]:4954.9098933431405

w[4]:0.0

y[1]:0.0

y[2]:70.52519026353832

{37: [557.3641701335071, 688.5086807531557, 118.03005955768384, -0.0]}

➤ w = [-59982.55443451],theta =-67544.2295785088

➤ e = [21196.36267341]

➤ E = [[119.83165235 177.63280231 203.00891693]]

✓ Added optimality cut to master problem.

optimality_cut : Size=197, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] +
166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] +
97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] +
35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] +
6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] +
116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] +
208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] +
203.00891692981068*x[3] + theta[37] : +Inf : True

=====

scenario38

229.0130062237069

w[1]:441.0074044201556

w[2]:0.0

w[3]:4941.228392791467

w[4]:0.0

y[1]:0.0

y[2]:70.99314422290684

{38: [570.7297629333085, 705.0191189176163, 120.86042038587709, -0.0]}

► w = [-35149.11259916],theta =-39578.963103387505

► e = [12467.42380958]

► E = [[70.28879468 104.19280153 119.07748746]]

✓ Added optimality cut to master problem.

optimality_cut : Size=198, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

=====

scenario39

229.0130062237069

w[1]:441.4081771810471

w[2]:0.0

w[3]:4944.317763883781

w[4]:0.0

y[1]:0.0

y[2]:70.88747719982362

{39: [597.9764018157447, 738.6767316547433, 126.63029685509886, -0.0]}

➤ w = [-13821.87130065],theta =-15563.952345895845

➤ e = [4898.48198334]

➤ E = [[27.63390972 40.96320735 46.81509412]]

✓ Added optimality cut to master problem.

optimality_cut : Size=199, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167,

168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

=====

scenario40

229.0130062237069

w[1]:433.16370895699356

w[2]:0.0

w[3]:4880.76498712762

w[4]:0.0

y[1]:0.0

y[2]:73.06119881753543

{40: [642.3582103585092, 793.5013186781583, 136.0287974876843, -0.0]}

➤ w = [-2122.0720783], theta = -2389.1884925809386

➤ e = [765.38870053]

➤ E = [[4.26230367 6.31823838 7.22084386]]

✓ Added optimality cut to master problem.

optimality_cut : Size=200, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

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scenario41

229.0130062237069

w[1]:582.3656825117387

w[2]:0.0

w[3]:6000.0

w[4]:30.893710923132858

y[1]:0.0

y[2]:33.72287565255644

{41: [564.9411988461017, 697.8685397510667, 33.231835226241294, 86.4027715882273]}

► w = [-1946.17480406],theta =-2196.2166028664988

► e = [-452.08588642]

► E = [[3.6669627 5.43573295 1.72562951]]

✓ Added optimality cut to master problem.

optimality_cut : Size=201, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.13219975857664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

=====

scenario42

229.0130062237069

w[1]:582.9954682788539

w[2]:0.0

w[3]:6000.0

w[4]:35.748436925339774

y[1]:0.0

y[2]:33.55682747342567

{42: [510.6672978729567, 630.8243091371817, 30.03925281605629, 78.1020573217463]}

➤ w = [-7967.64178596],theta =-8991.787304337286

➤ e = [-1849.6970314]

➤ E = [[15.01535483 22.25805539 7.06604933]]

✓ Added optimality cut to master problem.

optimality_cut : Size=202, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558x[1] + 1.7951177719954141x[2] + 2.0515631679947584x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879x[1] + 0.5364170633387517x[2] + 0.6130480723871449x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502x[1] + 1.5977782649966292x[2] + 1.8260323028532908x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

=====

scenario43

229.0130062237069

w[1]:576.4685976014783

w[2]:0.0

w[3]:5985.43582199338

w[4]:0.0

y[1]:0.0

y[2]:35.27769042078086

{43: [474.04223926950823, 585.581589685863, 100.38541537471939, -0.0]}

➤ w = [-22275.38325673],theta =-25128.326910032716

➤ e = [6142.58356678]

➤ E = [[41.94896954 62.18317838 71.06648957]]

✓ Added optimality cut to master problem.

optimality_cut : Size=203, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338x[1] + 9.398975156030474x[2] + 10.741685892606256x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213x[1] + 26.87638886545576x[2] + 30.71587298909229x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946x[1] + 50.11311033219171x[2] + 57.27212609393337x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534x[1] + 53.57819091568716x[2] + 61.23221818935675x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148x[1] + 30.746168556431453x[2] + 35.13847835020737x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824x[1] + 11.304306750736403x[2] + 12.919207715127316x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283x[1] + 2.169638814535971x[2] + 2.479587216612538x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185x[1] + 5.754724869656774x[2] + 6.5768284224648825x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003x[1] + 28.588208839421302x[2] + 32.67223867362434x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918x[1] + 77.6447742988156x[2] + 88.7368849129321x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557x[1] + 142.90334258183543x[2] + 163.3181058078119x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286x[1] + 145.33164507947927x[2] + 166.09330866226202x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654x[1] + 85.46493416657988x[2] + 97.67421047609123x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856x[1] + 30.981470101388307x[2] + 35.407394401586636x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864x[1] + 5.278085671444797x[2] + 6.0320979102226255x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544x[1] + 7.446134046055281x[2] + 8.509867481206035x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976x[1] + 39.247908426314446x[2] + 44.85475248721651x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

=====

scenario44

229.0130062237069

w[1]:578.3579549028237

w[2]:0.0

w[3]:5999.999999999999

w[4]:0.0

y[1]:0.0

y[2]:34.77954588338855

{44: [451.38302023242574, 557.5907896988788, 95.58699251980781, -0.0]}

► w = [-37987.86150837],theta =-42854.03781133602

► e = [10442.98514056]

► E = [[71.49083266 105.97464607 121.11388122]]

✓ Added optimality cut to master problem.

optimality_cut : Size=204, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] +
97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] +
35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] +
6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] +
116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] +
208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] +
203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] +
119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] +
46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] +
7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

=====

scenario45

229.0130062237069

w[1]:578.7587276637153

w[2]:0.0

w[3]:6000.0

w[4]:3.0893710923137405

y[1]:0.0

y[2]:34.67387886030534

{45: [440.50920327502655, 544.1584275750328, 25.912306075001577, 67.37199579500405]}

➤ w = [-34063.07877687],theta =-38427.797884590254

➤ e = [-7940.77437207]

➤ E = [[64.11232616 95.03709525 30.17050643]]

✓ Added optimality cut to master problem.

optimality_cut : Size=205, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

=====

scenario46

229.0130062237069

w[1]:577.7854223872646

w[2]:0.0

w[3]:5995.586612725267

w[4]:0.0

y[1]:0.0

y[2]:34.93049877350745

{46: [440.30656904152, 543.9081146983483, 93.24139109114543, -0.0]}

► w = [-20337.6786283],theta =-22942.762450113023

► e = [5596.1410902]

► E = [[38.28201434 56.74745655 64.85423606]]

✓ Added optimality cut to master problem.

optimality_cut : Size=206, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

=====

scenario47

229.0130062237069

w[1]:577.8426756388205

w[2]:0.0

w[3]:5996.02795145274

w[4]:0.0

y[1]:0.0

y[2]:34.915403484495556

{47: [450.66257954537656, 556.7008335560535, 95.43442860960917, -0.0]}

► w = [-7156.17881462], theta = -8072.8293037487365

► e = [1968.91830047]

► E = [[13.46994397 19.96721107 22.81966979]]

✓ Added optimality cut to master problem.

optimality_cut : Size=207, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

=====

scenario48

229.0130062237069

w[1]:576.0678248405867

w[2]:0.0

w[3]:5982.346450901066

w[4]:0.0

y[1]:0.0

y[2]:35.3833574438641

{48: [471.920333422532, 582.9604118748925, 99.93607060712444, -0.0]}

► w = [-1018.89986479],theta =-1149.391762375686

► e = [281.15347864]

► E = [[1.91906404 2.84473022 3.25112025]]

✓ Added optimality cut to master problem.

optimality_cut : Size=208, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208},

Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

=====

scenario49

229.0130062237069

w[1]:714.6779468574853

w[2]:1.162337253915723

w[3]:6000.0

w[4]:1050.8275101140116

y[1]:0.0

y[2]:0.0

{49: [404.1714503563933, 356.62186796152344, 23.774791197434915, 61.814457113330725]}

► w = [-496.67137902],theta =-565.9959603789207

► e = [-122.67792258]

► E = [[0.96855646 1.02553037 0.45579128]]

✓ Added optimality cut to master problem.

optimality_cut : Size=209, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188,

189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] +
5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] +
2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] +
0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

=====

scenario50

229.0130062237069

w[1]:712.5595765499161

w[2]:0.60381156047589

w[3]:6000.0

w[4]:1034.4979771974986

y[1]:0.0

y[2]:0.0

{50: [356.0103803319255, 314.1268061752283, 20.941787078348572, 54.448646403706235]}

➤ w = [-2329.20037286],theta =-2653.908988352555

➤ e = [-576.3179804]

➤ E = [[4.53955956 4.80659248 2.13626332]]

✓ Added optimality cut to master problem.

optimality_cut : Size=210, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] +
64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] +
0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

=====

scenario51

229.0130062237069

w[1]:713.0176025623634

w[2]:0.7245738725709998

w[3]:6000.0

w[4]:1038.028687017285

y[1]:0.0

y[2]:0.0

{51: [322.1324325395395, 284.23449929959355, 18.94896661997292, 49.26731321192955]}

► w = [-5995.61957159], theta = -6831.676956503037

► e = [-1482.94612768]

► E = [[11.68677943 12.37423704 5.49966091]]

✓ Added optimality cut to master problem.

optimality_cut : Size=211, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

=====

scenario52

229.0130062237069

w[1]:710.269446487679

w[2]:3.979039320256561e-13

w[3]:6000.0

w[4]:1016.8444280985659

y[1]:0.0

y[2]:0.0

{52: [299.1772753967719, 263.9799488795045, 17.598663258633653, 45.75652447244744]}

► w = [-9218.8928736],theta =-10502.371599496055

► e = [-2285.36241077]

► E = [[17.95623862 19.01248795 8.44999464]]

✓ Added optimality cut to master problem.

optimality_cut : Size=212, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

=====

scenario53

229.0130062237069

w[1]:712.50232329836

w[2]:0.5887162714639942

w[3]:6000.0

w[4]:1034.056638470025

y[1]:0.0

y[2]:0.0

{53: [284.93344531513156, 251.4118635133513, 16.760790900890107, 43.57805634231422]}

➤ w = [-8388.4095732], theta = -9557.780245813457

➤ e = [-2075.65634517]

➤ E = [[16.34856931 17.31024985 7.69344438]]

✓ Added optimality cut to master problem.

optimality_cut : Size=213, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] +
64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] +
0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] +
5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] +
8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

=====

scenario54

229.0130062237069

w[1]:715.765758637048

w[2]:1.4491477451416301

w[3]:6000.0

w[4]:1059.2129459360067

y[1]:0.0

y[2]:0.0

{54: [277.90983588811355, 245.2145610777472, 16.347637405183168, 42.50385725347617]}

► w = [-5184.25447333], theta = -5908.317247371356

► e = [-1279.36610333]

► E = [[10.1127568 10.70762485 4.75894438]]

✓ Added optimality cut to master problem.

optimality_cut : Size=214, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] +
5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] +
8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] +
4.7589443773854665*x[3] + theta[54] : +Inf : True

=====

scenario55

229.0130062237069

w[1]:705.8036928663167

w[2]:0.0

w[3]:6000.0

w[4]:982.4200073556449

y[1]:0.0

y[2]:1.1774325429268515

{55: [278.5379121172207, 344.07624438009617, 16.384583065718882, 42.599915970869034]}

➤ w = [-1641.95534085],theta =-1870.0033217384266

➤ e = [-340.20948278]

➤ E = [[3.19489252 4.73595833 1.50347883]]

✓ Added optimality cut to master problem.

optimality_cut : Size=215, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] +
64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] +
0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] +
5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] +
8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] +
4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] +
1.5034788343598828*x[3] + theta[55] : +Inf : True

=====

scenario56

229.0130062237069

w[1]:705.7464396147608

w[2]:0.0

w[3]:6000.0

w[4]:981.9786686281723

y[1]:0.0

y[2]:1.192527831938719

{56: [285.22839276627633, 352.3409557701061, 16.77814075095745, 43.6231659524893]}

➤ w = [-289.88118141],theta =-330.1409659409251

➤ e = [-60.06574389]

➤ E = [[0.56403915 0.83610509 0.26543019]]

✓ Added optimality cut to master problem.

optimality_cut : Size=216, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] +
0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] +
5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] +
8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] +
4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] +
1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] +
0.2654301866801469*x[3] + theta[56] : +Inf : True

=====

scenario57

229.0130062237069

w[1]:813.0390330305675

w[2]:27.096043776338547

w[3]:6000.0

w[4]:1809.047443913203

y[1]:0.0

y[2]:0.0

{57: [237.95749123312137, 209.9624922645188, 13.997499484301272, 36.39349865918325]}

► $w = [-52.6826011], \theta = -60.42135024103268$

► $e = [-12.03784956]$

► $E = [[0.1052605 \ 0.11145229 \ 0.04953435]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=217, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

=====

scenario58

229.0130062237069

w[1]:868.4029272851487

w[2]:41.69318825083383

w[3]:6000.0

w[4]:2235.821993379919

y[1]:0.0

y[2]:0.0

{58: [202.5184820637726, 178.692778291564, 11.912851886104283, 30.973414903871088]}

➤ w = [-186.90821252],theta =-215.04703533999654

➤ e = [-40.98021049]

➤ E = [[0.37791974 0.40015031 0.17784458]]

✓ Added optimality cut to master problem.

optimality_cut : Size=218, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + theta[62] : +Inf : True

63 : 48.12916769597741 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + theta[63] : +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] +
71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] +
121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] +
64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] +
22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] +
3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] +
0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] +
2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] +
5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] +
8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] +
7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] +
4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] +
1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] +
0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] +
0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] +
0.17784458323727365*x[3] + theta[58] : +Inf : True

=====

scenario59

229.0130062237069

w[1]:856.8950237224074

w[2]:38.65903515944444

w[3]:6000.0

w[4]:2147.1129091577786

y[1]:0.0

y[2]:0.0

{59: [177.8679324269702, 156.94229331791482, 10.46281955452767, 27.2033308417719]}

► w = [-651.10902579],theta =-748.6540793471092

► e = [-143.96839707]

► E = [[1.31337681 1.39063427 0.61805968]]

✓ Added optimality cut to master problem.

optimality_cut : Size=219, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \theta[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

=====

scenario60

229.0130062237069

w[1]:850.7689258059233

w[2]:37.04383923517247

w[3]:6000.0

w[4]:2099.889665318132

y[1]:0.0

y[2]:0.0

{60: [160.88154488019455, 141.95430430605396, 9.463620287070277, 24.605412746382683]}

► w = [-1062.61309374], theta = -1221.3857959159743

► e = [-236.02268996]

► E = [[2.14067777 2.26659999 1.00737777]]

✓ Added optimality cut to master problem.

optimality_cut : Size=220, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

=====

scenario61

229.0130062237069

w[1]:846.4176786876728

w[2]:35.89659727026901

w[3]:6000.0

w[4]:2066.3479220301588

y[1]:0.0

y[2]:0.0

{61: [149.37047034401664, 131.7974738329558, 8.786498255530399, 22.844895464379]}

► w = [-915.58407333],theta =-1052.1282646442057

► e = [-204.02248949]

► E = [[1.84277976 1.95117857 0.86719047]]

✓ Added optimality cut to master problem.

optimality_cut : Size=221, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

=====

scenario62

229.0130062237069

w[1]:851.1696985668145

w[2]:37.149506258255656

w[3]:6000.0

w[4]:2102.9790364104447

y[1]:0.0

y[2]:0.0

{62: [142.3067408014481, 125.56477129539532, 8.370984753026367, 21.76456035786852]}

► w = [-518.73460967], theta = -596.2560085007074

► e = [-115.1847502]

► E = [[1.0451007 1.10657722 0.4918121]]

✓ Added optimality cut to master problem.

optimality_cut : Size=222, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

=====

scenario63

229.0130062237069

w[1]:841.2076327960833

w[2]:34.52292597018726

w[3]:6000.0

w[4]:2026.1860978300847

y[1]:0.0

y[2]:0.0

{63: [139.14895422306395, 122.7784890203505, 8.185232601356711, 21.28160476352742]}

► w = [-188.80659826], theta = -216.8993572344649

► e = [-42.23580022]

► E = [[0.37958443 0.40191293 0.17862797]]

✓ Added optimality cut to master problem.

optimality_cut : Size=223, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

=====

scenario64

229.0130062237069

w[1]:817.2185203941501

w[2]:28.197999874206346

w[3]:6000.0

w[4]:1841.2651710187565

y[1]:0.0

y[2]:0.0

{64: [139.67493727002713, 123.24259170884743, 8.21617278058984, 21.36204922953355]}

► w = [-31.0217625],theta =-35.587560322695595

► e = [-7.06590859]

► E = [[0.06204012 0.06568953 0.02919535]]

✓ Added optimality cut to master problem.

optimality_cut : Size=224, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

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[Thetatheta[1] from previous iteration: 4.56]

[Thetatheta[2] from previous iteration: 31.12]

[Thetatheta[3] from previous iteration: 88.12]

[Thetatheta[4] from previous iteration: 151.38]

[Thetatheta[5] from previous iteration: 124.16]

[Thetatheta[6] from previous iteration: 97.95]

[Thetatheta[7] from previous iteration: 45.35]

[Thetatheta[8] from previous iteration: 7.47]

[Thetatheta[9] from previous iteration: -34.73]

[Thetatheta[10] from previous iteration: -157.96]

[Thetatheta[11] from previous iteration: -585.36]

[Thetatheta[12] from previous iteration: -1,068.39]

[Thetatheta[13] from previous iteration: -852.93]

[Thetatheta[14] from previous iteration: -628.01]

[Thetatheta[15] from previous iteration: -247.90]

[Thetatheta[16] from previous iteration: -76.35]

[Thetatheta[17] from previous iteration: -444.13]

[Thetatheta[18] from previous iteration: -2,622.02]

[Thetatheta[19] from previous iteration: -7,448.60]

[Thetatheta[20] from previous iteration: -13,854.53]

[Thetatheta[21] from previous iteration: -14,722.89]

[Thetatheta[22] from previous iteration: -8,416.91]

[Thetatheta[23] from previous iteration: -3,160.22]

[Thetatheta[24] from previous iteration: -595.95]

[Thetatheta[25] from previous iteration: -1,967.50]

[Thetatheta[26] from previous iteration: -9,787.87]

[Thetatheta[27] from previous iteration: -26,555.56]

[Thetatheta[28] from previous iteration: -48,894.95]

[Thetatheta[29] from previous iteration: -49,682.00]

[Thetatheta[30] from previous iteration: -29,259.30]

[Thetatheta[31] from previous iteration: -10,600.43]

[Thetatheta[32] from previous iteration: -1,790.75]

[Thetatheta[33] from previous iteration: -2,818.81]

[Thetatheta[34] from previous iteration: -14,933.55]

[Thetatheta[35] from previous iteration: -38,686.92]

[Thetatheta[36] from previous iteration: -69,370.40]

[Thetatheta[37] from previous iteration: -67,544.23]

[Thetatheta[38] from previous iteration: -39,578.96]

[Thetatheta[39] from previous iteration: -15,563.95]

[Thetatheta[40] from previous iteration: -2,389.19]

[Thetatheta[41] from previous iteration: -2,196.22]

[Thetatheta[42] from previous iteration: -8,991.79]

[Thetatheta[43] from previous iteration: -25,128.33]

[Thetatheta[44] from previous iteration: -42,854.04]

[Thetatheta[45] from previous iteration: -38,427.80]

[Thetatheta[46] from previous iteration: -22,942.76]

[Thetatheta[47] from previous iteration: -8,072.83]

[Thetatheta[48] from previous iteration: -1,149.39]

[Thetatheta[49] from previous iteration: -566.00]

[Thetatheta[50] from previous iteration: -2,653.91]

[Thetatheta[51] from previous iteration: -6,831.68]

[Thetatheta[52] from previous iteration: -10,502.37]

[Thetatheta[53] from previous iteration: -9,557.78]

[Thetatheta[54] from previous iteration: -5,908.32]

[Thetatheta[55] from previous iteration: -1,870.00]

[Thetatheta[56] from previous iteration: -330.14]

[Thetatheta[57] from previous iteration: -60.42]

[Thetatheta[58] from previous iteration: -215.05]

[Thetatheta[59] from previous iteration: -748.65]

[Thetatheta[60] from previous iteration: -1,221.39]

[Thetatheta[61] from previous iteration: -1,052.13]

[Thetatheta[62] from previous iteration: -596.26]

[Thetatheta[63] from previous iteration: -216.90]

[Thetatheta[64] from previous iteration: -35.59]

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↻ Iteration 5

Master Problem Objective Value : -551,024.33

Decision Variables:

x[1]: 110.99291920026997

x[2]: 108.79880110346946

x[3]: 280.20827969626055

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-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (110.99291920026997, 108.79880110346946,
280.20827969626055)-----

scenario1

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:897.7873281468189

w[4]:0.0

y[1]:155.54733586029187

y[2]:187.7112961896726

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

► w = [4.40539978],theta =4.405399776569969

► e = [9.785496]

► E = [[0.00951779 0.01007766 0.01151733]]

scenario2

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:812.6040111191552

w[4]:0.0

y[1]:159.76506678990216

y[2]:192.67252151999077

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

► w = [30.2715393],theta =30.271539295194447

► e = [60.25830153]

► E = [[0.05304883 0.05616935 0.06419354]]

scenario3

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:853.5144199548095

w[4]:0.0

y[1]:157.73944601449722

y[2]:190.2898277758248

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ w = [85.48259954],theta =85.4825995416264

➤ e = [179.09259324]

➤ E = [[0.16560308 0.17534444 0.20039365]]

scenario4

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:875.9310823305108

w[4]:0.0

y[1]:156.6295168224945

y[2]:188.98424216258314

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

➤ w = [146.59387122],theta =146.5938712196522

➤ e = [316.22021003]

➤ E = [[0.30008169 0.31773355 0.36312406]]

scenario5

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:912.9185752504176

w[4]:0.0

y[1]:154.79813365569

y[2]:186.83002590073446

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

➤ w = [119.86855848],theta =119.8685584779233

➤ e = [271.85361107]

➤ E = [[0.26887293 0.28468899 0.32535884]]

scenario6

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:830.5373410197159

w[4]:0.0

y[1]:158.8771234363

y[2]:191.6280530293975

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

➤ w = [95.17127154],theta =95.17127154135443

➤ e = [193.68136379]

➤ E = [[0.17427172 0.184523 0.21088343]]

scenario7

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:877.6123320086874

w[4]:0.0

y[1]:156.54627213309433

y[2]:188.88632324159008

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [43.90974913],theta =43.90974912811777

➤ e = [94.92932352]

➤ E = [[0.09025744 0.09556671 0.10921909]]

scenario8

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:997.5414757186882

w[4]:0.0

y[1]:150.60815095587984

y[2]:181.9014402107473

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

► w = [7.15680745],theta =7.156807451435885

► e = [18.39287286]

► E = [[0.01987744 0.0210467 0.02405337]]

scenario9

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2221.4912414319533

w[4]:0.0

y[1]:90.00601707253247

y[2]:110.6164657277541

{9: [436.5264598399147, 385.17040574110115, 66.02921241276017, -0.0]}

► w = [-38.87227204],theta =-38.87227204237652

► e = [107.84771361]

► E = [[0.25955863 0.27482679 0.31408776]]

scenario10

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2296.0266438311587

w[4]:0.0

y[1]:86.3155025091235

y[2]:106.27539356372571

{10: [436.2208913180268, 384.90078645708235, 65.98299196407125, -0.0]}

➤ w = [-175.05743901],theta =-175.05743900735428

➤ e = [431.08888083]

➤ E = [[1.0723182 1.13539574 1.29759513]]

scenario11

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2281.4558132869533

w[4]:0.0

y[1]:87.03695648392525

y[2]:107.12402421233276

{11: [446.4982555174795, 393.9690489860112, 67.53755125474477, -0.0]}

➤ w = [-649.86500375],theta =-649.8650037500126

➤ e = [1636.28478346]

➤ E = [[4.04437004 4.28227416 4.89402761]]

scenario12

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2289.8620616778408

w[4]:0.0

y[1]:86.62073303692425

y[2]:106.63442960736714

{12: [467.5104634221321, 412.5092324312929, 70.71586841679306, -0.0]}

► w = [-1184.92071696],theta =-1184.920716959357

► e = [2945.31591956]

► E = [[7.30669766 7.7365034 8.84171817]]

scenario13

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2227.095407025879

w[4]:0.0

y[1]:89.72853477453178

y[2]:110.29006932444372

{13: [501.2553686719416, 442.2841488281836, 75.82013979911719, -0.0]}

► w = [-953.79826883],theta =-953.7982688348754

► e = [2621.27072206]

► E = [[6.32456455 6.69659776 7.65325458]]

scenario14

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2222.6120745507383

w[4]:0.0

y[1]:89.95052061293235

y[2]:110.55118644709205

{14: [549.5312732287364, 484.880535201826, 83.12237746317017, -0.0]}

➤ w = [-702.74089229],theta =-702.7408922926462

➤ e = [1945.98721461]

➤ E = [[4.68579821 4.9614334 5.6702096]]

scenario15

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2289.8620616778408

w[4]:0.0

y[1]:86.62073303692425

y[2]:106.63442960736714

{15: [614.7056822336644, 542.3873666767626, 92.98069143030214, -0.0]}

➤ w = [-274.93974047],theta =-274.9397404660965

➤ e = [683.40808201]

➤ E = [[1.69538901 1.79511777 2.05156317]]

scenario16

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:2315.641223409897

w[4]:0.0

y[1]:85.34431446612113

y[2]:105.13300615213925

{16: [700.6169483826413, 618.1914250435068, 105.97567286460118, -0.0]}

➤ w = [-84.4309485],theta =-84.43094849595772

➤ e = [201.94253218]

➤ E = [[0.50661612 0.53641706 0.61304807]]

scenario17

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:3619.7305571162938

w[4]:0.0

y[1]:20.774183721364068

y[2]:29.180563101807223

{17: [667.5127975715615, 588.9818802102011, 100.96832232174877, -0.0]}

➤ w = [-468.19397127],theta =-468.1939712721502

➤ e = [384.80149507]

➤ E = [[1.50901281 1.59777826 1.8260323]]

scenario18

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:3634.8618042198923

w[4]:0.0

y[1]:20.02498151676224

y[2]:28.29929281286914

{18: [651.7194447810183, 575.0465689244278, 98.5794118156162, -0.0]}

➤ w = [-2763.58704401],theta =-2763.587044009376

➤ e = [2254.18255018]

➤ E = [[8.87680987 9.39897516 10.74168589]]

scenario19

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:3607.401392809658

w[4]:0.0

y[1]:21.384644776965587

y[2]:29.89863518909013

{19: [651.7911339199442, 575.1098240470095, 98.59025555091591, -0.0]}

➤ w = [-7853.41556893],theta =-7853.415568932361

➤ e = [6494.90694624]

► E = [[25.38325615 26.87638887 30.71587299]]

scenario20

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:3597.3138947405932

w[4]:0.0

y[1]:21.88411291336675

y[2]:30.48614871504887

{20: [667.2646554392036, 588.7629312698856, 100.93078821769465, -0.0]}

► w = [-14609.34279435],theta =-14609.342794353106

► e = [12144.21672899]

► E = [[47.32904865 50.11311033 57.27212609]]

scenario21

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:3572.655566127322

w[4]:0.0

y[1]:23.10503502456973

y[2]:31.922292889614653

{21: [699.3400674261661, 617.064765376029, 105.78253120731924, -0.0]}

➤ $w = [-15529.89400967], \theta = -15529.894009668798$

➤ $e = [13073.54549577]$

➤ $E = [[50.60162475 \ 53.57819092 \ 61.23221819]]$

scenario22

110.99291920026997

$w[1]:0.0$

$w[2]:0.0$

$w[3]:3557.524319023724$

$w[4]:0.0$

$y[1]:23.85423722917156$

$y[2]:32.803563178552764$

{22: [749.8953609004036, 661.6723772650621, 113.42955038829635, -0.0]}

➤ $w = [-8880.01376906], \theta = -8880.013769058878$

➤ $e = [7534.24280246]$

➤ $E = [[29.03804808 \ 30.74616856 \ 35.13847835]]$

scenario23

110.99291920026997

$w[1]:0.0$

$w[2]:0.0$

$w[3]:3643.828469170172$

$w[4]:0.0$

$y[1]:19.581009839961183$

y[2]:27.777058567572453

{23: [821.0004390209799, 724.4121520773353, 124.18494035611461, -0.0]}

➤ w = [-3330.48451729],theta =-3330.4845172913174

➤ e = [2704.47203442]

➤ E = [[10.67628971 11.30430675 12.91920772]]

scenario24

110.99291920026997

w[1]:0.0

w[2]:0.0

w[3]:3570.974316449144

w[4]:0.0

y[1]:23.188279713969933

y[2]:32.020211810607776

{24: [918.7980113171591, 810.7041276327875, 138.97785045133497, -0.0]}

➤ w = [-628.63089989],theta =-628.6308998864017

➤ e = [529.66003005]

➤ E = [[2.04910332 2.16963881 2.47958722]]

scenario25

110.99291920026997

w[1]:43.99018463199343

w[2]:47.00035743084209

w[3]:4927.742806738437

w[4]:0.0

y[1]:0.0

y[2]:0.0

{25: [608.9727590580751, 537.3289050512425, 128.95893721229825, -0.0]}

► w = [-1993.80785475],theta =-2023.1032494431772

► e = [727.18511817]

► E = [[3.88215567 4.11051776 6.57682842]]

✓ Added optimality cut to master problem.

optimality_cut : Size=225, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

=====

scenario26

110.99291920026997

w[1]:44.656142147195084

w[2]:47.783708798787075

w[3]:4941.192804163858

w[4]:0.0

y[1]:0.0

y[2]:0.0

{26: [579.4254007885773, 511.2577065781562, 122.70184957875753, -0.0]}

► w = [-9914.63264831], theta = -10062.188576117405

► e = [3602.66263902]

► E = [[19.28569644 20.42014917 32.67223867]]

✓ Added optimality cut to master problem.

optimality_cut : Size=226, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] +
32.67223867362434*x[3] + theta[26] : +Inf : True

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scenario27

110.99291920026997

w[1]:44.156674010793836

w[2]:47.196195272828334

w[3]:4931.1053060947925

w[4]:0.0

y[1]:0.0

y[2]:0.0

{27: [565.594516471754, 499.0539851221356, 119.77295642931259, -0.0]}

➤ w = [-26907.84801408],theta =-27304.48810086101

➤ e = [9804.74729437]

➤ E = [[52.37941123 55.46055307 88.73688491]]

✓ Added optimality cut to master problem.

optimality_cut : Size=227, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227}, Active=True

Key : Lower

: Body

: Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879x[1] + 0.38315504524196553x[2] + 0.6130480723871449x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502x[1] + 1.1412701892833068x[2] + 1.8260323028532908x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338x[1] + 6.71355368287891x[2] + 10.741685892606256x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213x[1] + 19.197420618182683x[2] + 30.71587298909229x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946x[1] + 35.79507880870836x[2] + 57.27212609393337x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534x[1] + 38.27013636834797x[2] + 61.23221818935675x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148x[1] + 21.961548968879608x[2] + 35.13847835020737x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824x[1] + 8.074504821954571x[2] + 12.919207715127316x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283x[1] + 1.5497420103828359x[2] + 2.479587216612538x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185x[1] + 4.110517764040551x[2] + 6.5768284224648825x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003x[1] + 20.420149171015204x[2] + 32.67223867362434x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918x[1] + 55.46055307058254x[2] + 88.7368849129321x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557x[1] + 102.07381612988239x[2] + 163.3181058078119x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286x[1] + 103.80831791391374x[2] + 166.09330866226202x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654x[1] + 61.046381547557026x[2] + 97.67421047609123x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856x[1] + 22.12962150099164x[2] + 35.407394401586636x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864x[1] + 3.770061193889139x[2] + 6.0320979102226255x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

=====

scenario28

110.99291920026997

w[1]:44.35091161939437

w[2]:47.42467275514559

w[3]:4935.028222010541

w[4]:0.0

y[1]:0.0

y[2]:0.0

{28: [565.7585388815307, 499.198710777821, 119.80769058667708, -0.0]}

➤ w = [-49537.59263762],theta =-50270.55006465155

➤ e = [18031.05743329]

➤ E = [[96.40304857 102.07381613 163.31810581]]

✓ Added optimality cut to master problem.

optimality_cut : Size=228, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228},

Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

=====

scenario29

110.99291920026997

w[1]:43.93468817239332

w[2]:46.935078150180004

w[3]:4926.6219736196535

w[4]:0.0

y[1]:0.0

y[2]:0.0

{29: [579.3480589854651, 511.1894638107043, 122.68547131456907, -0.0]}

► w = [-50348.07720793], theta = -51087.05290488438

► e = [18368.7413996]

► E = [[98.04118914 103.80831791 166.09330866]]

✓ Added optimality cut to master problem.

optimality_cut : Size=229, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229},

Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] +
163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] +
166.09330866226202*x[3] + theta[29] : +Inf : True

=====

scenario30

110.99291920026997

w[1]:44.62839391739499

w[2]:47.751069158456005

w[3]:4940.632387604465

w[4]:0.0

y[1]:0.0

y[2]:0.0

{30: [606.9424070349428, 535.5374179720081, 128.52898031328198, -0.0]}

► w = [-29638.74043479],theta =-30079.6100638181

► e = [10771.44260014]

► E = [[57.65491591 61.04638155 97.67421048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=230, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209,

210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] +
7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] +
5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] +
2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] +
0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

=====

scenario31

110.99291920026997

w[1]:44.35091161939437

w[2]:47.42467275514559

w[3]:4935.028222010541

w[4]:0.0

y[1]:0.0

y[2]:0.0

{31: [650.247747776886, 573.7480127443109, 137.69952305863467, -0.0]}

► w = [-10739.75889905], theta = -10898.664199659635

► e = [3909.13646016]

► E = [[20.90019808 22.1296215 35.4073944]]

✓ Added optimality cut to master problem.

optimality_cut : Size=231, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] +
97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] +
35.407394401586636*x[3] + theta[31] : +Inf : True

=====

scenario32

110.99291920026997

w[1]:40.438411217584786

w[2]:42.822483468468874

w[3]:4856.009487136194

w[4]:0.0

y[1]:0.0

y[2]:0.0

{32: [714.641782239231, 630.56627844638, 151.33590682713128, -0.0]}

► w = [-1818.81698067],theta =-1843.659201580075

► e = [676.80780553]

► E = [[3.56061335 3.77006119 6.03209791]]

✓ Added optimality cut to master problem.

optimality_cut : Size=232, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,
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126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146,
147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167,
168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188,
189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209,

210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

=====

scenario33

110.99291920026997

w[1]:107.78336494234861

w[2]:122.03889055190496

w[3]:6000.0

w[4]:216.14047678184397

y[1]:0.0

y[2]:0.0

{33: [646.950912625531, 570.8390405519389, 38.05593603679596, 98.94543369566944]}

► w = [-2714.96051697],theta =-2852.8659447500177

► e = [-916.38693977]

► E = [[5.02318567 5.31866718 2.36385208]]

✓ Added optimality cut to master problem.

optimality_cut : Size=233, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

=====

scenario34

110.99291920026997

w[1]:112.08434056135906

w[2]:127.09803480321631

w[3]:6000.0

w[4]:303.0050434876848

y[1]:0.0

y[2]:0.0

{34: [599.7752520768773, 529.2134577148915, 35.2808971809928, 91.73033267058125]}

➤ w = [-14243.74764571],theta =-15088.328543337877

➤ e = [-4763.62673738]

➤ E = [[26.47676362 28.0342203 12.45965347]]

✓ Added optimality cut to master problem.

optimality_cut : Size=234, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + theta[64] : +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 6.579587216612538 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] +
12.459653468671258*x[3] + theta[34] : +Inf : True

=====

scenario35

110.99291920026997

w[1]:110.47494323295518

w[2]:125.20493566401592

w[3]:6000.0

w[4]:270.50088304291876

y[1]:0.0

y[2]:0.0

{35: [571.327911870871, 504.1128634154743, 33.60752422769831, 87.37956299201556]}

➤ w = [-37033.67369914],theta =-39135.65200441829

➤ e = [-12428.0624594]

➤ E = [[68.72032106 72.76269289 32.33897462]]

✓ Added optimality cut to master problem.

optimality_cut : Size=235, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

=====

scenario36

110.99291920026997

w[1]:110.78017376075587

w[2]:125.56397170765734

w[3]:6000.0

w[4]:276.66546519623625

y[1]:0.0

y[2]:0.0

{36: [557.5760490321393, 491.9788667930638, 32.79859111953762, 85.27633691079775]}

► w = [-66360.18612321],theta =-70171.08500536402

► e = [-22255.15601825]

► E = [[123.17970075 130.4255655 57.966918]]

✓ Added optimality cut to master problem.

optimality_cut : Size=236, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

=====

scenario37

110.99291920026997

w[1]:111.52937596535776

w[2]:126.44524199659548

w[3]:6000.0

w[4]:291.7967122998343

y[1]:0.0

y[2]:0.0

{37: [557.3641701335071, 491.7919148236825, 32.7861276549122, 85.24393190277164]}

► w = [-64504.43620032],theta =-68314.43519767933

► e = [-21598.18945395]

► E = [[119.83165235 126.88057308 56.39136581]]

✓ Added optimality cut to master problem.

optimality_cut : Size=237, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25] : +\text{Inf} : \text{True}$

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26] : +\text{Inf} : \text{True}$

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27] : +\text{Inf} : \text{True}$

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28] : +\text{Inf} : \text{True}$

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29] : +\text{Inf} : \text{True}$

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30] : +\text{Inf} : \text{True}$

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31] : +\text{Inf} : \text{True}$

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32] : +\text{Inf} : \text{True}$

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33] : +\text{Inf} : \text{True}$

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34] : +\text{Inf} : \text{True}$

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35] : +\text{Inf} : \text{True}$

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36] : +\text{Inf} : \text{True}$

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37] : +\text{Inf} : \text{True}$

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38] : +\text{Inf} : \text{True}$

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39] : +\text{Inf} : \text{True}$

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40] : +\text{Inf} : \text{True}$

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41] : +\text{Inf} : \text{True}$

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

=====

scenario38

110.99291920026997

w[1]:110.6691808415556

w[2]:125.43341314633324

w[3]:6000.0

w[4]:274.42379895866634

y[1]:0.0

y[2]:0.0

{38: [570.7297629333085, 503.5850849411544, 33.572338996076994, 87.2880813898001]}

► $w = [-37870.98314921], \theta = -40036.61057083889$

► $e = [-12703.77307612]$

► $E = [[70.28879468 \ 74.42342967 \ 33.07707985]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=238, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

=====

scenario39

110.99291920026997

w[1]:110.86341845015608

w[2]:125.6618906286505

w[3]:6000.0

w[4]:278.3467148744139

y[1]:0.0

y[2]:0.0

{39: [597.9764018157447, 527.6262368962451, 35.17508245974971, 91.45521439534915]}

► w = [-14885.78632712],theta =-15743.353065303414

► e = [-4991.34420104]

► E = [[27.63390972 29.25943382 13.00419281]]

✓ Added optimality cut to master problem.

optimality_cut : Size=239, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.13219975857664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

=====

scenario40

110.99291920026997

w[1]:106.86767335894638

w[2]:120.96178242098063

w[3]:6000.0

w[4]:197.64673032189057

y[1]:0.0

y[2]:0.0

{40: [642.3582103585092, 566.7866561986845, 37.78577707991233, 98.24302040777198]}

► w = [-2306.03489234],theta =-2418.4538237853703

► $e = [-779.89843893]$

► $E = [[4.26230367 \ 4.51302741 \ 2.00578996]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=240, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : $4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + theta[61] : +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \text{theta}[30]$: +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641 \cdot x[1] + 28.0342203045103 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

235 : -12428.06245940283 : $68.72032106367413 \cdot x[1] + 72.76269289094907 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

236 : -22255.156018251044 : $123.1797007521802 \cdot x[1] + 130.42556550230836 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

237 : -21598.18945394994 : $119.83165235440212 \cdot x[1] + 126.8805730811316 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

238 : -12703.773076115522 : $70.28879468381454 \cdot x[1] + 74.42342966521537 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

239 : -4991.344201038478 : $27.633909721859872 \cdot x[1] + 29.25943382314574 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

240 : -779.8984389293898 : $4.2623036690128515 \cdot x[1] + 4.5130274142489 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

=====

scenario41

110.99291920026997

w[1]:179.17956021792224

w[2]:206.02068512367305

w[3]:6000.0

w[4]:1658.0922840988005

y[1]:0.0

y[2]:0.0

{41: [564.9411988461017, 498.477528393619, 33.231835226241294, 86.4027715882273]}

► $w = [-1855.98020708], \theta = -1932.9119029074502$

► $e = [-543.0081876]$

► $E = [[3.6669627 \ 3.88266639 \ 1.72562951]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=241, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

=====

scenario42

110.99291920026997

w[1]:179.48479074572305

w[2]:206.37972116731453

w[3]:6000.0

w[4]:1664.256866252119

y[1]:0.0

y[2]:0.0

{42: [510.6672978729567, 450.58879224084404, 30.03925281605629, 78.1020573217463]}

► w = [-7598.01654548],theta =-7913.316670543419

► e = [-2221.70313828]

► E = [[15.01535483 15.89861099 7.06604933]]

✓ Added optimality cut to master problem.

optimality_cut : Size=242, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

19 : 6494.906946237561 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : $89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

=====

scenario43

110.99291920026997

w[1]:176.32149254851538

w[2]:202.65880216957584

w[3]:6000.0

w[4]:1600.3693784813713

y[1]:0.0

y[2]:0.0

{43: [474.04223926950823, 418.2725640613307, 27.884837604088734, 72.50057777063066]}

► w = [-21279.04249404], theta = -23036.62121262763

► e = [-6259.03064861]

► E = [[41.94896954 44.41655598 19.74069155]]

✓ Added optimality cut to master problem.

optimality_cut : Size=243, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \theta[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \theta[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \theta[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \theta[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \theta[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \theta[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \theta[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \theta[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \theta[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \theta[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \theta[25]$: +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] +
19.740691548023207*x[3] + theta[43] : +Inf : True

=====

scenario44

110.99291920026997

w[1]:177.2371841319175

w[2]:203.73591030050017

w[3]:6000.0

w[4]:1618.8631249413238

y[1]:0.0

y[2]:0.0

{44: [451.38302023242574, 398.27913549919907, 26.551942366613293, 69.03505015319452]}

➤ w = [-36238.56144746],theta =-39242.31576130607

➤ e = [-10640.95642284]

➤ E = [[71.49083266 75.69617576 33.64274478]]

✓ Added optimality cut to master problem.

optimality_cut : Size=244, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 3.7008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

=====

scenario45

110.99291920026997

w[1]:177.43142174051803

w[2]:203.96438778281754

w[3]:6000.0

w[4]:1622.7860408570723

y[1]:0.0

y[2]:0.0

{45: [440.50920327502655, 388.68459112502336, 25.912306075001577, 67.37199579500405]}

➤ w = [-32493.50014841],theta =-33831.60233117809

➤ e = [-9537.80162009]

➤ E = [[64.11232616 67.88363946 30.17050643]]

✓ Added optimality cut to master problem.

optimality_cut : Size=245, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \text{theta}[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \text{theta}[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \text{theta}[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \text{theta}[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \text{theta}[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \text{theta}[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

98 : -388.6795681741489 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

99 : -1000.126458202169 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

100 : -1541.290928191132 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

101 : -1399.8612560423385 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

102 : -862.8283022455653 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

103 : -275.588687165391 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] +
12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] +
32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] +
57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] +
56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] +
33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] +
13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] +
2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] +
19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] +
33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

=====

scenario46

110.99291920026997

w[1]:176.95970183391688

w[2]:203.40951389718975

w[3]:6000.0

w[4]:1613.2589593473995

y[1]:0.0

y[2]:0.0

{46: [440.30656904152, 388.50579621310584, 25.900386414207077, 67.34100467693835]}

► w = [-19409.27158546],theta =-21016.360679373392

► e = [-5702.22907295]

► E = [[38.28201434 40.53389754 18.01506557]]

✓ Added optimality cut to master problem.

optimality_cut : Size=246, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.10510325409077247*x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641 \cdot x[1] + 28.0342203045103 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

235 : -12428.06245940283 : $68.72032106367413 \cdot x[1] + 72.76269289094907 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

236 : -22255.156018251044 : $123.1797007521802 \cdot x[1] + 130.42556550230836 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

237 : -21598.18945394994 : $119.83165235440212 \cdot x[1] + 126.8805730811316 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

238 : -12703.773076115522 : $70.28879468381454 \cdot x[1] + 74.42342966521537 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

239 : -4991.344201038478 : $27.633909721859872 \cdot x[1] + 29.25943382314574 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

240 : -779.8984389293898 : $4.2623036690128515 \cdot x[1] + 4.5130274142489 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

241 : -543.0081875967824 : $3.66696270406019 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

242 : -2221.7031382755217 : $15.015354826277694 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

243 : -6259.030648613752 : $41.94896953954929 \cdot x[1] + 44.41655598305218 \cdot x[2] + 19.740691548023207 \cdot x[3] + \theta[43]$: +Inf : True

244 : -10640.956422843938 : $71.49083266469698 \cdot x[1] + 75.69617576262033 \cdot x[2] + 33.64274478338684 \cdot x[3] + \theta[44]$: +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] +
18.015065571174564*x[3] + theta[46] : +Inf : True

=====

scenario47

110.99291920026997

w[1]:176.98745006371695

w[2]:203.44215353752082

w[3]:6000.0

w[4]:1613.8193759067917

y[1]:0.0

y[2]:0.0

{47: [450.66257954537656, 397.6434525400381, 26.509563502669227, 68.92486510693995]}

► w = [-6829.21606402],theta =-7394.73595915906

► e = [-2006.24376588]

► E = [[13.46994397 14.26229362 6.33879716]]

✓ Added optimality cut to master problem.

optimality_cut : Size=247, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209,

210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641 \cdot x[1] + 28.0342203045103 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

235 : -12428.06245940283 : $68.72032106367413 \cdot x[1] + 72.76269289094907 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

236 : -22255.156018251044 : $123.1797007521802 \cdot x[1] + 130.42556550230836 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

237 : -21598.18945394994 : $119.83165235440212 \cdot x[1] + 126.8805730811316 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

=====

scenario48

110.99291920026997

w[1]:176.12725493991485

w[2]:202.43032468725852

w[3]:6000.0

w[4]:1596.4464625656228

y[1]:0.0

y[2]:0.0

{48: [471.920333422532, 416.4002941963517, 27.760019613090133, 72.1760509940343]}

► w = [-973.61266606], theta = -1053.96950428431

► e = [-286.48340241]

► E = [[1.91906404 2.03195016 0.90308896]]

✓ Added optimality cut to master problem.

optimality_cut : Size=248, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : 96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] +
6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] +
0.903088958053048*x[3] + theta[48] : +Inf : True

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scenario49

110.99291920026997

w[1]:243.30571928587824

w[2]:281.4508939287084

w[3]:6000.0

w[4]:2953.214952854916

y[1]:0.0

y[2]:0.0

{49: [404.1714503563933, 356.62186796152344, 23.774791197434915, 61.814457113330725]}

► w = [-469.47379662],theta =-469.47379662356246

► e = [-122.67792258]

► E = [[0.96855646 1.02553037 0.45579128]]

scenario50

110.99291920026997

w[1]:242.27903478327573

w[2]:280.2432272364599

w[3]:6000.0

w[4]:2932.479540157394

y[1]:0.0

y[2]:0.0

{50: [356.0103803319255, 314.1268061752283, 20.941787078348572, 54.448646403706235]}

➤ w = [-2201.72711769],theta =-2201.7271176882955

➤ e = [-576.3179804]

➤ E = [[4.53955956 4.80659248 2.13626332]]

scenario51

110.99291920026997

w[1]:242.50102062167628

w[2]:280.50434435910825

w[3]:6000.0

w[4]:2936.9628726325336

y[1]:0.0

y[2]:0.0

{51: [322.1324325395395, 284.23449929959355, 18.94896661997292, 49.26731321192955]}

➤ w = [-5667.44856829],theta =-5667.448568285886

➤ e = [-1482.94612768]

➤ E = [[11.68677943 12.37423704 5.49966091]]

scenario52

110.99291920026997

w[1]:241.16910559127308

w[2]:278.93764162321827

w[3]:6000.0

w[4]:2910.062877781693

y[1]:0.0

y[2]:0.0

{52: [299.1772753967719, 263.9799488795045, 17.598663258633653, 45.75652447244744]}

➤ w = [-8714.6721103],theta =-8714.672110297452

➤ e = [-2285.36241077]

➤ E = [[17.95623862 19.01248795 8.44999464]]

scenario53

110.99291920026997

w[1]:242.25128655347567

w[2]:280.2105875961288

w[3]:6000.0

w[4]:2931.9191235979997

y[1]:0.0

y[2]:0.0

{53: [284.93344531513156, 251.4118635133513, 16.760790900890107, 43.57805634231422]}

➤ w = [-7929.3330222],theta =-7929.333022198924

➤ e = [-2075.65634517]

➤ E = [[16.34856931 17.31024985 7.69344438]]

scenario54

110.99291920026997

w[1]:243.83293565207958

w[2]:282.0710470949982

w[3]:6000.0

w[4]:2963.8628674833762

y[1]:0.0

y[2]:0.0

{54: [277.90983588811355, 245.2145610777472, 16.347637405183168, 42.50385725347617]}

► w = [-4900.28286535],theta =-4900.282865346848

► e = [-1279.36610333]

► E = [[10.1127568 10.70762485 4.75894438]]

scenario55

110.99291920026997

w[1]:239.0047436668678

w[2]:276.3917496773971

w[3]:6000.0

w[4]:2866.3503861490753

y[1]:0.0

y[2]:0.0

{55: [278.5379121172207, 245.7687459857829, 16.384583065718882, 42.599915970869034]}

► w = [-1552.57673005],theta =-1629.8001134995793

► e = [-408.63150166]

► E = [[3.19489252 3.38282738 1.50347883]]

✓ Added optimality cut to master problem.

optimality_cut : Size=249, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] +
0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] +
0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] +
0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] +
0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] +
0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] +
0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] +
0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] +
1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] +
4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] +
8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

83 : -8381.716542387952 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

84 : -15009.291268122797 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

85 : -14566.220794524372 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

86 : -8567.660911798837 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

87 : -3366.255391398042 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

88 : -525.9780169523792 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

89 : -366.21482419317874 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

90 : -1498.357930464886 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

91 : -4221.2067165069475 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 19.740691548023207 \cdot x[3] + \theta[43]$: +Inf : True

92 : -7176.458982848237 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 33.64274478338684 \cdot x[3] + \theta[44]$: +Inf : True

93 : -6432.470860058384 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

94 : -3845.689374781463 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 18.015065571174564 \cdot x[3] + \theta[46]$: +Inf : True

95 : -1353.0481211762367 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 6.338797163551849 \cdot x[3] + \theta[47]$: +Inf : True

96 : -193.20973650710715 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 0.903088958053048 \cdot x[3] + \theta[48]$: +Inf : True

97 : -82.73627336707335 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

=====

scenario56

110.99291920026997

w[1]:238.97699543706773

w[2]:276.3591100370661

w[3]:6000.0

w[4]:2865.789969589683

y[1]:0.0

y[2]:0.0

{56: [285.22839276627633, 251.67211126436143, 16.77814075095745, 43.6231659524893]}

► w = [-274.10268635],theta =-287.7352912132934

► e = [-72.14600523]

► E = [[0.56403915 0.59721792 0.26543019]]

✓ Added optimality cut to master problem.

optimality_cut : Size=250, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : $0.05304882616689596 \cdot x[1] + 0.056169345353184 \cdot x[2] + 0.06419353754649597 \cdot x[3] + \text{theta}[2]$: +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \text{theta}[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \text{theta}[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \text{theta}[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : 98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

190 : 12987.282220744644 : 57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

191 : 4713.301674826999 : 20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

192 : 816.036839813409 : 3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641 \cdot x[1] + 28.0342203045103 \cdot x[2] + 12.459653468671258 \cdot x[3] + \text{theta}[34]$: +Inf : True

235 : -12428.06245940283 : $68.72032106367413 \cdot x[1] + 72.76269289094907 \cdot x[2] + 32.33897461819961 \cdot x[3] + \text{theta}[35]$: +Inf : True

236 : -22255.156018251044 : $123.1797007521802 \cdot x[1] + 130.42556550230836 \cdot x[2] + 57.966918001025995 \cdot x[3] + \text{theta}[36]$: +Inf : True

237 : -21598.18945394994 : $119.83165235440212 \cdot x[1] + 126.8805730811316 \cdot x[2] + 56.39136581383633 \cdot x[3] + \text{theta}[37]$: +Inf : True

238 : -12703.773076115522 : $70.28879468381454 \cdot x[1] + 74.42342966521537 \cdot x[2] + 33.07707985120687 \cdot x[3] + \text{theta}[38]$: +Inf : True

239 : -4991.344201038478 : $27.633909721859872 \cdot x[1] + 29.25943382314574 \cdot x[2] + 13.004192810287009 \cdot x[3] + \text{theta}[39]$: +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] +
2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] +
19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] +
33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] +
18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] +
6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] +
0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] +
1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] +
0.2654301866801469*x[3] + theta[56] : +Inf : True

=====

scenario57

110.99291920026997

w[1]:290.9771780823942

w[2]:337.5257960174365

w[3]:6000.0

w[4]:3916.0106018912684

y[1]:0.0

y[2]:0.0

{57: [237.95749123312137, 209.9624922645188, 13.997499484301272, 36.39349865918325]}

➤ w = [-49.72683019],theta =-49.72683018793648

➤ e = [-12.03784956]

➤ E = [[0.1052605 0.11145229 0.04953435]]

scenario58

110.99291920026997

w[1]:317.80971629905946

w[2]:369.0883282175531

w[3]:6000.0

w[4]:4457.933414823838

y[1]:0.0

y[2]:0.0

{58: [202.5184820637726, 178.692778291564, 11.912851886104283, 30.973414903871088]}

➤ w = [-176.29602455],theta =-176.29602454518025

➤ e = [-40.98021049]

➤ E = [[0.37791974 0.40015031 0.17784458]]

scenario59

110.99291920026997

w[1]:312.23232210924596

w[2]:362.52776051101387

w[3]:6000.0

w[4]:4345.2896863859405

y[1]:0.0

y[2]:0.0

{59: [177.8679324269702, 156.94229331791482, 10.46281955452767, 27.2033308417719]}

➤ w = [-614.22870396],theta =-614.2287039600853

➤ e = [-143.96839707]

➤ E = [[1.31337681 1.39063427 0.61805968]]

scenario60

110.99291920026997

w[1]:309.2632615206387

w[2]:359.03531899559255

w[3]:6000.0

w[4]:4285.3251145309405

y[1]:0.0

y[2]:0.0

{60: [160.88154488019455, 141.95430430605396, 9.463620287070277, 24.605412746382683]}

➤ w = [-1002.50171952],theta =-1002.5017195153287

➤ e = [-236.02268996]

➤ E = [[2.14067777 2.26659999 1.00737777]]

scenario61

110.99291920026997

w[1]:307.15439605583356

w[2]:356.5547063304334

w[3]:6000.0

w[4]:4242.733456017109

y[1]:0.0

y[2]:0.0

{61: [149.37047034401664, 131.7974738329558, 8.786498255530399, 22.844895464379]}

➤ w = [-863.83783426],theta =-863.83783425814

➤ e = [-204.02248949]

➤ E = [[1.84277976 1.95117857 0.86719047]]

scenario62

110.99291920026997

w[1]:309.4574991292391

w[2]:359.26379647790975

w[3]:6000.0

w[4]:4289.248030446686

y[1]:0.0

y[2]:0.0

{62: [142.3067408014481, 125.56477129539532, 8.370984753026367, 21.76456035786852]}

➤ w = [-489.38762414],theta =-489.3876241396801

➤ e = [-115.1847502]

➤ E = [[1.0451007 1.10657722 0.4918121]]

scenario63

110.99291920026997

w[1]:304.6293071440274

w[2]:353.58449906030864

w[3]:6000.0

w[4]:4191.735549112389

y[1]:0.0

y[2]:0.0

{63: [139.14895422306395, 122.7784890203505, 8.185232601356711, 21.28160476352742]}

➤ w = [-178.14766483],theta =-178.14766482755732

➤ e = [-42.23580022]

➤ E = [[0.37958443 0.40191293 0.17862797]]

scenario64

110.99291920026997

w[1]:293.0027988577991

w[2]:339.9084897616026

w[3]:6000.0

w[4]:3956.9210107269228

y[1]:0.0

y[2]:0.0

{64: [139.67493727002713, 123.24259170884743, 8.21617278058984, 21.36204922953355]}

➤ w = [-29.27964295],theta =-29.27964295364845

➤ e = [-7.06590859]

➤ E = [[0.06204012 0.06568953 0.02919535]]

[Thetatheta[1] from previous iteration: 4.41]

[Thetatheta[2] from previous iteration: 30.27]

[Thetatheta[3] from previous iteration: 85.48]

[Thetatheta[4] from previous iteration: 146.59]

[Thetatheta[5] from previous iteration: 119.87]

[Thetatheta[6] from previous iteration: 95.17]

[Thetatheta[7] from previous iteration: 43.91]

[Thetatheta[8] from previous iteration: 7.16]

[Thetatheta[9] from previous iteration: -38.87]

[Thetatheta[10] from previous iteration: -175.06]

[Thetatheta[11] from previous iteration: -649.87]

[Thetatheta[12] from previous iteration: -1,184.92]

[Thetatheta[13] from previous iteration: -953.80]

[Thetatheta[14] from previous iteration: -702.74]

[Thetatheta[15] from previous iteration: -274.94]

[Thetatheta[16] from previous iteration: -84.43]

[Thetatheta[17] from previous iteration: -468.19]

[Thetatheta[18] from previous iteration: -2,763.59]

[Thetatheta[19] from previous iteration: -7,853.42]

[Thetatheta[20] from previous iteration: -14,609.34]

[Thetatheta[21] from previous iteration: -15,529.89]

[Thetatheta[22] from previous iteration: -8,880.01]

[Thetatheta[23] from previous iteration: -3,330.48]

[Thetatheta[24] from previous iteration: -628.63]

[Thetatheta[25] from previous iteration: -2,023.10]

[Thetatheta[26] from previous iteration: -10,062.19]

[Thetatheta[27] from previous iteration: -27,304.49]

[Thetatheta[28] from previous iteration: -50,270.55]

[Thetatheta[29] from previous iteration: -51,087.05]

[Thetatheta[30] from previous iteration: -30,079.61]

[Thetatheta[31] from previous iteration: -10,898.66]

[Thetatheta[32] from previous iteration: -1,843.66]

[Thetatheta[33] from previous iteration: -2,852.87]

[Thetatheta[34] from previous iteration: -15,088.33]

[Thetatheta[35] from previous iteration: -39,135.65]

[Thetatheta[36] from previous iteration: -70,171.09]

[Thetatheta[37] from previous iteration: -68,314.44]

[Thetatheta[38] from previous iteration: -40,036.61]

[Thetatheta[39] from previous iteration: -15,743.35]

[Thetatheta[40] from previous iteration: -2,418.45]

[Thetatheta[41] from previous iteration: -1,932.91]

[Thetatheta[42] from previous iteration: -7,913.32]

[Thetatheta[43] from previous iteration: -23,036.62]

[Thetatheta[44] from previous iteration: -39,242.32]

[Thetatheta[45] from previous iteration: -33,831.60]

[Thetatheta[46] from previous iteration: -21,016.36]

[Thetatheta[47] from previous iteration: -7,394.74]

[Thetatheta[48] from previous iteration: -1,053.97]

[Thetatheta[49] from previous iteration: -469.47]

[Thetatheta[50] from previous iteration: -2,201.73]

[Thetatheta[51] from previous iteration: -5,667.45]

[Thetatheta[52] from previous iteration: -8,714.67]

[Thetatheta[53] from previous iteration: -7,929.33]

[Thetatheta[54] from previous iteration: -4,900.28]

[Thetatheta[55] from previous iteration: -1,629.80]

[Thetatheta[56] from previous iteration: -287.74]

[Thetatheta[57] from previous iteration: -49.73]

[Thetatheta[58] from previous iteration: -176.30]

[Thetatheta[59] from previous iteration: -614.23]

[Thetatheta[60] from previous iteration: -1,002.50]

[Thetatheta[61] from previous iteration: -863.84]

[Thetatheta[62] from previous iteration: -489.39]

[Thetatheta[63] from previous iteration: -178.15]

[Thetatheta[64] from previous iteration: -29.28]

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↻ Iteration 6

Master Problem Objective Value : -532,468.83

Decision Variables:

x[1]: 126.62525374422897

x[2]: 123.5626726172085

x[3]: 249.8120736385625

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-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [-0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (126.62525374422897, 123.5626726172085,
249.8120736385625)-----

scenario1

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:800.3978839379545

w[4]:0.0

y[1]:149.2865858754363

y[2]:180.6157795401696

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

► w = [4.45791224],theta =4.457912242798946

► e = [9.785496]

► E = [[0.00951779 0.01007766 0.01151733]]

scenario2

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:724.4550135518309

w[4]:0.0

y[1]:154.098345517717

y[2]:186.2502374115143

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

► w = [30.56422529],theta =30.56422529441982

➤ $e = [60.25830153]$

➤ $E = [[0.05304883 \ 0.05616935 \ 0.06419354]]$

scenario3

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:760.9275763030613

w[4]:0.0

y[1]:151.78743463688483

y[2]:183.54421488119743

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ $w = [86.39628053], \theta = 86.39628052956195$

➤ $e = [179.09259324]$

➤ $E = [[0.16560308 \ 0.17534444 \ 0.20039365]]$

scenario4

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:780.9125421941467

w[4]:0.0

y[1]:150.5211820994425

y[2]:182.06146280979092

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

► w = [148.24951027],theta =148.24951026739615

► e = [316.22021003]

► E = [[0.30008169 0.31773355 0.36312406]]

scenario5

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:813.8877359144374

w[4]:0.0

y[1]:148.43186541266272

y[2]:179.6149218919702

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

► w = [121.35200963],theta =121.35200963014768

► e = [271.85361107]

► E = [[0.26887293 0.28468899 0.32535884]]

scenario6

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:740.4429862646989

w[4]:0.0

y[1]:153.0853434877632

y[2]:185.06403575438912

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

➤ w = [96.13277997],theta =96.13277997495925

➤ e = [193.68136379]

➤ E = [[0.17427172 0.184523 0.21088343]]

scenario7

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:782.4114146359773

w[4]:0.0

y[1]:150.4262131591344

y[2]:181.95025640443546

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [44.40772604],theta =44.40772603524393

➤ e = [94.92932352]

➤ E = [[0.09025744 0.09556671 0.10921909]]

scenario8

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:889.3309821532831

w[4]:0.0

y[1]:143.65176208381808

y[2]:174.01753282241063

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

➤ w = [7.26647715],theta =7.266477145246946

➤ e = [18.39287286]

➤ E = [[0.01987744 0.0210467 0.02405337]]

scenario9

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:1980.5101198065233

w[4]:0.0

y[1]:74.5143735394691

y[2]:93.05926972361564

{9: [436.5264598399147, 385.17040574110115, 66.02921241276017, -0.0]}

➤ w = [-37.44021061],theta =-37.44021061232216

➤ e = [107.84771361]

➤ E = [[0.25955863 0.27482679 0.31408776]]

scenario10

126.62525374422897

w[1]:0.0
w[2]:0.0
w[3]:2046.960131394381
w[4]:0.0
y[1]:70.3040838524735
y[2]:88.12911908618904
{10: [436.2208913180268, 384.90078645708235, 65.98299196407125, -0.0]}
 ➤ w = [-169.14114368],theta =-169.14114367944563
 ➤ e = [431.08888083]
 ➤ E = [[1.0723182 1.13539574 1.29759513]]

scenario11

126.62525374422897
w[1]:0.0
w[2]:0.0
w[3]:2033.9699035651759
w[4]:0.0
y[1]:71.12714800181098
y[2]:89.09290793260325
{11: [446.4982555174795, 393.9690489860112, 67.53755125474477, -0.0]}
 ➤ w = [-627.55102301],theta =-627.5510230126094
 ➤ e = [1636.28478346]
 ➤ E = [[4.04437004 4.28227416 4.89402761]]

scenario12

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:2041.4642657743325

w[4]:0.0

y[1]:70.65230330027012

y[2]:88.53687590582581

{12: [467.5104634221321, 412.5092324312929, 70.71586841679306, -0.0]}

► w = [-1144.60751383],theta =-1144.6075138333001

► e = [2945.31591956]

► E = [[7.30669766 7.7365034 8.84171817]]

scenario13

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:1985.5063612792949

w[4]:0.0

y[1]:74.19781040510851

y[2]:92.68858170576402

{13: [501.2553686719416, 442.2841488281836, 75.82013979911719, -0.0]}

► w = [-918.90378334],theta =-918.9037833361251

► e = [2621.27072206]

► E = [[6.32456455 6.69659776 7.65325458]]

scenario14

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:1981.5093681010774

w[4]:0.0

y[1]:74.451060912597

y[2]:92.98513212004534

{14: [549.5312732287364, 484.880535201826, 83.12237746317017, -0.0]}

➤ w = [-676.88796337],theta =-676.8879633696043

➤ e = [1945.98721461]

➤ E = [[4.68579821 4.9614334 5.6702096]]

scenario15

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:2041.4642657743325

w[4]:0.0

y[1]:70.65230330027012

y[2]:88.53687590582581

{15: [614.7056822336644, 542.3873666767626, 92.98069143030214, -0.0]}

➤ w = [-265.58577995],theta =-265.5857799468837

➤ e = [683.40808201]

► E = [[1.69538901 1.79511777 2.05156317]]

scenario16

126.62525374422897

w[1]:0.0

w[2]:0.0

w[3]:2064.4469765490803

w[4]:0.0

y[1]:69.19611288221148

y[2]:86.83171102370832

{16: [700.6169483826413, 618.1914250435068, 105.97567286460118, -0.0]}

► w = [-81.63579817],theta =-81.6357981662246

► e = [201.94253218]

► E = [[0.50661612 0.53641706 0.61304807]]

scenario17

126.62525374422897

w[1]:4.468128483493729

w[2]:0.0

w[3]:3227.07236726295

w[4]:0.0

y[1]:0.0

y[2]:0.5726092696350804

{17: [476.7948554082581, 588.9818802102011, 100.96832232174877, -0.0]}

► w = [-458.67528997],theta =-458.6752899727486

► e = [331.40047126]

► E = [[1.07786629 1.59777826 1.8260323]]

scenario18

126.62525374422897

w[1]:5.3228489462673

w[2]:0.4282483785643194

w[3]:3240.562219239433

w[4]:0.0

y[1]:0.0

y[2]:0.0

{18: [465.5138891292987, 410.7475492317342, 98.5794118156162, -0.0]}

► w = [-2705.69442873],theta =-2706.2854593557727

► e = [1610.13039299]

► E = [[6.34057848 6.71355368 10.74168589]]

✓ Added optimality cut to master problem.

optimality_cut : Size=251, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251},
Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : 4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

15 : 683.408082012721 : 1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

16 : 201.94253218087894 : 0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

17 : 384.80149507066477 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

38 : 14476.392575108388 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

39 : 5687.810833741524 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

40 : 888.7214769195373 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

41 : 618.7767719126125 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 6.212266228054911*x[3] + theta[41] : +Inf : True

42 : 2531.708227337223 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 25.43777758804692*x[3] + theta[42] : +Inf : True

43 : 7132.383762373813 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

104 : -48.65660817777649 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

105 : -8.118549700894718 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : 4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

174 : 1675.9318460518732 : 3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

175 : 588.5677767538126 : 1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

176 : 173.91785424557327 : 0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

177 : 331.4004712649398 : 1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

178 : 1941.3572166888684 : 6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

179 : 5593.572921045409 : 18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

180 : 10458.896856398489 : 33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

181 : 11259.25754941549 : 36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

182 : 6488.67441354561 : 20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

183 : 2329.157548012461 : 7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

184 : 456.15618914804844 : 1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

185 : 876.7774853356144 : 3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

186 : 4343.781810476438 : 19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

187 : 11821.723880635855 : 52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019 \cdot x[1] + 5.435732949548046 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694 \cdot x[1] + 22.25805538954105 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

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scenario19

126.62525374422897

w[1]:3.771689587900454

w[2]:0.0

w[3]:3216.0806360228535

w[4]:0.0

y[1]:0.0

y[2]:1.388122908908656

{19: [465.5650956571029, 575.1098240470095, 98.59025555091591, -0.0]}

➤ w = [-7696.37090745],theta =-7696.370907449393

➤ e = [5593.57292105]

➤ E = [[18.13089725 26.87638887 30.71587299]]

scenario20

126.62525374422897

w[1]:3.2018759460514445

w[2]:0.0

w[3]:3207.087401371866

w[4]:0.0

y[1]:0.0

y[2]:2.055361341041589

{20: [476.61761102800256, 588.7629312698856, 100.93078821769465, -0.0]}

➤ w = [-14321.23356663],theta =-14321.233566628733

➤ e = [10458.8968564]

➤ E = [[33.80646332 50.11311033 57.27212609]]

scenario21

126.62525374422897

w[1]:1.8089981548649234

w[2]:0.0

w[3]:3185.103938891672

w[4]:0.0

y[1]:0.0

y[2]:3.686388619588712

{21: [499.52861959011864, 617.064765376029, 105.78253120731924, -0.0]}

➤ w = [-15234.29972371],theta =-15234.29972371404

➤ e = [11259.25754942]

➤ E = [[36.14401768 53.57819092 61.23221819]]

scenario22

126.62525374422897

w[1]:0.9542776920913808

w[2]:0.0

w[3]:3171.6140869151895

w[4]:0.0

y[1]:0.0

y[2]:4.68724626778814

{22: [535.6395435002883, 661.6723772650621, 113.42955038829635, -0.0]}

➤ w = [-8714.81349184],theta =-8714.813491840374

➤ e = [6488.67441355]

➤ E = [[20.74146292 30.74616856 35.13847835]]

scenario23

126.62525374422897

w[1]:5.829349961244191

w[2]:1.0213492071269172

w[3]:3248.5562055958667

w[4]:0.0

y[1]:0.0

y[2]:0.0

{23: [586.4288850149857, 517.4372514838108, 124.18494035611461, -0.0]}

➤ w = [-3258.94993568],theta =-3260.641084885605

➤ e = [1931.76573887]

➤ E = [[7.62592122 8.07450482 12.91920772]]

✓ Added optimality cut to master problem.

optimality_cut : Size=252, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] +
0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] +
0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : $0.16560308326763487 \cdot x[1] + 0.1753444411069076 \cdot x[2] + 0.20039364697932285 \cdot x[3] + \theta[3]$: +Inf : True

4 : 316.22021002633863 : $0.300081685736066 \cdot x[1] + 0.31773354960289335 \cdot x[2] + 0.363124056689021 \cdot x[3] + \theta[4]$: +Inf : True

5 : 271.85361107332903 : $0.2688729304090609 \cdot x[1] + 0.28468898513900553 \cdot x[2] + 0.3253588401588636 \cdot x[3] + \theta[5]$: +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \theta[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \theta[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \theta[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

21 : 13073.5454957668 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

22 : 7534.242802458173 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

23 : 2704.472034422052 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

24 : 529.6600300534212 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

25 : 1018.0591654370878 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

26 : 5043.727694629039 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

27 : 13726.64621211272 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

28 : 25243.480406612867 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

29 : 25716.237959437174 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

30 : 15080.019640201128 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

31 : 5472.791044230403 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

32 : 947.5309277454276 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

33 : 1044.2548848496808 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

34 : 5428.31884026755 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

35 : 14162.210709552062 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

36 : 25360.52662544886 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

106 : -27.637816375761858 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

107 : -97.09496546601656 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

108 : -159.17809322852167 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

109 : -137.59656268160575 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

110 : -77.6827385080845 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

111 : -28.484609452721294 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

112 : -4.765380212742096 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

113 : 92.00070262844588 : $0.2595586330208133 \cdot x[1] + 0.19630484850313604 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \theta[9]$: +Inf : True

114 : 367.7452085464239 : $1.0723181950379732 \cdot x[1] + 0.8109969542303997 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \theta[10]$: +Inf : True

115 : 1395.851100988342 : $4.044370036020941 \cdot x[1] + 3.0587672541334845 \cdot x[2] + 4.894027606613574 \cdot x[3] + \theta[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : 4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

160 : -60.73686951846584 : 0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

161 : -10.134189626634102 : 0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

162 : -34.49961906215793 : 0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

163 : -121.20130171964831 : 1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

164 : -198.69817154732715 : 2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

165 : -171.75846789910793 : 2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

166 : -96.96948737905726 : 1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

167 : -35.556650420293494 : 0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

168 : -5.948509093147033 : 0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

169 : 92.88109212728267 : 0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

170 : 371.2643014511742 : 0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

171 : 1409.208527792058 : 2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

172 : 2536.578200110367 : 5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : $3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

202 : -1849.6970314014807 : $15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

203 : 6142.583566779079 : $41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

204 : 10442.985140558467 : $71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

205 : -7940.7743720720755 : $64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

206 : 5596.141090199235 : $38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

207 : 1968.9183004702477 : $13.46994397254767 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

208 : 281.1534786413767 : $1.919064035862726 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

209 : -122.67792257876403 : $0.9685564636340608 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

210 : -576.317980396152 : $4.539559561688448 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

211 : -1482.9461276790792 : $11.686779426385783 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

212 : -2285.3624107661626 : $17.956238618288122 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

213 : -2075.6563451662273 : $16.348569305157238 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

214 : -1279.3661033296326 : $10.112756801944107 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474 \cdot x[1] + 4.735958328233625 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] +
1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] +
7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] +
19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] +
33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] +
30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] +
18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] +
6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] +
0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] +
1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] +
0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] +
10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] +
12.919207715127316*x[3] + theta[23] : +Inf : True

=====

scenario24

126.62525374422897

w[1]:1.7140292145567457

w[2]:0.0

w[3]:3183.6050664498407

w[4]:0.0

y[1]:0.0

y[2]:3.797595024944229

{24: [656.2842937979708, 810.7041276327875, 138.97785045133497, -0.0]}

► w = [-616.69545464],theta =-616.6954546368038

► e = [456.15618915]

► E = [[1.46364523 2.16963881 2.47958722]]

scenario25

126.62525374422897

w[1]:78.35396404325132

w[2]:85.94597409693432

w[3]:4393.19512700776

w[4]:0.0

y[1]:0.0

y[2]:0.0

{25: [608.9727590580751, 537.3289050512425, 128.95893721229825, -0.0]}

► w = [-1915.27153506],theta =-1915.2715350636981

► e = [727.18511817]

► E = [[3.88215567 4.11051776 6.57682842]]

scenario26

126.62525374422897

w[1]:79.11371556571669

w[2]:86.83562533977823

w[3]:4405.186106542411

w[4]:0.0

y[1]:0.0

y[2]:0.0

{26: [579.4254007885773, 511.2577065781562, 122.70184957875753, -0.0]}

➤ w = [-9524.48146652],theta =-9524.481466523646

➤ e = [3602.66263902]

➤ E = [[19.28569644 20.42014917 32.67223867]]

scenario27

126.62525374422897

w[1]:78.54390192386768

w[2]:86.1683869076453

w[3]:4396.192871891423

w[4]:0.0

y[1]:0.0

y[2]:0.0

{27: [565.594516471754, 499.0539851221356, 119.77295642931259, -0.0]}

➤ w = [-25848.20833458],theta =-25848.208334582996

➤ e = [9804.74729437]

➤ E = [[52.37941123 55.46055307 88.73688491]]

scenario28

126.62525374422897

w[1]:78.76549611792012

w[2]:86.42786852014143

w[3]:4399.690240922363

w[4]:0.0

y[1]:0.0

y[2]:0.0

{28: [565.7585388815307, 499.198710777821, 119.80769058667708, -0.0]}

► w = [-47587.35125305],theta =-47587.351253050234

► e = [18031.05743329]

► E = [[96.40304857 102.07381613 163.31810581]]

scenario29

126.62525374422897

w[1]:78.2906514163792

w[2]:85.87183649336401

w[3]:4392.195878713206

w[4]:0.0

y[1]:0.0

y[2]:0.0

{29: [579.3480589854651, 511.1894638107043, 122.68547131456907, -0.0]}

► w = [-48364.69610851],theta =-48364.6961085063

► e = [18368.7413996]

► E = [[98.04118914 103.80831791 166.09330866]]

scenario30

126.62525374422897

w[1]:79.0820592522806

w[2]:86.79855653799308

w[3]:4404.686482395134

w[4]:0.0

y[1]:0.0

y[2]:0.0

{30: [606.9424070349428, 535.5374179720081, 128.52898031328198, -0.0]}

➤ w = [-28472.37687373],theta =-28472.376873726822

➤ e = [10771.44260014]

➤ E = [[57.65491591 61.04638155 97.67421048]]

scenario31

126.62525374422897

w[1]:78.76549611792012

w[2]:86.42786852014143

w[3]:4399.690240922363

w[4]:0.0

y[1]:0.0

y[2]:0.0

{31: [650.247747776886, 573.7480127443109, 137.69952305863467, -0.0]}

➤ w = [-10316.94621983],theta =-10316.946219829308

➤ e = [3909.13646016]

► E = [[20.90019808 22.1296215 35.4073944]]

scenario32

126.62525374422897

w[1]:74.30195592343597

w[2]:81.20116746843348

w[3]:4329.243236156288

w[4]:0.0

y[1]:0.0

y[2]:0.0

{32: [714.641782239231, 630.56627844638, 151.33590682713128, -0.0]}

► w = [-1746.78548776],theta =-1746.7854877599834

► e = [676.80780553]

► E = [[3.56061335 3.77006119 6.03209791]]

scenario33

126.62525374422897

w[1]:151.13182863274693

w[2]:171.16714940102298

w[3]:5541.831041597871

w[4]:0.0

y[1]:0.0

y[2]:0.0

{33: [646.950912625531, 570.8390405519389, 137.0013697324654, -0.0]}

► $w = [-2673.22218611], \theta = -2782.656144404784$

► $e = [745.89634632]$

► $E = [[5.02318567 \ 5.31866718 \ 8.50986748]]$

✓ Added optimality cut to master problem.

optimality_cut : Size=253, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : $10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

215 : -340.20948277658624 : $3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

216 : -60.06574388842755 : $0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

217 : -12.037849556499072 : $0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

=====

scenario34

126.62525374422897

w[1]:156.0385572153358

w[2]:176.9128136777232

w[3]:5619.272784425825

w[4]:0.0

y[1]:0.0

y[2]:0.0

{34: [599.7752520768773, 529.2134577148915, 127.01122985157404, -0.0]}

➤ w = [-14144.49822858],theta =-14692.808704595005

➤ e = [3877.37060019]

➤ E = [[26.47676362 28.0342203 44.85475249]]

✓ Added optimality cut to master problem.

optimality_cut : Size=254, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] +
10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] +
12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

254 : 3877.370600191106 : 26.47676362092641*x[1] + 28.0342203045103*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

=====

scenario35

126.62525374422897

w[1]:154.20249103604448

w[2]:174.76282317418378

w[3]:5590.294583883752

w[4]:0.0

y[1]:0.0

y[2]:0.0

{35: [571.327911870871, 504.1128634154743, 120.98708721971389, -0.0]}

➤ w = [-36659.81481135],theta =-38175.13803134173

➤ e = [10115.86479254]

➤ E = [[68.72032106 72.76269289 116.42030863]]

✓ Added optimality cut to master problem.

optimality_cut : Size=255, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + \text{theta}[34]$: +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] +
12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

254 : 3877.370600191106 : 26.47676362092641*x[1] + 28.0342203045103*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

255 : 10115.864792537184 : 68.72032106367413*x[1] + 72.76269289094907*x[2] +
116.42030862551856*x[3] + theta[35] : +Inf : True

=====

scenario36

126.62525374422897

w[1]:154.5507104838411

w[2]:175.17057999382058

w[3]:5595.7904495038

w[4]:0.0

y[1]:0.0

y[2]:0.0

{36: [557.5760490321393, 491.9788667930638, 118.07492803033537, -0.0]}

➤ w = [-65729.73999742],theta =-68449.38432148288

➤ e = [18114.66187532]

➤ E = [[123.17970075 130.4255655 208.6809048]]

✓ Added optimality cut to master problem.

optimality_cut : Size=256, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,
51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78,
79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104,

105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + \text{theta}[34]$: +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] +
12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] +
8.509867481206035*x[3] + theta[33] : +Inf : True

254 : 3877.370600191106 : 26.47676362092641*x[1] + 28.0342203045103*x[2] +
44.85475248721651*x[3] + theta[34] : +Inf : True

255 : 10115.864792537184 : 68.72032106367413*x[1] + 72.76269289094907*x[2] +
116.42030862551856*x[3] + theta[35] : +Inf : True

256 : 18114.661875320613 : 123.1797007521802*x[1] + 130.42556550230836*x[2] +
208.68090480369352*x[3] + theta[36] : +Inf : True

=====

scenario37

126.62525374422897

w[1]:155.40543094661467

w[2]:176.17143764201995

w[3]:5609.280301480283

w[4]:0.0

y[1]:0.0

y[2]:0.0

{37: [557.3641701335071, 491.7919148236825, 118.03005955768384, -0.0]}

➤ w = [-63985.57295586],theta =-66536.8495822892

➤ e = [17579.92164856]

➤ E = [[119.83165235 126.88057308 203.00891693]]

✓ Added optimality cut to master problem.

optimality_cut : Size=257, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50,

51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : 6.3245645514497895*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \text{theta}[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \text{theta}[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \text{theta}[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \text{theta}[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \theta[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \theta[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \theta[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \theta[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \theta[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \theta[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \theta[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : 1.5090128058301502*x[1] + 1.5977782649966292*x[2] + 0.5072311952370255*x[3] + theta[17] : +Inf : True

66 : -1334.108039904671 : 8.876809869584338*x[1] + 9.398975156030474*x[2] + 2.983801636835073*x[3] + theta[18] : +Inf : True

67 : -3843.9245192018207 : 25.383256150708213*x[1] + 26.87638886545576*x[2] + 8.532186941414528*x[3] + theta[19] : +Inf : True

68 : -7187.393574302278 : 47.329048647069946*x[1] + 50.11311033219171*x[2] + 15.908923914981502*x[3] + theta[20] : +Inf : True

69 : -7737.4044770864675 : 50.601624753704534*x[1] + 53.57819091568716*x[2] + 17.008949497043552*x[3] + theta[21] : +Inf : True

70 : -4459.041658597693 : 29.038048081074148*x[1] + 30.746168556431453*x[2] + 9.760688430613165*x[3] + theta[22] : +Inf : True

71 : -1600.6058979232544 : 10.676289709028824*x[1] + 11.304306750736403*x[2] + 3.5886688097575896*x[3] + theta[23] : +Inf : True

72 : -313.4722626846774 : 2.0491033248395283*x[1] + 2.169638814535971*x[2] + 0.6887742268368167*x[3] + theta[24] : +Inf : True

73 : -602.5248121974598 : 5.4350179324536185*x[1] + 5.754724869656774*x[2] + 1.826896784018024*x[3] + theta[25] : +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : $7.30669765730133 \cdot x[1] + 5.5260738584631905 \cdot x[2] + 8.8417181735411 \cdot x[3] + \theta[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895 \cdot x[1] + 4.783284114541855 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111 \cdot x[1] + 3.5438810019535283 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558 \cdot x[1] + 1.282226979996724 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879 \cdot x[1] + 0.38315504524196553 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502 \cdot x[1] + 1.1412701892833068 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338 \cdot x[1] + 6.71355368287891 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213 \cdot x[1] + 19.197420618182683 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946 \cdot x[1] + 35.79507880870836 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534 \cdot x[1] + 38.27013636834797 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148 \cdot x[1] + 21.961548968879608 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283 \cdot x[1] + 1.5497420103828359 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : $4.2623036690128515 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \text{theta}[40]$: +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : $0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + \text{theta}[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + \text{theta}[34]$: +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : $7.625921220734874 \cdot x[1] + 8.074504821954571 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

253 : 745.8963463212003 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

254 : 3877.370600191106 : $26.47676362092641 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

255 : 10115.864792537184 : $68.72032106367413 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

256 : 18114.661875320613 : $123.1797007521802 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

257 : 17579.921648563904 : $119.83165235440212 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

=====

scenario38

126.62525374422897

w[1]:154.42408523009686

w[2]:175.0223047866799

w[3]:5593.791952914691

w[4]:0.0

y[1]:0.0

y[2]:0.0

{38: [570.7297629333085, 503.5850849411544, 120.86042038587709, -0.0]}

➤ w = [-37503.00799333], theta = -39054.17381269341

➤ e = [10340.28041079]

➤ E = [[70.28879468 74.42342967 119.07748746]]

✓ Added optimality cut to master problem.

optimality_cut : Size=258, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : 1.0723181950379732*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

11 : 1636.2847834552335 : 4.044370036020941*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

12 : 2945.3159195594317 : 7.30669765730133*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \theta[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \theta[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \theta[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + \text{theta}[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + \text{theta}[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + \text{theta}[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + \text{theta}[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + \text{theta}[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + \text{theta}[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + \text{theta}[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + \text{theta}[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325*x[1] + 2.7316499947637554*x[2] + 3.1218857083014346*x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 1.7705235463552964*x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.6430606851813393*x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \theta[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \theta[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \theta[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \theta[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \theta[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \theta[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \theta[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \theta[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \theta[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \theta[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \theta[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \theta[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \theta[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \theta[29]$: +Inf : True

78 : -8924.90958297616 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 27.131725132247585 \cdot x[3] + \theta[30]$: +Inf : True

79 : -3238.998781279216 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 9.835387333774072 \cdot x[3] + \theta[31]$: +Inf : True

80 : -560.783610298314 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 1.6755827528396188 \cdot x[3] + \theta[32]$: +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : $4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

116 : 2512.534804848658 : $7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

117 : 2236.10441187552 : $6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

118 : 1660.04623613669 : $4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

119 : 582.9889352679946 : $1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

120 : 172.26934377879059 : $0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

121 : 328.25923457048543 : $1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

122 : 1922.9557264832868 : $8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

123 : 5540.553272504695 : $25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

124 : 10359.760393304665 : $47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

125 : 11152.534729041883 : $50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

126 : 6427.170390668401 : $29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

127 : 2307.0802252824847 : $10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

128 : 451.83243380067347 : $2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

129 : 868.4667982708218 : $5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + \text{theta}[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + \text{theta}[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + \text{theta}[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + \text{theta}[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + \text{theta}[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + \text{theta}[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + \text{theta}[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + \text{theta}[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + \text{theta}[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + \text{theta}[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + \text{theta}[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + \text{theta}[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + \text{theta}[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + \text{theta}[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + \text{theta}[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + \text{theta}[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + \text{theta}[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + \text{theta}[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : $119.83165235440212 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

198 : 12467.423809583142 : $70.28879468381454 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

199 : 4898.481983344741 : $27.633909721859872 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : $0.10526049624697124 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

218 : -40.98021048819866 : $0.3779197393792061 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

219 : -143.9683970703005 : $1.3133768130407482 \cdot x[1] + 1.3906342726313798 \cdot x[2] + 0.6180596767250586 \cdot x[3] + \theta[59]$: +Inf : True

220 : -236.02268995953227 : $2.1406777700600026 \cdot x[1] + 2.2665999918282376 \cdot x[2] + 1.0073777741458847 \cdot x[3] + \theta[60]$: +Inf : True

221 : -204.0224894934155 : $1.8427797583723748 \cdot x[1] + 1.9511785676883961 \cdot x[2] + 0.8671904745281772 \cdot x[3] + \theta[61]$: +Inf : True

222 : -115.1847502016426 : $1.0451007044458347 \cdot x[1] + 1.1065772164720598 \cdot x[2] + 0.4918120962098051 \cdot x[3] + \theta[62]$: +Inf : True

223 : -42.23580022300055 : $0.37958443222509614 \cdot x[1] + 0.4019129282383369 \cdot x[2] + 0.17862796810592776 \cdot x[3] + \theta[63]$: +Inf : True

224 : -7.065908591307251 : $0.0620401152619143 \cdot x[1] + 0.06568953380673276 \cdot x[2] + 0.029195348358547937 \cdot x[3] + \theta[64]$: +Inf : True

225 : 727.1851181693482 : $3.8821556660382988 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

226 : 3602.6626390207416 : $19.285696439292145 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

254 : 3877.370600191106 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

255 : 10115.864792537184 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

256 : 18114.661875320613 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

257 : 17579.921648563904 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

258 : 10340.280410791704 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

=====

scenario39

126.62525374422897

w[1]:154.64567942414925

w[2]:175.28178639917604

w[3]:5597.289321945631

w[4]:0.0

y[1]:0.0

y[2]:0.0

{39: [597.9764018157447, 527.6262368962451, 126.63029685509886, -0.0]}

► w = [-14746.77838779],theta =-15354.473245901818

► e = [4062.7220241]

► E = [[27.63390972 29.25943382 46.81509412]]

✓ Added optimality cut to master problem.

optimality_cut : Size=259, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : 0.17427172426184484*x[1] + 0.18452300215960046*x[2] + 0.2108834310395433*x[3] + theta[6] : +Inf : True

7 : 94.92932352217971 : 0.09025744467169525*x[1] + 0.09556670612297144*x[2] + 0.10921909271196732*x[3] + theta[7] : +Inf : True

8 : 18.392872863500727 : 0.019877440458911873*x[1] + 0.021046701662377272*x[2] + 0.024053373328431164*x[3] + theta[8] : +Inf : True

9 : 107.84771360750833 : 0.2595586330208133*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

45 : 10868.657660098655 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 108.6138231374659*x[3] + theta[45] : +Inf : True

46 : 6497.888943596268 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

47 : 2286.184756470193 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

48 : 326.4578306499398 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

49 : 139.79577224091724 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 1.64084859721535*x[3] + theta[49] : +Inf : True

50 : 656.7344427770108 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 7.690547963331018*x[3] + theta[50] : +Inf : True

51 : 1689.8688431691844 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 19.798779263524146*x[3] + theta[51] : +Inf : True

52 : 2604.2501890126064 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 30.419980718041042*x[3] + theta[52] : +Inf : True

53 : 2365.28281193361 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 27.696399764031085*x[3] + theta[53] : +Inf : True

54 : 1457.8823037942338 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 17.132199758587664*x[3] + theta[54] : +Inf : True

55 : 465.64985072773015 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 5.412523803695572*x[3] + theta[55] : +Inf : True

56 : 82.21288967969141 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.9555486720485278*x[3] + theta[56] : +Inf : True

57 : 13.717549494615234 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.17832366423016302*x[3] + theta[57] : +Inf : True

58 : 46.698379393528754 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.6402404996541844*x[3] + theta[58] : +Inf : True

59 : 164.05701061499371 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 2.2250148362102085*x[3] + theta[59] : +Inf : True

60 : 268.956088558537 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 3.6265599869251806*x[3] + theta[60] : +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 9.075621853784545 \cdot x[3] + \text{theta}[26]$: +Inf : True

75 : -8123.933472474864 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 24.64913469803671 \cdot x[3] + \text{theta}[27]$: +Inf : True

76 : -14940.019016158618 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 45.36614050217 \cdot x[3] + \text{theta}[28]$: +Inf : True

77 : -15219.814302524028 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 46.1370301839617 \cdot x[3] + \text{theta}[29]$: +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : $5.4350179324536185 \cdot x[1] + 4.110517764040551 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

130 : 4302.608523173343 : $26.999975015009003 \cdot x[1] + 20.420149171015204 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

131 : 11709.669625843093 : $73.33117572665918 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

132 : 21534.234306049337 : $134.9642679939557 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

133 : 21937.52544294844 : $137.257664797286 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

134 : 12864.180019600142 : $80.71688226843654 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

135 : 4668.625829567975 : $29.260277317977856 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

136 : 808.3018934644665 : $4.984858689697864 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

137 : 890.8133507493193 : $7.032459932385544 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 8.509867481206035 \cdot x[3] + \theta[33]$: +Inf : True

138 : 4630.688316799664 : $37.067469069296976 \cdot x[1] + 28.0342203045103 \cdot x[2] + 44.85475248721651 \cdot x[3] + \theta[34]$: +Inf : True

139 : 12081.232809372981 : $96.2084494891438 \cdot x[1] + 72.76269289094907 \cdot x[2] + 116.42030862551856 \cdot x[3] + \theta[35]$: +Inf : True

140 : 21634.081896811473 : $172.45158105305225 \cdot x[1] + 130.42556550230836 \cdot x[2] + 208.68090480369352 \cdot x[3] + \theta[36]$: +Inf : True

141 : 20995.449283142036 : $167.76431329616295 \cdot x[1] + 126.8805730811316 \cdot x[2] + 203.00891692981068 \cdot x[3] + \theta[37]$: +Inf : True

142 : 12349.249176316951 : $98.40431255734036 \cdot x[1] + 74.42342966521537 \cdot x[2] + 119.07748746434463 \cdot x[3] + \theta[38]$: +Inf : True

143 : 4852.0508744978715 : $38.687473610603824 \cdot x[1] + 29.25943382314574 \cdot x[2] + 46.81509411703319 \cdot x[3] + \theta[39]$: +Inf : True

144 : 758.1338313313604 : $5.9672251366179925 \cdot x[1] + 4.5130274142489 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

145 : -457.13712537217486 : $5.133747785684267 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

146 : -1870.3640373389271 : $21.021496756788775 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

147 : 6084.3600258617425 : $58.72855735536902 \cdot x[1] + 44.41655598305218 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

148 : 10343.999499415735 : $100.0871657305758 \cdot x[1] + 75.69617576262033 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

149 : -8029.498108072879 : $89.75725662054474 \cdot x[1] + 67.88363946091617 \cdot x[2] + 30.170506427073875 \cdot x[3] + \theta[45]$: +Inf : True

150 : 5543.097098822938 : $53.5948200742443 \cdot x[1] + 40.53389753514273 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

151 : 1950.2555677643686 : $18.857921561566744 \cdot x[1] + 14.262293617991649 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

152 : 278.48851675852006 : $2.6866896502078177 \cdot x[1] + 2.0319501556193567 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

153 : -103.27769296165714 : $1.3559790490876855 \cdot x[1] + 1.0255303732595935 \cdot x[2] + 0.45579127700426414 \cdot x[3] + \theta[49]$: +Inf : True

154 : -485.17932303117914 : $6.35538338636383 \cdot x[1] + 4.806592477081885 \cdot x[2] + 2.136263323147507 \cdot x[3] + \theta[50]$: +Inf : True

155 : -1248.433716790294 : $16.3614911969401 \cdot x[1] + 12.374237039702589 \cdot x[2] + 5.499660906534489 \cdot x[3] + \theta[51]$: +Inf : True

156 : -1923.9562620868624 : $25.138734065603373 \cdot x[1] + 19.012487948775647 \cdot x[2] + 8.449994643900297 \cdot x[3] + \theta[52]$: +Inf : True

157 : -1747.4130161631956 : $22.88799702722014 \cdot x[1] + 17.310249852519426 \cdot x[2] + 7.6934443788975315 \cdot x[3] + \theta[53]$: +Inf : True

158 : -1077.0477428030856 : $14.157859522721756 \cdot x[1] + 10.707624849117288 \cdot x[2] + 4.7589443773854665 \cdot x[3] + \theta[54]$: +Inf : True

159 : -344.01070604783297 : $4.472849532220648 \cdot x[1] + 3.3828273773097317 \cdot x[2] + 1.5034788343598828 \cdot x[3] + \theta[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362 \cdot x[1] + 0.5972179200303298 \cdot x[2] + 0.2654301866801469 \cdot x[3] + \theta[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975 \cdot x[1] + 0.11145229014385186 \cdot x[2] + 0.04953435117504534 \cdot x[3] + \theta[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886 \cdot x[1] + 0.4001503122838651 \cdot x[2] + 0.17784458323727365 \cdot x[3] + \theta[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341*x[1] + 0.5364170633387517*x[2] + 0.6130480723871449*x[3] + \text{theta}[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786*x[1] + 1.5977782649966292*x[2] + 1.8260323028532908*x[3] + \text{theta}[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265*x[1] + 9.398975156030474*x[2] + 10.741685892606256*x[3] + \text{theta}[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864*x[1] + 26.87638886545576*x[2] + 30.71587298909229*x[3] + \text{theta}[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567*x[1] + 50.11311033219171*x[2] + 57.27212609393337*x[3] + \text{theta}[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753*x[1] + 53.57819091568716*x[2] + 61.23221818935675*x[3] + \text{theta}[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963*x[1] + 30.746168556431453*x[2] + 35.13847835020737*x[3] + \text{theta}[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874*x[1] + 11.304306750736403*x[2] + 12.919207715127316*x[3] + \text{theta}[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345*x[1] + 2.169638814535971*x[2] + 2.479587216612538*x[3] + \text{theta}[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988*x[1] + 5.754724869656774*x[2] + 6.5768284224648825*x[3] + \text{theta}[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145*x[1] + 28.588208839421302*x[2] + 32.67223867362434*x[3] + \text{theta}[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986*x[1] + 77.6447742988156*x[2] + 88.7368849129321*x[3] + \text{theta}[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112*x[1] + 142.90334258183543*x[2] + 163.3181058078119*x[3] + \text{theta}[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855*x[1] + 145.33164507947927*x[2] + 166.09330866226202*x[3] + \text{theta}[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261*x[1] + 85.46493416657988*x[2] + 97.67421047609123*x[3] + \text{theta}[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892*x[1] + 30.981470101388307*x[2] + 35.407394401586636*x[3] + \text{theta}[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883*x[1] + 5.278085671444797*x[2] + 6.0320979102226255*x[3] + \text{theta}[32]$: +Inf : True

193 : 899.3378804215617 : $5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + \text{theta}[33]$: +Inf : True

194 : 4675.001123658993 : $26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + \text{theta}[34]$: +Inf : True

195 : 12196.842692716265 : $68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + \text{theta}[35]$: +Inf : True

196 : 21841.106603958004 : $123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + \text{theta}[36]$: +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : 52.379411233327986*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

228 : 18031.057433294896 : 96.4030485671112*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

229 : 18368.741399597977 : 98.04118914091855*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

230 : 10771.442600143659 : 57.6549159060261*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

231 : 3909.136460164572 : 20.900198084269892*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

232 : 676.807805532448 : 3.5606133497841883*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

233 : -916.3869397660462 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

234 : -4763.626737377647 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

235 : -12428.06245940283 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

236 : -22255.156018251044 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

237 : -21598.18945394994 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

238 : -12703.773076115522 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

239 : -4991.344201038478 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

240 : -779.8984389293898 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

241 : -543.0081875967824 : 3.66696270406019*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

242 : -2221.7031382755217 : 15.015354826277694*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

243 : -6259.030648613752 : 41.94896953954929*x[1] + 44.41655598305218*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

254 : 3877.370600191106 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

255 : 10115.864792537184 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

256 : 18114.661875320613 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

257 : 17579.921648563904 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

258 : 10340.280410791704 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

259 : 4062.722024101088 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

=====

scenario40

126.62525374422897

w[1]:150.087170289357

w[2]:169.94387894211263

w[3]:5525.343444737725

w[4]:0.0

y[1]:0.0

y[2]:0.0

{40: [642.3582103585092, 566.7866561986845, 136.0287974876843, -0.0]}

► w = [-2266.40993637],theta =-2358.878982337977

► e = [634.80105494]

► E = [[4.26230367 4.51302741 7.22084386]]

✓ Added optimality cut to master problem.

optimality_cut : Size=260, Index={1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260}, Active=True

Key : Lower : Body : Upper : Active

1 : 9.785496 : 0.009517792788000001*x[1] + 0.010077662951999999*x[2] + 0.011517329088000003*x[3] + theta[1] : +Inf : True

2 : 60.25830152831999 : 0.05304882616689596*x[1] + 0.056169345353184*x[2] + 0.06419353754649597*x[3] + theta[2] : +Inf : True

3 : 179.09259323679183 : 0.16560308326763487*x[1] + 0.1753444411069076*x[2] + 0.20039364697932285*x[3] + theta[3] : +Inf : True

4 : 316.22021002633863 : 0.300081685736066*x[1] + 0.31773354960289335*x[2] + 0.363124056689021*x[3] + theta[4] : +Inf : True

5 : 271.85361107332903 : 0.2688729304090609*x[1] + 0.28468898513900553*x[2] + 0.3253588401588636*x[3] + theta[5] : +Inf : True

6 : 193.68136379025395 : $0.17427172426184484 \cdot x[1] + 0.18452300215960046 \cdot x[2] + 0.2108834310395433 \cdot x[3] + \text{theta}[6]$: +Inf : True

7 : 94.92932352217971 : $0.09025744467169525 \cdot x[1] + 0.09556670612297144 \cdot x[2] + 0.10921909271196732 \cdot x[3] + \text{theta}[7]$: +Inf : True

8 : 18.392872863500727 : $0.019877440458911873 \cdot x[1] + 0.021046701662377272 \cdot x[2] + 0.024053373328431164 \cdot x[3] + \text{theta}[8]$: +Inf : True

9 : 107.84771360750833 : $0.2595586330208133 \cdot x[1] + 0.27482678790439047 \cdot x[2] + 0.3140877576050175 \cdot x[3] + \text{theta}[9]$: +Inf : True

10 : 431.0888808319322 : $1.0723181950379732 \cdot x[1] + 1.1353957359225595 \cdot x[2] + 1.2975951267686392 \cdot x[3] + \text{theta}[10]$: +Inf : True

11 : 1636.2847834552335 : $4.044370036020941 \cdot x[1] + 4.282274155786878 \cdot x[2] + 4.894027606613574 \cdot x[3] + \text{theta}[11]$: +Inf : True

12 : 2945.3159195594317 : $7.30669765730133 \cdot x[1] + 7.7365034018484655 \cdot x[2] + 8.8417181735411 \cdot x[3] + \text{theta}[12]$: +Inf : True

13 : 2621.2707220550355 : $6.3245645514497895 \cdot x[1] + 6.696597760358598 \cdot x[2] + 7.653254583266969 \cdot x[3] + \text{theta}[13]$: +Inf : True

14 : 1945.9872146099954 : $4.685798213694111 \cdot x[1] + 4.96143340273494 \cdot x[2] + 5.670209603125645 \cdot x[3] + \text{theta}[14]$: +Inf : True

15 : 683.408082012721 : $1.695389006884558 \cdot x[1] + 1.7951177719954141 \cdot x[2] + 2.0515631679947584 \cdot x[3] + \text{theta}[15]$: +Inf : True

16 : 201.94253218087894 : $0.5066161153754879 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \text{theta}[16]$: +Inf : True

17 : 384.80149507066477 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \text{theta}[17]$: +Inf : True

18 : 2254.1825501837575 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \text{theta}[18]$: +Inf : True

19 : 6494.906946237561 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \text{theta}[19]$: +Inf : True

20 : 12144.216728993506 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \text{theta}[20]$: +Inf : True

21 : 13073.5454957668 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \text{theta}[21]$: +Inf : True

22 : 7534.242802458173 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \text{theta}[22]$: +Inf : True

23 : 2704.472034422052 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \text{theta}[23]$: +Inf : True

24 : 529.6600300534212 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \text{theta}[24]$: +Inf : True

25 : 1018.0591654370878 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \text{theta}[25]$: +Inf : True

26 : 5043.727694629039 : $26.999975015009003 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \text{theta}[26]$: +Inf : True

27 : 13726.64621211272 : $73.33117572665918 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \text{theta}[27]$: +Inf : True

28 : 25243.480406612867 : $134.9642679939557 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \text{theta}[28]$: +Inf : True

29 : 25716.237959437174 : $137.257664797286 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \text{theta}[29]$: +Inf : True

30 : 15080.019640201128 : $80.71688226843654 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \text{theta}[30]$: +Inf : True

31 : 5472.791044230403 : $29.260277317977856 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \text{theta}[31]$: +Inf : True

32 : 947.5309277454276 : $4.984858689697864 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \text{theta}[32]$: +Inf : True

33 : 1044.2548848496808 : $7.032459932385544 \cdot x[1] + 7.446134046055281 \cdot x[2] + 8.509867481206035 \cdot x[3] + \text{theta}[33]$: +Inf : True

34 : 5428.31884026755 : $37.067469069296976 \cdot x[1] + 39.247908426314446 \cdot x[2] + 44.85475248721651 \cdot x[3] + \text{theta}[34]$: +Inf : True

35 : 14162.210709552062 : $96.2084494891438 \cdot x[1] + 101.86777004732873 \cdot x[2] + 116.42030862551856 \cdot x[3] + \text{theta}[35]$: +Inf : True

36 : 25360.52662544886 : $172.45158105305225 \cdot x[1] + 182.5957917032318 \cdot x[2] + 208.68090480369352 \cdot x[3] + \text{theta}[36]$: +Inf : True

37 : 24611.890307989474 : $167.76431329616295 \cdot x[1] + 177.6328023135843 \cdot x[2] + 203.00891692981068 \cdot x[3] + \text{theta}[37]$: +Inf : True

38 : 14476.392575108388 : $98.40431255734036 \cdot x[1] + 104.19280153130154 \cdot x[2] + 119.07748746434463 \cdot x[3] + \text{theta}[38]$: +Inf : True

39 : 5687.810833741524 : $38.687473610603824 \cdot x[1] + 40.96320735240404 \cdot x[2] + 46.81509411703319 \cdot x[3] + \text{theta}[39]$: +Inf : True

40 : 888.7214769195373 : $5.9672251366179925 \cdot x[1] + 6.318238379948461 \cdot x[2] + 7.220843862798243 \cdot x[3] + \theta[40]$: +Inf : True

41 : 618.7767719126125 : $5.133747785684267 \cdot x[1] + 5.435732949548046 \cdot x[2] + 6.212266228054911 \cdot x[3] + \theta[41]$: +Inf : True

42 : 2531.708227337223 : $21.021496756788775 \cdot x[1] + 22.25805538954105 \cdot x[2] + 25.43777758804692 \cdot x[3] + \theta[42]$: +Inf : True

43 : 7132.383762373813 : $58.72855735536902 \cdot x[1] + 62.183178376273055 \cdot x[2] + 71.0664895728835 \cdot x[3] + \theta[43]$: +Inf : True

44 : 12125.741039984952 : $100.0871657305758 \cdot x[1] + 105.97464606766847 \cdot x[2] + 121.11388122019257 \cdot x[3] + \theta[44]$: +Inf : True

45 : 10868.657660098655 : $89.75725662054474 \cdot x[1] + 95.03709524528266 \cdot x[2] + 108.6138231374659 \cdot x[3] + \theta[45]$: +Inf : True

46 : 6497.888943596268 : $53.5948200742443 \cdot x[1] + 56.74745654919984 \cdot x[2] + 64.8542360562284 \cdot x[3] + \theta[46]$: +Inf : True

47 : 2286.184756470193 : $18.857921561566744 \cdot x[1] + 19.967211065188312 \cdot x[2] + 22.819669788786644 \cdot x[3] + \theta[47]$: +Inf : True

48 : 326.4578306499398 : $2.6866896502078177 \cdot x[1] + 2.8447302178671 \cdot x[2] + 3.2511202489909716 \cdot x[3] + \theta[48]$: +Inf : True

49 : 139.79577224091724 : $1.3559790490876855 \cdot x[1] + 1.4357425225634315 \cdot x[2] + 1.64084859721535 \cdot x[3] + \theta[49]$: +Inf : True

50 : 656.7344427770108 : $6.35538338636383 \cdot x[1] + 6.729229467914641 \cdot x[2] + 7.690547963331018 \cdot x[3] + \theta[50]$: +Inf : True

51 : 1689.8688431691844 : $16.3614911969401 \cdot x[1] + 17.32393185558363 \cdot x[2] + 19.798779263524146 \cdot x[3] + \theta[51]$: +Inf : True

52 : 2604.2501890126064 : $25.138734065603373 \cdot x[1] + 26.617483128285915 \cdot x[2] + 30.419980718041042 \cdot x[3] + \theta[52]$: +Inf : True

53 : 2365.28281193361 : $22.88799702722014 \cdot x[1] + 24.234349793527205 \cdot x[2] + 27.696399764031085 \cdot x[3] + \theta[53]$: +Inf : True

54 : 1457.8823037942338 : $14.157859522721756 \cdot x[1] + 14.99067478876421 \cdot x[2] + 17.132199758587664 \cdot x[3] + \theta[54]$: +Inf : True

55 : 465.64985072773015 : $4.472849532220648 \cdot x[1] + 4.735958328233625 \cdot x[2] + 5.412523803695572 \cdot x[3] + \theta[55]$: +Inf : True

56 : 82.21288967969141 : $0.7896548053734362 \cdot x[1] + 0.8361050880424619 \cdot x[2] + 0.9555486720485278 \cdot x[3] + \theta[56]$: +Inf : True

57 : 13.717549494615234 : $0.14736469474575975 \cdot x[1] + 0.15603320620139266 \cdot x[2] + 0.17832366423016302 \cdot x[3] + \text{theta}[57]$: +Inf : True

58 : 46.698379393528754 : $0.5290876351308886 \cdot x[1] + 0.5602104371974115 \cdot x[2] + 0.6402404996541844 \cdot x[3] + \text{theta}[58]$: +Inf : True

59 : 164.05701061499371 : $1.8387275382570476 \cdot x[1] + 1.9468879816839328 \cdot x[2] + 2.2250148362102085 \cdot x[3] + \text{theta}[59]$: +Inf : True

60 : 268.956088558537 : $2.996948878084004 \cdot x[1] + 3.173239988559534 \cdot x[2] + 3.6265599869251806 \cdot x[3] + \text{theta}[60]$: +Inf : True

61 : 232.4907438413341 : $2.579891661721325 \cdot x[1] + 2.7316499947637554 \cdot x[2] + 3.1218857083014346 \cdot x[3] + \text{theta}[61]$: +Inf : True

62 : 131.2570409274533 : $1.4631409862241684 \cdot x[1] + 1.5492081030608844 \cdot x[2] + 1.7705235463552964 \cdot x[3] + \text{theta}[62]$: +Inf : True

63 : 48.12916769597741 : $0.5314182051151346 \cdot x[1] + 0.5626780995336719 \cdot x[2] + 0.6430606851813393 \cdot x[3] + \text{theta}[63]$: +Inf : True

64 : 8.051849324978035 : $0.08685616136668002 \cdot x[1] + 0.09196534732942589 \cdot x[2] + 0.10510325409077247 \cdot x[3] + \text{theta}[64]$: +Inf : True

65 : -227.73966034794427 : $1.5090128058301502 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 0.5072311952370255 \cdot x[3] + \text{theta}[17]$: +Inf : True

66 : -1334.108039904671 : $8.876809869584338 \cdot x[1] + 9.398975156030474 \cdot x[2] + 2.983801636835073 \cdot x[3] + \text{theta}[18]$: +Inf : True

67 : -3843.9245192018207 : $25.383256150708213 \cdot x[1] + 26.87638886545576 \cdot x[2] + 8.532186941414528 \cdot x[3] + \text{theta}[19]$: +Inf : True

68 : -7187.393574302278 : $47.329048647069946 \cdot x[1] + 50.11311033219171 \cdot x[2] + 15.908923914981502 \cdot x[3] + \text{theta}[20]$: +Inf : True

69 : -7737.4044770864675 : $50.601624753704534 \cdot x[1] + 53.57819091568716 \cdot x[2] + 17.008949497043552 \cdot x[3] + \text{theta}[21]$: +Inf : True

70 : -4459.041658597693 : $29.038048081074148 \cdot x[1] + 30.746168556431453 \cdot x[2] + 9.760688430613165 \cdot x[3] + \text{theta}[22]$: +Inf : True

71 : -1600.6058979232544 : $10.676289709028824 \cdot x[1] + 11.304306750736403 \cdot x[2] + 3.5886688097575896 \cdot x[3] + \text{theta}[23]$: +Inf : True

72 : -313.4722626846774 : $2.0491033248395283 \cdot x[1] + 2.169638814535971 \cdot x[2] + 0.6887742268368167 \cdot x[3] + \text{theta}[24]$: +Inf : True

73 : -602.5248121974598 : $5.4350179324536185 \cdot x[1] + 5.754724869656774 \cdot x[2] + 1.826896784018024 \cdot x[3] + \text{theta}[25]$: +Inf : True

74 : -2985.0633294743257 : 26.999975015009003*x[1] + 28.588208839421302*x[2] + 9.075621853784545*x[3] + theta[26] : +Inf : True

75 : -8123.933472474864 : 73.33117572665918*x[1] + 77.6447742988156*x[2] + 24.64913469803671*x[3] + theta[27] : +Inf : True

76 : -14940.019016158618 : 134.9642679939557*x[1] + 142.90334258183543*x[2] + 45.36614050217*x[3] + theta[28] : +Inf : True

77 : -15219.814302524028 : 137.257664797286*x[1] + 145.33164507947927*x[2] + 46.1370301839617*x[3] + theta[29] : +Inf : True

78 : -8924.90958297616 : 80.71688226843654*x[1] + 85.46493416657988*x[2] + 27.131725132247585*x[3] + theta[30] : +Inf : True

79 : -3238.998781279216 : 29.260277317977856*x[1] + 30.981470101388307*x[2] + 9.835387333774072*x[3] + theta[31] : +Inf : True

80 : -560.783610298314 : 4.984858689697864*x[1] + 5.278085671444797*x[2] + 1.6755827528396188*x[3] + theta[32] : +Inf : True

81 : -618.0284012375657 : 7.032459932385544*x[1] + 7.446134046055281*x[2] + 2.3638520781127887*x[3] + theta[33] : +Inf : True

82 : -3212.6784973012036 : 37.067469069296976*x[1] + 39.247908426314446*x[2] + 12.459653468671258*x[3] + theta[34] : +Inf : True

83 : -8381.716542387952 : 96.2084494891438*x[1] + 101.86777004732873*x[2] + 32.33897461819961*x[3] + theta[35] : +Inf : True

84 : -15009.291268122797 : 172.45158105305225*x[1] + 182.5957917032318*x[2] + 57.966918001025995*x[3] + theta[36] : +Inf : True

85 : -14566.220794524372 : 167.76431329616295*x[1] + 177.6328023135843*x[2] + 56.39136581383633*x[3] + theta[37] : +Inf : True

86 : -8567.660911798837 : 98.40431255734036*x[1] + 104.19280153130154*x[2] + 33.07707985120687*x[3] + theta[38] : +Inf : True

87 : -3366.255391398042 : 38.687473610603824*x[1] + 40.96320735240404*x[2] + 13.004192810287009*x[3] + theta[39] : +Inf : True

88 : -525.9780169523792 : 5.9672251366179925*x[1] + 6.318238379948461*x[2] + 2.005789961888402*x[3] + theta[40] : +Inf : True

89 : -366.21482419317874 : 5.133747785684267*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

90 : -1498.357930464886 : 21.021496756788775*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

91 : -4221.2067165069475 : 58.72855735536902*x[1] + 62.183178376273055*x[2] + 19.740691548023207*x[3] + theta[43] : +Inf : True

92 : -7176.458982848237 : 100.0871657305758*x[1] + 105.97464606766847*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

93 : -6432.470860058384 : 89.75725662054474*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

94 : -3845.689374781463 : 53.5948200742443*x[1] + 56.74745654919984*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

95 : -1353.0481211762367 : 18.857921561566744*x[1] + 19.967211065188312*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

96 : -193.20973650710715 : 2.6866896502078177*x[1] + 2.8447302178671*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

97 : -82.73627336707335 : 1.3559790490876855*x[1] + 1.4357425225634315*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

98 : -388.6795681741489 : 6.35538338636383*x[1] + 6.729229467914641*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

99 : -1000.126458202169 : 16.3614911969401*x[1] + 17.32393185558363*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

100 : -1541.290928191132 : 25.138734065603373*x[1] + 26.617483128285915*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

101 : -1399.8612560423385 : 22.88799702722014*x[1] + 24.234349793527205*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

102 : -862.8283022455653 : 14.157859522721756*x[1] + 14.99067478876421*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

103 : -275.588687165391 : 4.472849532220648*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

104 : -48.65660817777649 : 0.7896548053734362*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

105 : -8.118549700894718 : 0.14736469474575975*x[1] + 0.15603320620139266*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

106 : -27.637816375761858 : 0.5290876351308886*x[1] + 0.5602104371974115*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

107 : -97.09496546601656 : 1.8387275382570476*x[1] + 1.9468879816839328*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

108 : -159.17809322852167 : 2.996948878084004*x[1] + 3.173239988559534*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

109 : -137.59656268160575 : 2.579891661721325*x[1] + 2.7316499947637554*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

110 : -77.6827385080845 : 1.4631409862241684*x[1] + 1.5492081030608844*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

111 : -28.484609452721294 : 0.5314182051151346*x[1] + 0.5626780995336719*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

112 : -4.765380212742096 : 0.08685616136668002*x[1] + 0.09196534732942589*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

113 : 92.00070262844588 : 0.2595586330208133*x[1] + 0.19630484850313604*x[2] + 0.3140877576050175*x[3] + theta[9] : +Inf : True

114 : 367.7452085464239 : 1.0723181950379732*x[1] + 0.8109969542303997*x[2] + 1.2975951267686392*x[3] + theta[10] : +Inf : True

115 : 1395.851100988342 : 4.044370036020941*x[1] + 3.0587672541334845*x[2] + 4.894027606613574*x[3] + theta[11] : +Inf : True

116 : 2512.534804848658 : 7.30669765730133*x[1] + 5.5260738584631905*x[2] + 8.8417181735411*x[3] + theta[12] : +Inf : True

117 : 2236.10441187552 : 6.3245645514497895*x[1] + 4.783284114541855*x[2] + 7.653254583266969*x[3] + theta[13] : +Inf : True

118 : 1660.04623613669 : 4.685798213694111*x[1] + 3.5438810019535283*x[2] + 5.670209603125645*x[3] + theta[14] : +Inf : True

119 : 582.9889352679946 : 1.695389006884558*x[1] + 1.282226979996724*x[2] + 2.0515631679947584*x[3] + theta[15] : +Inf : True

120 : 172.26934377879059 : 0.5066161153754879*x[1] + 0.38315504524196553*x[2] + 0.6130480723871449*x[3] + theta[16] : +Inf : True

121 : 328.25923457048543 : 1.5090128058301502*x[1] + 1.1412701892833068*x[2] + 1.8260323028532908*x[3] + theta[17] : +Inf : True

122 : 1922.9557264832868 : 8.876809869584338*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

123 : 5540.553272504695 : 25.383256150708213*x[1] + 19.197420618182683*x[2] + 30.71587298909229*x[3] + theta[19] : +Inf : True

124 : 10359.760393304665 : 47.329048647069946*x[1] + 35.79507880870836*x[2] + 57.27212609393337*x[3] + theta[20] : +Inf : True

125 : 11152.534729041883 : 50.601624753704534*x[1] + 38.27013636834797*x[2] + 61.23221818935675*x[3] + theta[21] : +Inf : True

126 : 6427.170390668401 : 29.038048081074148*x[1] + 21.961548968879608*x[2] + 35.13847835020737*x[3] + theta[22] : +Inf : True

127 : 2307.0802252824847 : 10.676289709028824*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

128 : 451.83243380067347 : 2.0491033248395283*x[1] + 1.5497420103828359*x[2] + 2.479587216612538*x[3] + theta[24] : +Inf : True

129 : 868.4667982708218 : 5.4350179324536185*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

130 : 4302.608523173343 : 26.999975015009003*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

131 : 11709.669625843093 : 73.33117572665918*x[1] + 55.46055307058254*x[2] + 88.7368849129321*x[3] + theta[27] : +Inf : True

132 : 21534.234306049337 : 134.9642679939557*x[1] + 102.07381612988239*x[2] + 163.3181058078119*x[3] + theta[28] : +Inf : True

133 : 21937.52544294844 : 137.257664797286*x[1] + 103.80831791391374*x[2] + 166.09330866226202*x[3] + theta[29] : +Inf : True

134 : 12864.180019600142 : 80.71688226843654*x[1] + 61.046381547557026*x[2] + 97.67421047609123*x[3] + theta[30] : +Inf : True

135 : 4668.625829567975 : 29.260277317977856*x[1] + 22.12962150099164*x[2] + 35.407394401586636*x[3] + theta[31] : +Inf : True

136 : 808.3018934644665 : 4.984858689697864*x[1] + 3.770061193889139*x[2] + 6.0320979102226255*x[3] + theta[32] : +Inf : True

137 : 890.8133507493193 : 7.032459932385544*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

138 : 4630.688316799664 : 37.067469069296976*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

139 : 12081.232809372981 : 96.2084494891438*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

140 : 21634.081896811473 : 172.45158105305225*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

141 : 20995.449283142036 : 167.76431329616295*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

142 : 12349.249176316951 : 98.40431255734036*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

143 : 4852.0508744978715 : 38.687473610603824*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

144 : 758.1338313313604 : 5.9672251366179925*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

145 : -457.13712537217486 : 5.133747785684267*x[1] + 3.882666392534318*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

146 : -1870.3640373389271 : 21.021496756788775*x[1] + 15.89861099252932*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

147 : 6084.3600258617425 : 58.72855735536902*x[1] + 44.41655598305218*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

148 : 10343.999499415735 : 100.0871657305758*x[1] + 75.69617576262033*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

149 : -8029.498108072879 : 89.75725662054474*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

150 : 5543.097098822938 : 53.5948200742443*x[1] + 40.53389753514273*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

151 : 1950.2555677643686 : 18.857921561566744*x[1] + 14.262293617991649*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

152 : 278.48851675852006 : 2.6866896502078177*x[1] + 2.0319501556193567*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

153 : -103.27769296165714 : 1.3559790490876855*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

154 : -485.17932303117914 : 6.35538338636383*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

155 : -1248.433716790294 : 16.3614911969401*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

156 : -1923.9562620868624 : 25.138734065603373*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

157 : -1747.4130161631956 : 22.88799702722014*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

158 : -1077.0477428030856 : 14.157859522721756*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

159 : -344.01070604783297 : $4.472849532220648*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + \text{theta}[55]$: +Inf : True

160 : -60.73686951846584 : $0.7896548053734362*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + \text{theta}[56]$: +Inf : True

161 : -10.134189626634102 : $0.14736469474575975*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + \text{theta}[57]$: +Inf : True

162 : -34.49961906215793 : $0.5290876351308886*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + \text{theta}[58]$: +Inf : True

163 : -121.20130171964831 : $1.8387275382570476*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + \text{theta}[59]$: +Inf : True

164 : -198.69817154732715 : $2.996948878084004*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + \text{theta}[60]$: +Inf : True

165 : -171.75846789910793 : $2.579891661721325*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + \text{theta}[61]$: +Inf : True

166 : -96.96948737905726 : $1.4631409862241684*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + \text{theta}[62]$: +Inf : True

167 : -35.556650420293494 : $0.5314182051151346*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + \text{theta}[63]$: +Inf : True

168 : -5.948509093147033 : $0.08685616136668002*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + \text{theta}[64]$: +Inf : True

169 : 92.88109212728267 : $0.18539902358629512*x[1] + 0.27482678790439047*x[2] + 0.3140877576050175*x[3] + \text{theta}[9]$: +Inf : True

170 : 371.2643014511742 : $0.7659415678842664*x[1] + 1.1353957359225595*x[2] + 1.2975951267686392*x[3] + \text{theta}[10]$: +Inf : True

171 : 1409.208527792058 : $2.8888357400149576*x[1] + 4.282274155786878*x[2] + 4.894027606613574*x[3] + \text{theta}[11]$: +Inf : True

172 : 2536.578200110367 : $5.2190697552152345*x[1] + 7.7365034018484655*x[2] + 8.8417181735411*x[3] + \text{theta}[12]$: +Inf : True

173 : 2257.5025402188257 : $4.517546108178419*x[1] + 6.696597760358598*x[2] + 7.653254583266969*x[3] + \text{theta}[13]$: +Inf : True

174 : 1675.9318460518732 : $3.3469987240672205*x[1] + 4.96143340273494*x[2] + 5.670209603125645*x[3] + \text{theta}[14]$: +Inf : True

175 : 588.5677767538126 : $1.210992147774684*x[1] + 1.7951177719954141*x[2] + 2.0515631679947584*x[3] + \text{theta}[15]$: +Inf : True

176 : 173.91785424557327 : $0.3618686538396341 \cdot x[1] + 0.5364170633387517 \cdot x[2] + 0.6130480723871449 \cdot x[3] + \theta[16]$: +Inf : True

177 : 331.4004712649398 : $1.0778662898786786 \cdot x[1] + 1.5977782649966292 \cdot x[2] + 1.8260323028532908 \cdot x[3] + \theta[17]$: +Inf : True

178 : 1941.3572166888684 : $6.3405784782745265 \cdot x[1] + 9.398975156030474 \cdot x[2] + 10.741685892606256 \cdot x[3] + \theta[18]$: +Inf : True

179 : 5593.572921045409 : $18.130897250505864 \cdot x[1] + 26.87638886545576 \cdot x[2] + 30.71587298909229 \cdot x[3] + \theta[19]$: +Inf : True

180 : 10458.896856398489 : $33.80646331933567 \cdot x[1] + 50.11311033219171 \cdot x[2] + 57.27212609393337 \cdot x[3] + \theta[20]$: +Inf : True

181 : 11259.25754941549 : $36.14401768121753 \cdot x[1] + 53.57819091568716 \cdot x[2] + 61.23221818935675 \cdot x[3] + \theta[21]$: +Inf : True

182 : 6488.67441354561 : $20.741462915052963 \cdot x[1] + 30.746168556431453 \cdot x[2] + 35.13847835020737 \cdot x[3] + \theta[22]$: +Inf : True

183 : 2329.157548012461 : $7.625921220734874 \cdot x[1] + 11.304306750736403 \cdot x[2] + 12.919207715127316 \cdot x[3] + \theta[23]$: +Inf : True

184 : 456.15618914804844 : $1.4636452320282345 \cdot x[1] + 2.169638814535971 \cdot x[2] + 2.479587216612538 \cdot x[3] + \theta[24]$: +Inf : True

185 : 876.7774853356144 : $3.8821556660382988 \cdot x[1] + 5.754724869656774 \cdot x[2] + 6.5768284224648825 \cdot x[3] + \theta[25]$: +Inf : True

186 : 4343.781810476438 : $19.285696439292145 \cdot x[1] + 28.588208839421302 \cdot x[2] + 32.67223867362434 \cdot x[3] + \theta[26]$: +Inf : True

187 : 11821.723880635855 : $52.379411233327986 \cdot x[1] + 77.6447742988156 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

188 : 21740.303533858427 : $96.4030485671112 \cdot x[1] + 142.90334258183543 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

189 : 22147.453916086706 : $98.04118914091855 \cdot x[1] + 145.33164507947927 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

190 : 12987.282220744644 : $57.6549159060261 \cdot x[1] + 85.46493416657988 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

191 : 4713.301674826999 : $20.900198084269892 \cdot x[1] + 30.981470101388307 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

192 : 816.036839813409 : $3.5606133497841883 \cdot x[1] + 5.278085671444797 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

193 : 899.3378804215617 : 5.023185665989674*x[1] + 7.446134046055281*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

194 : 4675.001123658993 : 26.47676362092641*x[1] + 39.247908426314446*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

195 : 12196.842692716265 : 68.72032106367413*x[1] + 101.86777004732873*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

196 : 21841.106603958004 : 123.1797007521802*x[1] + 182.5957917032318*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

197 : 21196.362673411342 : 119.83165235440212*x[1] + 177.6328023135843*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

198 : 12467.423809583142 : 70.28879468381454*x[1] + 104.19280153130154*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

199 : 4898.481983344741 : 27.633909721859872*x[1] + 40.96320735240404*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

200 : 765.3887005307035 : 4.2623036690128515*x[1] + 6.318238379948461*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

201 : -452.0858864177861 : 3.66696270406019*x[1] + 5.435732949548046*x[2] + 1.7256295077930317*x[3] + theta[41] : +Inf : True

202 : -1849.6970314014807 : 15.015354826277694*x[1] + 22.25805538954105*x[2] + 7.066049330013037*x[3] + theta[42] : +Inf : True

203 : 6142.583566779079 : 41.94896953954929*x[1] + 62.183178376273055*x[2] + 71.0664895728835*x[3] + theta[43] : +Inf : True

204 : 10442.985140558467 : 71.49083266469698*x[1] + 105.97464606766847*x[2] + 121.11388122019257*x[3] + theta[44] : +Inf : True

205 : -7940.7743720720755 : 64.11232615753194*x[1] + 95.03709524528266*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

206 : 5596.141090199235 : 38.282014338745924*x[1] + 56.74745654919984*x[2] + 64.8542360562284*x[3] + theta[46] : +Inf : True

207 : 1968.9183004702477 : 13.46994397254767*x[1] + 19.967211065188312*x[2] + 22.819669788786644*x[3] + theta[47] : +Inf : True

208 : 281.1534786413767 : 1.919064035862726*x[1] + 2.8447302178671*x[2] + 3.2511202489909716*x[3] + theta[48] : +Inf : True

209 : -122.67792257876403 : 0.9685564636340608*x[1] + 1.0255303732595935*x[2] + 0.45579127700426414*x[3] + theta[49] : +Inf : True

210 : -576.317980396152 : 4.539559561688448*x[1] + 4.806592477081885*x[2] + 2.136263323147507*x[3] + theta[50] : +Inf : True

211 : -1482.9461276790792 : 11.686779426385783*x[1] + 12.374237039702589*x[2] + 5.499660906534489*x[3] + theta[51] : +Inf : True

212 : -2285.3624107661626 : 17.956238618288122*x[1] + 19.012487948775647*x[2] + 8.449994643900297*x[3] + theta[52] : +Inf : True

213 : -2075.6563451662273 : 16.348569305157238*x[1] + 17.310249852519426*x[2] + 7.6934443788975315*x[3] + theta[53] : +Inf : True

214 : -1279.3661033296326 : 10.112756801944107*x[1] + 10.707624849117288*x[2] + 4.7589443773854665*x[3] + theta[54] : +Inf : True

215 : -340.20948277658624 : 3.1948925230147474*x[1] + 4.735958328233625*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

216 : -60.06574388842755 : 0.5640391466953114*x[1] + 0.8361050880424619*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

217 : -12.037849556499072 : 0.10526049624697124*x[1] + 0.11145229014385186*x[2] + 0.04953435117504534*x[3] + theta[57] : +Inf : True

218 : -40.98021048819866 : 0.3779197393792061*x[1] + 0.4001503122838651*x[2] + 0.17784458323727365*x[3] + theta[58] : +Inf : True

219 : -143.9683970703005 : 1.3133768130407482*x[1] + 1.3906342726313798*x[2] + 0.6180596767250586*x[3] + theta[59] : +Inf : True

220 : -236.02268995953227 : 2.1406777700600026*x[1] + 2.2665999918282376*x[2] + 1.0073777741458847*x[3] + theta[60] : +Inf : True

221 : -204.0224894934155 : 1.8427797583723748*x[1] + 1.9511785676883961*x[2] + 0.8671904745281772*x[3] + theta[61] : +Inf : True

222 : -115.1847502016426 : 1.0451007044458347*x[1] + 1.1065772164720598*x[2] + 0.4918120962098051*x[3] + theta[62] : +Inf : True

223 : -42.23580022300055 : 0.37958443222509614*x[1] + 0.4019129282383369*x[2] + 0.17862796810592776*x[3] + theta[63] : +Inf : True

224 : -7.065908591307251 : 0.0620401152619143*x[1] + 0.06568953380673276*x[2] + 0.029195348358547937*x[3] + theta[64] : +Inf : True

225 : 727.1851181693482 : 3.8821556660382988*x[1] + 4.110517764040551*x[2] + 6.5768284224648825*x[3] + theta[25] : +Inf : True

226 : 3602.6626390207416 : 19.285696439292145*x[1] + 20.420149171015204*x[2] + 32.67223867362434*x[3] + theta[26] : +Inf : True

227 : 9804.747294366227 : $52.379411233327986 \cdot x[1] + 55.46055307058254 \cdot x[2] + 88.7368849129321 \cdot x[3] + \theta[27]$: +Inf : True

228 : 18031.057433294896 : $96.4030485671112 \cdot x[1] + 102.07381612988239 \cdot x[2] + 163.3181058078119 \cdot x[3] + \theta[28]$: +Inf : True

229 : 18368.741399597977 : $98.04118914091855 \cdot x[1] + 103.80831791391374 \cdot x[2] + 166.09330866226202 \cdot x[3] + \theta[29]$: +Inf : True

230 : 10771.442600143659 : $57.6549159060261 \cdot x[1] + 61.046381547557026 \cdot x[2] + 97.67421047609123 \cdot x[3] + \theta[30]$: +Inf : True

231 : 3909.136460164572 : $20.900198084269892 \cdot x[1] + 22.12962150099164 \cdot x[2] + 35.407394401586636 \cdot x[3] + \theta[31]$: +Inf : True

232 : 676.807805532448 : $3.5606133497841883 \cdot x[1] + 3.770061193889139 \cdot x[2] + 6.0320979102226255 \cdot x[3] + \theta[32]$: +Inf : True

233 : -916.3869397660462 : $5.023185665989674 \cdot x[1] + 5.3186671757537685 \cdot x[2] + 2.3638520781127887 \cdot x[3] + \theta[33]$: +Inf : True

234 : -4763.626737377647 : $26.47676362092641 \cdot x[1] + 28.0342203045103 \cdot x[2] + 12.459653468671258 \cdot x[3] + \theta[34]$: +Inf : True

235 : -12428.06245940283 : $68.72032106367413 \cdot x[1] + 72.76269289094907 \cdot x[2] + 32.33897461819961 \cdot x[3] + \theta[35]$: +Inf : True

236 : -22255.156018251044 : $123.1797007521802 \cdot x[1] + 130.42556550230836 \cdot x[2] + 57.966918001025995 \cdot x[3] + \theta[36]$: +Inf : True

237 : -21598.18945394994 : $119.83165235440212 \cdot x[1] + 126.8805730811316 \cdot x[2] + 56.39136581383633 \cdot x[3] + \theta[37]$: +Inf : True

238 : -12703.773076115522 : $70.28879468381454 \cdot x[1] + 74.42342966521537 \cdot x[2] + 33.07707985120687 \cdot x[3] + \theta[38]$: +Inf : True

239 : -4991.344201038478 : $27.633909721859872 \cdot x[1] + 29.25943382314574 \cdot x[2] + 13.004192810287009 \cdot x[3] + \theta[39]$: +Inf : True

240 : -779.8984389293898 : $4.2623036690128515 \cdot x[1] + 4.5130274142489 \cdot x[2] + 2.005789961888402 \cdot x[3] + \theta[40]$: +Inf : True

241 : -543.0081875967824 : $3.66696270406019 \cdot x[1] + 3.882666392534318 \cdot x[2] + 1.7256295077930317 \cdot x[3] + \theta[41]$: +Inf : True

242 : -2221.7031382755217 : $15.015354826277694 \cdot x[1] + 15.89861099252932 \cdot x[2] + 7.066049330013037 \cdot x[3] + \theta[42]$: +Inf : True

243 : -6259.030648613752 : $41.94896953954929 \cdot x[1] + 44.41655598305218 \cdot x[2] + 19.740691548023207 \cdot x[3] + \theta[43]$: +Inf : True

244 : -10640.956422843938 : 71.49083266469698*x[1] + 75.69617576262033*x[2] + 33.64274478338684*x[3] + theta[44] : +Inf : True

245 : -9537.801620086571 : 64.11232615753194*x[1] + 67.88363946091617*x[2] + 30.170506427073875*x[3] + theta[45] : +Inf : True

246 : -5702.229072951826 : 38.282014338745924*x[1] + 40.53389753514273*x[2] + 18.015065571174564*x[3] + theta[46] : +Inf : True

247 : -2006.2437658820063 : 13.46994397254767*x[1] + 14.262293617991649*x[2] + 6.338797163551849*x[3] + theta[47] : +Inf : True

248 : -286.48340240709 : 1.919064035862726*x[1] + 2.0319501556193567*x[2] + 0.903088958053048*x[3] + theta[48] : +Inf : True

249 : -408.63150165902823 : 3.1948925230147474*x[1] + 3.3828273773097317*x[2] + 1.5034788343598828*x[3] + theta[55] : +Inf : True

250 : -72.1460052291169 : 0.5640391466953114*x[1] + 0.5972179200303298*x[2] + 0.2654301866801469*x[3] + theta[56] : +Inf : True

251 : 1610.130392988398 : 6.3405784782745265*x[1] + 6.71355368287891*x[2] + 10.741685892606256*x[3] + theta[18] : +Inf : True

252 : 1931.765738872894 : 7.625921220734874*x[1] + 8.074504821954571*x[2] + 12.919207715127316*x[3] + theta[23] : +Inf : True

253 : 745.8963463212003 : 5.023185665989674*x[1] + 5.3186671757537685*x[2] + 8.509867481206035*x[3] + theta[33] : +Inf : True

254 : 3877.370600191106 : 26.47676362092641*x[1] + 28.0342203045103*x[2] + 44.85475248721651*x[3] + theta[34] : +Inf : True

255 : 10115.864792537184 : 68.72032106367413*x[1] + 72.76269289094907*x[2] + 116.42030862551856*x[3] + theta[35] : +Inf : True

256 : 18114.661875320613 : 123.1797007521802*x[1] + 130.42556550230836*x[2] + 208.68090480369352*x[3] + theta[36] : +Inf : True

257 : 17579.921648563904 : 119.83165235440212*x[1] + 126.8805730811316*x[2] + 203.00891692981068*x[3] + theta[37] : +Inf : True

258 : 10340.280410791704 : 70.28879468381454*x[1] + 74.42342966521537*x[2] + 119.07748746434463*x[3] + theta[38] : +Inf : True

259 : 4062.722024101088 : 27.633909721859872*x[1] + 29.25943382314574*x[2] + 46.81509411703319*x[3] + theta[39] : +Inf : True

260 : 634.8010549425267 : 4.2623036690128515*x[1] + 4.5130274142489*x[2] + 7.220843862798243*x[3] + theta[40] : +Inf : True

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scenario41

126.62525374422897

w[1]:232.5835231037222

w[2]:266.54517639424625

w[3]:6000.0

w[4]:827.3639725419125

y[1]:0.0

y[2]:0.0

{41: [564.9411988461017, 498.477528393619, 33.231835226241294, 86.4027715882273]}

► w = [-1918.17399249],theta =-1918.1739924852532

► e = [-543.0081876]

► E = [[3.6669627 3.88266639 1.72562951]]

scenario42

126.62525374422897

w[1]:232.93174255151888

w[2]:266.95293321388306

w[3]:6000.0

w[4]:832.8598381619622

y[1]:0.0

y[2]:0.0

{42: [510.6672978729567, 450.58879224084404, 30.03925281605629, 78.1020573217463]}

➤ $w = [-7852.68555391], \theta = -7852.685553913753$

➤ $e = [-2221.70313828]$

➤ $E = [[15.01535483 \ 15.89861099 \ 7.06604933]]$

scenario43

126.62525374422897

$w[1]: 229.32292281980835$

$w[2]: 262.72708981037454$

$w[3]: 6000.0$

$w[4]: 775.9026853723699$

$y[1]: 0.0$

$y[2]: 0.0$

{43: [474.04223926950823, 418.2725640613307, 27.884837604088734, 72.50057777063066]}

➤ $w = [-21990.52101726], \theta = -21990.521017256822$

➤ $e = [-6259.03064861]$

➤ $E = [[41.94896954 \ 44.41655598 \ 19.74069155]]$

scenario44

126.62525374422897

$w[1]: 230.36758116319817$

$w[2]: 263.9503602692849$

$w[3]: 6000.0$

$w[4]: 792.3902822325144$

$y[1]: 0.0$

y[2]:0.0

{44: [451.38302023242574, 398.27913549919907, 26.551942366613293, 69.03505015319452]}

► w = [-37451.08687076],theta =-37451.08687075954

► e = [-10640.95642284]

► E = [[71.49083266 75.69617576 33.64274478]]

scenario45

126.62525374422897

w[1]:230.5891753572506

w[2]:264.2098418817811

w[3]:6000.0

w[4]:795.8876512634542

y[1]:0.0

y[2]:0.0

{45: [440.50920327502655, 388.68459112502336, 25.912306075001577, 67.37199579500405]}

► w = [-33580.88187996],theta =-33580.88187996353

► e = [-9537.80162009]

► E = [[64.11232616 67.88363946 30.17050643]]

scenario46

126.62525374422897

w[1]:230.0510180288377

w[2]:263.57967225143324

w[3]:6000.0

w[4]:787.3940407597438

y[1]:0.0

y[2]:0.0

{46: [440.30656904152, 388.50579621310584, 25.900386414207077, 67.34100467693835]}

➤ w = [-20058.55645054],theta =-20058.55645053985

➤ e = [-5702.22907295]

➤ E = [[38.28201434 40.53389754 18.01506557]]

scenario47

126.62525374422897

w[1]:230.08267434227372

w[2]:263.61674105321845

w[3]:6000.0

w[4]:787.893664907021

y[1]:0.0

y[2]:0.0

{47: [450.66257954537656, 397.6434525400381, 26.509563502669227, 68.92486510693995]}

➤ w = [-7057.67402022],theta =-7057.674020217933

➤ e = [-2006.24376588]

➤ E = [[13.46994397 14.26229362 6.33879716]]

scenario48

126.62525374422897

w[1]:229.1013286257559

w[2]:262.46760819787835

w[3]:6000.0

w[4]:772.4053163414292

y[1]:0.0

y[2]:0.0

{48: [471.920333422532, 416.4002941963517, 27.760019613090133, 72.1760509940343]}

➤ w = [-1006.16109004],theta =-1006.1610900442329

➤ e = [-286.48340241]

➤ E = [[1.91906404 2.03195016 0.90308896]]

scenario49

126.62525374422897

w[1]:305.74126345445046

w[2]:352.21117731975687

w[3]:6000.0

w[4]:1981.9953768993491

y[1]:0.0

y[2]:0.0

{49: [404.1714503563933, 356.62186796152344, 23.774791197434915, 61.814457113330725]}

➤ w = [-485.90106838],theta =-485.9010683769225

➤ e = [-122.67792258]

➤ E = [[0.96855646 1.02553037 0.45579128]]

scenario50

126.62525374422897

w[1]:304.5699798573164

w[2]:350.83963165370585

w[3]:6000.0

w[4]:1963.5092834500956

y[1]:0.0

y[2]:0.0

{50: [356.0103803319255, 314.1268061752283, 20.941787078348572, 54.448646403706235]}

➤ w = [-2278.72064503],theta =-2278.720645025484

➤ e = [-576.3179804]

➤ E = [[4.53955956 4.80659248 2.13626332]]

scenario51

126.62525374422897

w[1]:304.82323036480483

w[2]:351.1361820679873

w[3]:6000.0

w[4]:1967.5062766283127

y[1]:0.0

y[2]:0.0

{51: [322.1324325395395, 284.23449929959355, 18.94896661997292, 49.26731321192955]}

➤ w = [-5865.66303359],theta =-5865.663033592837

➤ e = [-1482.94612768]

➤ E = [[11.68677943 12.37423704 5.49966091]]

scenario52

126.62525374422897

w[1]:303.30372731987416

w[2]:349.3568795822995

w[3]:6000.0

w[4]:1943.5243175590113

y[1]:0.0

y[2]:0.0

{52: [299.1772753967719, 263.9799488795045, 17.598663258633653, 45.75652447244744]}

➤ w = [-9019.22019038],theta =-9019.220190379485

➤ e = [-2285.36241077]

➤ E = [[17.95623862 19.01248795 8.44999464]]

scenario53

126.62525374422897

w[1]:304.5383235438803

w[2]:350.8025628519207

w[3]:6000.0

w[4]:1963.0096593028184

y[1]:0.0

y[2]:0.0

{53: [284.93344531513156, 251.4118635133513, 16.760790900890107, 43.57805634231422]}

➤ w = [-8206.61411095],theta =-8206.61411095115

➤ e = [-2075.65634517]

➤ E = [[16.34856931 17.31024985 7.69344438]]

scenario54

126.62525374422897

w[1]:306.3427334097356

w[2]:352.9154845536751

w[3]:6000.0

w[4]:1991.488235697615

y[1]:0.0

y[2]:0.0

{54: [277.90983588811355, 245.2145610777472, 16.347637405183168, 42.50385725347617]}

➤ w = [-5071.80100641],theta =-5071.80100641409

➤ e = [-1279.36610333]

➤ E = [[10.1127568 10.70762485 4.75894438]]

scenario55

126.62525374422897

w[1]:300.83453487186165

w[2]:346.4655130430567

w[3]:6000.0

w[4]:1904.5536340713952

y[1]:0.0

y[2]:0.0

{55: [278.5379121172207, 245.7687459857829, 16.384583065718882, 42.599915970869034]}

➤ w = [-1606.7639351],theta =-1606.7639350974941

➤ $e = [-408.63150166]$

➤ $E = [[3.19489252 \ 3.38282738 \ 1.50347883]]$

scenario56

126.62525374422897

w[1]:300.80287855842556

w[2]:346.4284442412717

w[3]:6000.0

w[4]:1904.054009924118

y[1]:0.0

y[2]:0.0

{56: [285.22839276627633, 251.67211126436143, 16.77814075095745, 43.6231659524893]}

➤ $w = [-283.66911298], \theta = -283.6691129757652$

➤ $e = [-72.14600523]$

➤ $E = [[0.56403915 \ 0.59721792 \ 0.26543019]]$

scenario57

126.62525374422897

w[1]:360.1268099375968

w[2]:415.89537878666613

w[3]:6000.0

w[4]:2840.3496619214493

y[1]:0.0

y[2]:0.0

{57: [237.95749123312137, 209.9624922645188, 13.997499484301272, 36.39349865918325]}

► w = [-51.51210843],theta =-51.51210842587702

► e = [-12.03784956]

► E = [[0.1052605 0.11145229 0.04953435]]

scenario58

126.62525374422897

w[1]:390.73846503026425

w[2]:451.7409101129184

w[3]:6000.0

w[4]:3323.48621233843

y[1]:0.0

y[2]:0.0

{58: [202.5184820637726, 178.692778291564, 11.912851886104283, 30.973414903871088]}

► w = [-182.70575954],theta =-182.70575954033774

► e = [-40.98021049]

► E = [[0.37791974 0.40015031 0.17784458]]

scenario59

126.62525374422897

w[1]:384.37554602961677

w[2]:444.29008095410074

w[3]:6000.0

w[4]:3223.0617587357283

y[1]:0.0

y[2]:0.0

{59: [177.8679324269702, 156.94229331791482, 10.46281955452767, 27.2033308417719]}

➤ w = [-636.50432612],theta =-636.5043261178596

➤ e = [-143.96839707]

➤ E = [[1.31337681 1.39063427 0.61805968]]

scenario60

126.62525374422897

w[1]:380.9883204919586

w[2]:440.3237191630884

w[3]:6000.0

w[4]:3169.6019749770767

y[1]:0.0

y[2]:0.0

{60: [160.88154488019455, 141.95430430605396, 9.463620287070277, 24.605412746382683]}

➤ w = [-1038.80883922],theta =-1038.8088392192333

➤ e = [-236.02268996]

➤ E = [[2.14067777 2.26659999 1.00737777]]

scenario61

126.62525374422897

w[1]:378.5824406708183

w[2]:437.50649022741595

w[3]:6000.0

w[4]:3131.630539784015

y[1]:0.0

y[2]:0.0

{61: [149.37047034401664, 131.7974738329558, 8.786498255530399, 22.844895464379]}

➤ w = [-895.09243325],theta =-895.0924332505344

➤ e = [-204.02248949]

➤ E = [[1.84277976 1.95117857 0.86719047]]

scenario62

126.62525374422897

w[1]:381.20991468601096

w[2]:440.5832007755845

w[3]:6000.0

w[4]:3173.0993440080147

y[1]:0.0

y[2]:0.0

{62: [142.3067408014481, 125.56477129539532, 8.370984753026367, 21.76456035786852]}

➤ w = [-507.11313001],theta =-507.1131300096675

➤ e = [-115.1847502]

➤ E = [[1.0451007 1.10657722 0.4918121]]

scenario63

126.62525374422897

w[1]:375.70171614813705

w[2]:434.1332292649662

w[3]:6000.0

w[4]:3086.164742381796

y[1]:0.0

y[2]:0.0

{63: [139.14895422306395, 122.7784890203505, 8.185232601356711, 21.28160476352742]}

➤ w = [-184.58563397],theta =-184.5856339657845

➤ e = [-42.23580022]

➤ E = [[0.37958443 0.40191293 0.17862797]]

scenario64

126.62525374422897

w[1]:362.43772081842906

w[2]:418.601401316983

w[3]:6000.0

w[4]:2876.82222467268

y[1]:0.0

y[2]:0.0

{64: [139.67493727002713, 123.24259170884743, 8.21617278058984, 21.36204922953355]}

➤ w = [-30.3318788],theta =-30.331878802855794

➤ e = [-7.06590859]

➤ E = [[0.06204012 0.06568953 0.02919535]]

[Thetatheta[1] from previous iteration: 4.46]

[Thetatheta[2] from previous iteration: 30.56]

[Thetatheta[3] from previous iteration: 86.40]

[Thetatheta[4] from previous iteration: 148.25]

[Thetatheta[5] from previous iteration: 121.35]

[Thetatheta[6] from previous iteration: 96.13]

[Thetatheta[7] from previous iteration: 44.41]

[Thetatheta[8] from previous iteration: 7.27]

[Thetatheta[9] from previous iteration: -37.44]

[Thetatheta[10] from previous iteration: -169.14]

[Thetatheta[11] from previous iteration: -627.55]

[Thetatheta[12] from previous iteration: -1,144.61]

[Thetatheta[13] from previous iteration: -918.90]

[Thetatheta[14] from previous iteration: -676.89]

[Thetatheta[15] from previous iteration: -265.59]

[Thetatheta[16] from previous iteration: -81.64]

[Thetatheta[17] from previous iteration: -458.68]

[Thetatheta[18] from previous iteration: -2,706.29]

[Thetatheta[19] from previous iteration: -7,696.37]

[Thetatheta[20] from previous iteration: -14,321.23]

[Thetatheta[21] from previous iteration: -15,234.30]

[Thetatheta[22] from previous iteration: -8,714.81]

[Thetatheta[23] from previous iteration: -3,260.64]

[Thetatheta[24] from previous iteration: -616.70]

[Thetatheta[25] from previous iteration: -1,915.27]

[Thetatheta[26] from previous iteration: -9,524.48]

[Thetatheta[27] from previous iteration: -25,848.21]

[Thetatheta[28] from previous iteration: -47,587.35]

[Thetatheta[29] from previous iteration: -48,364.70]

[Thetatheta[30] from previous iteration: -28,472.38]

[Thetatheta[31] from previous iteration: -10,316.95]

[Thetatheta[32] from previous iteration: -1,746.79]

[Thetatheta[33] from previous iteration: -2,782.66]

[Thetatheta[34] from previous iteration: -14,692.81]

[Thetatheta[35] from previous iteration: -38,175.14]

[Thetatheta[36] from previous iteration: -68,449.38]

[Thetatheta[37] from previous iteration: -66,536.85]

[Thetatheta[38] from previous iteration: -39,054.17]

[Thetatheta[39] from previous iteration: -15,354.47]

[Thetatheta[40] from previous iteration: -2,358.88]

[Thetatheta[41] from previous iteration: -1,918.17]

[Thetatheta[42] from previous iteration: -7,852.69]

[Thetatheta[43] from previous iteration: -21,990.52]

[Thetatheta[44] from previous iteration: -37,451.09]

[Thetatheta[45] from previous iteration: -33,580.88]

[Thetatheta[46] from previous iteration: -20,058.56]

[Thetatheta[47] from previous iteration: -7,057.67]

[Thetatheta[48] from previous iteration: -1,006.16]

[Thetatheta[49] from previous iteration: -485.90]

[Thetatheta[50] from previous iteration: -2,278.72]

[Thetatheta[51] from previous iteration: -5,865.66]

[Thetatheta[52] from previous iteration: -9,019.22]

[Thetatheta[53] from previous iteration: -8,206.61]

[Thetatheta[54] from previous iteration: -5,071.80]

[Thetatheta[55] from previous iteration: -1,606.76]

[Thetatheta[56] from previous iteration: -283.67]

[Thetatheta[57] from previous iteration: -51.51]

[Thetatheta[58] from previous iteration: -182.71]

[Thetatheta[59] from previous iteration: -636.50]

[Thetatheta[60] from previous iteration: -1,038.81]

[Thetatheta[61] from previous iteration: -895.09]

[Thetatheta[62] from previous iteration: -507.11]

[Thetatheta[63] from previous iteration: -184.59]

[Thetatheta[64] from previous iteration: -30.33]

=====

↻ Iteration 7

Master Problem Objective Value : -528,449.18

Decision Variables:

x[1]: 123.03906490310668

x[2]: 109.10379223975046

x[3]: 267.85714285714283

=====

-----feasibility check-----

scenario1

◆ Scenario 1 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario2

◆ Scenario 2 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario3

◆ Scenario 3 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario4

◆ Scenario 4 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario5

◆ Scenario 5 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario6

◆ Scenario 6 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario7

◆ Scenario 7 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario8

◆ Scenario 8 feasibility check:

$$\pi \text{ (Dual values)} = [0.0, 0.0, -0.0, -0.0]$$

current x is feasible.

scenario9

◆ Scenario 9 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario10

◆ Scenario 10 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario11

◆ Scenario 11 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario12

◆ Scenario 12 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario13

◆ Scenario 13 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario14

◆ Scenario 14 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario15

◆ Scenario 15 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario16

◆ Scenario 16 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario17

◆ Scenario 17 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario18

◆ Scenario 18 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario19

◆ Scenario 19 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario20

◆ Scenario 20 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario21

◆ Scenario 21 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario22

◆ Scenario 22 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario23

◆ Scenario 23 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario24

◆ Scenario 24 feasibility check:

π (Dual values) = [0.0, 0.0, -0.0, -0.0]

current x is feasible.

scenario25

◆ Scenario 25 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario26

◆ Scenario 26 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario27

◆ Scenario 27 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario28

◆ Scenario 28 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario29

◆ Scenario 29 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario30

◆ Scenario 30 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario31

◆ Scenario 31 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario32

◆ Scenario 32 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario33

◆ Scenario 33 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario34

◆ Scenario 34 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario35

◆ Scenario 35 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario36

◆ Scenario 36 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario37

◆ Scenario 37 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario38

◆ Scenario 38 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario39

◆ Scenario 39 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario40

◆ Scenario 40 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario41

◆ Scenario 41 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario42

◆ Scenario 42 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario43

◆ Scenario 43 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario44

◆ Scenario 44 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario45

◆ Scenario 45 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario46

◆ Scenario 46 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario47

◆ Scenario 47 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario48

◆ Scenario 48 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario49

◆ Scenario 49 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario50

◆ Scenario 50 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario51

◆ Scenario 51 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario52

◆ Scenario 52 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario53

◆ Scenario 53 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario54

◆ Scenario 54 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario55

◆ Scenario 55 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario56

◆ Scenario 56 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario57

◆ Scenario 57 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario58

◆ Scenario 58 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario59

◆ Scenario 59 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario60

◆ Scenario 60 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario61

◆ Scenario 61 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario62

◆ Scenario 62 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario63

◆ Scenario 63 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

scenario64

◆ Scenario 64 feasibility check:

π (Dual values) = [-0.0, -0.0, -0.0, -0.0]

current x is feasible.

-----sub problem solving for feasible x = (123.03906490310668, 109.10379223975046,
267.85714285714283)-----

scenario1

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:858.2142857142858

w[4]:0.0

y[1]:150.72285450630577

y[2]:187.56471744957594

{1: [237.64775999999998, 209.6892, 35.94672, -0.0]}

➤ w = [4.42992557],theta =4.429925567588566

➤ e = [9.785496]

➤ E = [[0.00951779 0.01007766 0.01151733]]

scenario2

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:776.7857142857139

w[4]:0.0

y[1]:155.39833897262386

y[2]:192.53985037570854

{2: [243.90264904319997, 215.208219744, 36.8928376704, -0.0]}

➤ w = [30.40823742],theta =30.40823741978394

➤ $e = [60.25830153]$

➤ $E = [[0.05304883 \ 0.05616935 \ 0.06419354]]$

scenario3

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:815.8928571428569

w[4]:0.0

y[1]:153.15287603814215

y[2]:190.15047732565802

{3: [255.84656176684547, 225.7469662648637, 38.699479931119484, -0.0]}

➤ $w = [85.90933153], \theta = 85.90933152694319$

➤ $e = [179.09259324]$

➤ $E = [[0.16560308 \ 0.17534444 \ 0.20039365]]$

scenario4

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:837.3214285714288

w[4]:0.0

y[1]:151.92248538911105

y[2]:188.841231818781

{4: [274.2726311452937, 242.00526277525918, 41.48661647575871, -0.0]}

➤ w = [147.36713251],theta =147.36713250793628

➤ e = [316.22021003]

➤ E = [[0.30008169 0.31773355 0.36312406]]

scenario5

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:872.678571428572

w[4]:0.0

y[1]:149.89234081820976

y[2]:186.68097673243392

{5: [300.0981420939346, 264.7924783181776, 45.39299628311615, -0.0]}

➤ w = [120.56139992],theta =120.56139992222158

➤ e = [271.85361107]

➤ E = [[0.26887293 0.28468899 0.32535884]]

scenario6

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:793.928571428571

w[4]:0.0

y[1]:154.414026453399

y[2]:191.49245397020695

{6: [335.9778759626854, 296.4510670258989, 50.820182918725514, -0.0]}

➤ w = [95.62034119],theta =95.62034119278505

➤ e = [193.68136379]

➤ E = [[0.17427172 0.184523 0.21088343]]

scenario7

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:838.9285714285708

w[4]:0.0

y[1]:151.83020609043376

y[2]:188.74303840576525

{7: [384.2377380659656, 339.033298293499, 58.11999399317124, -0.0]}

➤ w = [44.14232776],theta =44.14232776007553

➤ e = [94.92932352]

➤ E = [[0.09025744 0.09556671 0.10921909]]

scenario8

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:953.5714285714291

w[4]:0.0

y[1]:145.2476161181175

y[2]:181.73857494397322

{8: [446.6840552564463, 394.13298993215847, 67.56565541694142, -0.0]}

► w = [7.20802836],theta =7.208028355435235

► e = [18.39287286]

► E = [[0.01987744 0.0210467 0.02405337]]

scenario9

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:2123.571428571428

w[4]:0.0

y[1]:78.06828668102129

y[2]:110.25377026848875

{9: [436.5264598399147, 385.17040574110115, 66.02921241276017, -0.0]}

► w = [-38.20343201],theta =-38.20343201482657

► e = [107.84771361]

► E = [[0.25955863 0.27482679 0.31408776]]

scenario10

123.03906490310668

w[1]:0.0
w[2]:0.0
w[3]:2194.821428571428
w[4]:0.0
y[1]:73.97723777299299
y[2]:105.90052895812272
{10: [436.2208913180268, 384.90078645708235, 65.98299196407125, -0.0]}

- w = [-172.29425089],theta =-172.2942508877204
- e = [431.08888083]
- E = [[1.0723182 1.13539574 1.29759513]]

scenario11

123.03906490310668
w[1]:0.0
w[2]:0.0
w[3]:2180.892857142857
w[4]:0.0
y[1]:74.77699169486318
y[2]:106.75153853759275
{11: [446.4982555174795, 393.9690489860112, 67.53755125474477, -0.0]}

- w = [-639.44332548],theta =-639.4433254770443
- e = [1636.28478346]
- E = [[4.04437004 4.28227416 4.89402761]]

scenario12

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:2188.928571428571

w[4]:0.0

y[1]:74.31559520147654

y[2]:106.26057147251387

{12: [467.5104634221321, 412.5092324312929, 70.71586841679306, -0.0]}

➤ w = [-1166.09255545],theta =-1166.0925554548362

➤ e = [2945.31591956]

➤ E = [[7.30669766 7.7365034 8.84171817]]

scenario13

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:2128.9285714285716

w[4]:0.0

y[1]:77.76068901876351

y[2]:109.9264588917695

{13: [501.2553686719416, 442.2841488281836, 75.82013979911719, -0.0]}

➤ w = [-937.50090327],theta =-937.5009032662495

➤ e = [2621.27072206]

➤ E = [[6.32456455 6.69659776 7.65325458]]

scenario14

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:2124.6428571428564

w[4]:0.0

y[1]:78.00676714856975

y[2]:110.18830799314492

{14: [549.5312732287364, 484.880535201826, 83.12237746317017, -0.0]}

➤ w = [-690.66635881],theta =-690.666358805296

➤ e = [1945.98721461]

➤ E = [[4.68579821 4.9614334 5.6702096]]

scenario15

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:2188.928571428571

w[4]:0.0

y[1]:74.31559520147654

y[2]:106.26057147251387

{15: [614.7056822336644, 542.3873666767626, 92.98069143030214, -0.0]}

➤ w = [-270.57100105],theta =-270.57100105305767

➤ e = [683.40808201]

► E = [[1.69538901 1.79511777 2.05156317]]

scenario16

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:2213.571428571428

w[4]:0.0

y[1]:72.90064595509081

y[2]:104.75493913960531

{16: [700.6169483826413, 618.1914250435068, 105.97567286460118, -0.0]}

► w = [-83.12548186],theta =-83.12548185583323

► e = [201.94253218]

► E = [[0.50661612 0.53641706 0.61304807]]

scenario17

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:3460.178571428571

w[4]:0.0

y[1]:1.3226699477085049

y[2]:28.589581777035534

{17: [667.5127975715615, 588.9818802102011, 100.96832232174877, -0.0]}

➤ $w = [-464.30549276], \theta = -464.3054927620028$

➤ $e = [384.80149507]$

➤ $E = [[1.50901281 \ 1.59777826 \ 1.8260323 \]]$

scenario18

123.03906490310668

$w[1]:0.0$

$w[2]:0.0$

$w[3]:3474.6428571428573$

$w[4]:0.0$

$y[1]:0.4921562596125$

$y[2]:27.705841059893544$

{18: [651.7194447810183, 575.0465689244278, 98.5794118156162, -0.0]}

➤ $w = [-2740.71296085], \theta = -2740.7129608450823$

➤ $e = [2254.18255018]$

➤ $E = [[8.87680987 \ 9.39897516 \ 10.74168589 \]]$

scenario19

123.03906490310668

$w[1]:0.0$

$w[2]:0.0$

$w[3]:3448.3928571428564$

$w[4]:0.0$

$y[1]:1.9993848046756$

y[2]:29.30966680581787

{19: [651.7911339199442, 575.1098240470095, 98.59025555091591, -0.0]}

➤ w = [-7788.00708089],theta =-7788.007080894184

➤ e = [6494.90694624]

➤ E = [[25.38325615 26.87638887 30.71587299]]

scenario20

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:3438.75

w[4]:0.0

y[1]:2.553060596739556

y[2]:29.89882728391254

{20: [667.2646554392036, 588.7629312698856, 100.93078821769465, -0.0]}

➤ w = [-14487.38359834],theta =-14487.38359834177

➤ e = [12144.21672899]

➤ E = [[47.32904865 50.11311033 57.27212609]]

scenario21

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:3415.178571428571

w[4]:0.0

y[1]:3.906490310673746

y[2]:31.33899734147724

{21: [699.3400674261661, 617.064765376029, 105.78253120731924, -0.0]}

➤ w = [-15399.50192176],theta =-15399.501921759987

➤ e = [13073.54549577]

➤ E = [[50.60162475 53.57819092 61.23221819]]

scenario22

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:3400.7142857142853

w[4]:0.0

y[1]:4.737003998769694

y[2]:32.22273805861923

{22: [749.8953609004036, 661.6723772650621, 113.42955038829635, -0.0]}

➤ w = [-8805.18748163],theta =-8805.187481632003

➤ e = [7534.24280246]

➤ E = [[29.03804808 30.74616856 35.13847835]]

scenario23

123.03906490310668

w[1]:-1.1368683772161603e-13

w[2]:0.0

w[3]:3483.2142857142853

w[4]:0.0

y[1]:0.0

y[2]:27.182142857142736

{23: [586.4288850149857, 724.4121520773353, 124.18494035611461, -0.0]}

➤ w = [-3302.97346971],theta =-3302.9734697102276

➤ e = [2329.15754801]

➤ E = [[7.62592122 11.30430675 12.91920772]]

scenario24

123.03906490310668

w[1]:0.0

w[2]:0.0

w[3]:3413.571428571428

w[4]:0.0

y[1]:3.998769609351058

y[2]:31.43719075449303

{24: [918.7980113171591, 810.7041276327875, 138.97785045133497, -0.0]}

➤ w = [-623.35069669],theta =-623.350696688042

➤ e = [529.66003005]

➤ E = [[2.04910332 2.16963881 2.47958722]]

scenario25

123.03906490310668

w[1]:70.47062442325421

w[2]:47.80489354923776

w[3]:4710.535714285714

w[4]:0.0

y[1]:0.0

y[2]:0.0

{25: [608.9727590580751, 537.3289050512425, 128.95893721229825, -0.0]}

► w = [-1960.59523122],theta =-1960.5952312170787

► e = [727.18511817]

► E = [[3.88215567 4.11051776 6.57682842]]

scenario26

123.03906490310668

w[1]:71.20885881267287

w[2]:48.59044085336399

w[3]:4723.392857142857

w[4]:0.0

y[1]:0.0

y[2]:0.0

{26: [579.4254007885773, 511.2577065781562, 122.70184957875753, -0.0]}

► w = [-9749.6396314],theta =-9749.639631397757

► e = [3602.66263902]

► E = [[19.28569644 20.42014917 32.67223867]]

scenario27

123.03906490310668

w[1]:70.65518302060889

w[2]:48.00128037526929

w[3]:4713.749999999999

w[4]:0.0

y[1]:0.0

y[2]:0.0

{27: [565.594516471754, 499.0539851221356, 119.77295642931259, -0.0]}

➤ w = [-26459.73160249],theta =-26459.731602493317

➤ e = [9804.74729437]

➤ E = [[52.37941123 55.46055307 88.73688491]]

scenario28

123.03906490310668

w[1]:70.87050138418937

w[2]:48.23039833897275

w[3]:4717.5

w[4]:0.0

y[1]:0.0

y[2]:0.0

{28: [565.7585388815307, 499.198710777821, 119.80769058667708, -0.0]}

➤ w = [-48712.84514289],theta =-48712.84514288546

➤ e = [18031.05743329]

➤ E = [[96.40304857 102.07381613 163.31810581]]

scenario29

123.03906490310668

w[1]:70.40910489080272

w[2]:47.7394312738939

w[3]:4709.464285714285

w[4]:0.0

y[1]:0.0

y[2]:0.0

{29: [579.3480589854651, 511.1894638107043, 122.68547131456907, -0.0]}

➤ w = [-49509.31509069],theta =-49509.31509068991

➤ e = [18368.7413996]

➤ E = [[98.04118914 103.80831791 166.09330866]]

scenario30

123.03906490310668

w[1]:71.17809904644707

w[2]:48.557709715692

w[3]:4722.857142857142

w[4]:0.0

y[1]:0.0

y[2]:0.0

{30: [606.9424070349428, 535.5374179720081, 128.52898031328198, -0.0]}

➤ w = [-29145.49101831],theta =-29145.49101830725

➤ e = [10771.44260014]

➤ E = [[57.65491591 61.04638155 97.67421048]]

scenario31

123.03906490310668

w[1]:70.87050138418937

w[2]:48.23039833897275

w[3]:4717.5

w[4]:0.0

y[1]:0.0

y[2]:0.0

{31: [650.247747776886, 573.7480127443109, 137.69952305863467, -0.0]}

➤ w = [-10560.95349543],theta =-10560.953495427195

➤ e = [3909.13646016]

➤ E = [[20.90019808 22.1296215 35.4073944]]

scenario32

123.03906490310668

w[1]:66.53337434635483

w[2]:43.6153079272313

w[3]:4641.964285714284

w[4]:0.0

y[1]:0.0

y[2]:0.0

{32: [714.641782239231, 630.56627844638, 151.33590682713128, -0.0]}

► w = [-1788.3552164],theta =-1788.3552164025177

► e = [676.80780553]

► E = [[3.56061335 3.77006119 6.03209791]]

scenario33

123.03906490310668

w[1]:141.1873269763148

w[2]:123.05377905699362

w[3]:5942.142857142857

w[4]:0.0

y[1]:0.0

y[2]:0.0

{33: [646.950912625531, 570.8390405519389, 137.0013697324654, -0.0]}

► w = [-2731.867269],theta =-2731.867269001437

► e = [745.89634632]

► E = [[5.02318567 5.31866718 8.50986748]]

scenario34

123.03906490310668

w[1]:145.9550907413102

w[2]:128.12710539614204

w[3]:6000.0

w[4]:25.17857142857065

y[1]:0.0

y[2]:0.0

{34: [599.7752520768773, 529.2134577148915, 35.2808971809928, 91.73033267058125]}

➤ w = [-14417.34990177],theta =-14417.349901772082

➤ e = [-4763.62673738]

➤ E = [[26.47676362 28.0342203 12.45965347]]

scenario35

123.03906490310668

w[1]:144.17102430021515

w[2]:126.2286994111704

w[3]:5994.107142857142

w[4]:0.0

y[1]:0.0

y[2]:0.0

{35: [571.327911870871, 504.1128634154743, 120.98708721971389, -0.0]}

➤ w = [-37462.11621794],theta =-37462.1162179356

➤ e = [10115.86479254]

➤ E = [[68.72032106 72.76269289 116.42030863]]

scenario36

123.03906490310668

w[1]:144.50938172869866

w[2]:126.58874192556152

w[3]:5999.999999999999

w[4]:0.0

y[1]:0.0

y[2]:0.0

{36: [557.5760490321393, 491.9788667930638, 118.07492803033537, -0.0]}

➤ w = [-67167.84805115],theta =-67167.84805114873

➤ e = [18114.66187532]

➤ E = [[123.17970075 130.4255655 208.6809048]]

scenario37

123.03906490310668

w[1]:145.33989541679466

w[2]:127.47248264270354

w[3]:6000.0

w[4]:14.464285714285325

y[1]:0.0

y[2]:0.0

{37: [557.3641701335071, 491.7919148236825, 32.7861276549122, 85.24393190277164]}

➤ w = [-65290.14571884],theta =-65290.145718840155

➤ e = [-21598.18945395]

➤ E = [[119.83165235 126.88057308 56.39136581]]

scenario38

123.03906490310668

w[1]:144.38634266379557

w[2]:126.45781737487385

w[3]:5997.857142857142

w[4]:0.0

y[1]:0.0

y[2]:0.0

{38: [570.7297629333085, 503.5850849411544, 120.86042038587709, -0.0]}

➤ w = [-38323.62113904],theta =-38323.62113904123

➤ e = [10340.28041079]

➤ E = [[70.28879468 74.42342967 119.07748746]]

scenario39

123.03906490310668

w[1]:144.601661027376

w[2]:126.68693533857731

w[3]:6000.0

w[4]:1.6071428571422075

y[1]:0.0

y[2]:0.0

{39: [597.9764018157447, 527.6262368962451, 35.17508245974971, 91.45521439534915]}

➤ w = [-15066.97573305],theta =-15066.975733053081

➤ e = [-4991.34420104]

➤ E = [[27.63390972 29.25943382 13.00419281]]

scenario40

123.03906490310668

w[1]:140.17225469086418

w[2]:121.97365151382007

w[3]:5924.464285714284

w[4]:0.0

y[1]:0.0

y[2]:0.0

{40: [642.3582103585092, 566.7866561986845, 136.0287974876843, -0.0]}

➤ w = [-2316.17181431],theta =-2316.1718143090775

➤ e = [634.80105494]

➤ E = [[4.26230367 4.51302741 7.22084386]]

scenario41

123.03906490310668

w[1]:220.33220547523814

w[2]:207.27099628685698

w[3]:6000.0

w[4]:1320.5357142857129

y[1]:0.0

y[2]:0.0

{41: [564.9411988461017, 498.477528393619, 33.231835226241294, 86.4027715882273]}

➤ w = [-1880.02366675],theta =-1880.0236667536606

➤ e = [-543.0081876]

➤ E = [[3.6669627 3.88266639 1.72562951]]

scenario42

123.03906490310668

w[1]:220.67056290372176

w[2]:207.63103880124822

w[3]:6000.0

w[4]:1326.4285714285706

y[1]:0.0

y[2]:0.0

{42: [510.6672978729567, 450.58879224084404, 30.03925281605629, 78.1020573217463]}

➤ w = [-7696.46889074],theta =-7696.468890743531

➤ e = [-2221.70313828]

➤ E = [[15.01535483 15.89861099 7.06604933]]

scenario43

123.03906490310668

w[1]:217.16394955398323

w[2]:203.89968910664874

w[3]:6000.0

w[4]:1265.3571428571431

y[1]:0.0

y[2]:0.0

{43: [474.04223926950823, 418.2725640613307, 27.884837604088734, 72.50057777063066]}

➤ w = [-21554.09256647],theta =-21554.09256646662

➤ e = [-6259.03064861]

► E = [[41.94896954 44.41655598 19.74069155]]

scenario44

123.03906490310668

w[1]:218.1790218394338

w[2]:204.97981664982228

w[3]:6000.0

w[4]:1283.0357142857129

y[1]:0.0

y[2]:0.0

{44: [451.38302023242574, 398.27913549919907, 26.551942366613293, 69.03505015319452]}

► w = [-36707.31095235],theta =-36707.31095235134

► e = [-10640.95642284]

► E = [[71.49083266 75.69617576 33.64274478]]

scenario45

123.03906490310668

w[1]:218.39434020301428

w[2]:205.2089346135258

w[3]:6000.0

w[4]:1286.7857142857138

y[1]:0.0

y[2]:0.0

{45: [440.50920327502655, 388.68459112502336, 25.912306075001577, 67.37199579500405]}

► $w = [-32913.8704256], \theta = -32913.87042560328$

► $e = [-9537.80162009]$

► $E = [[64.11232616 \ 67.88363946 \ 30.17050643]]$

scenario46

123.03906490310668

$w[1]: 217.8714241771761$

$w[2]: 204.65250527310297$

$w[3]: 6000.0$

$w[4]: 1277.6785714285716$

$y[1]: 0.0$

$y[2]: 0.0$

{46: [440.30656904152, 388.50579621310584, 25.900386414207077, 67.34100467693835]}

► $w = [-19660.27824742], \theta = -19660.27824741889$

► $e = [-5702.22907295]$

► $E = [[38.28201434 \ 40.53389754 \ 18.01506557]]$

scenario47

123.03906490310668

$w[1]: 217.90218394340184$

$w[2]: 204.68523641077496$

$w[3]: 6000.0$

$w[4]: 1278.2142857142853$

$y[1]: 0.0$

y[2]:0.0

{47: [450.66257954537656, 397.6434525400381, 26.509563502669227, 68.92486510693995]}

➤ w = [-6917.5354937],theta =-6917.535493701149

➤ e = [-2006.24376588]

➤ E = [[13.46994397 14.26229362 6.33879716]]

scenario48

123.03906490310668

w[1]:216.94863119040275

w[2]:203.67057114294522

w[3]:6000.0

w[4]:1261.6071428571422

y[1]:0.0

y[2]:0.0

{48: [471.920333422532, 416.4002941963517, 27.760019613090133, 72.1760509940343]}

➤ w = [-986.19554254],theta =-986.1955425389682

➤ e = [-286.48340241]

➤ E = [[1.91906404 2.03195016 0.90308896]]

scenario49

123.03906490310668

w[1]:291.41802522300804

w[2]:282.912655446676

w[3]:6000.0

w[4]:2558.5714285714275

y[1]:0.0

y[2]:0.0

{49: [404.1714503563933, 356.62186796152344, 23.774791197434915, 61.814457113330725]}

➤ w = [-475.82440615],theta =-475.8244061473979

➤ e = [-122.67792258]

➤ E = [[0.96855646 1.02553037 0.45579128]]

scenario50

123.03906490310668

w[1]:290.27991387265433

w[2]:281.7016033528148

w[3]:6000.0

w[4]:2538.75

y[1]:0.0

y[2]:0.0

{50: [356.0103803319255, 314.1268061752283, 20.941787078348572, 54.448646403706235]}

➤ w = [-2231.49200107],theta =-2231.4920010677415

➤ e = [-576.3179804]

➤ E = [[4.53955956 4.80659248 2.13626332]]

scenario51

123.03906490310668

w[1]:290.5259920024605

w[2]:281.96345245419025

w[3]:6000.0

w[4]:2543.0357142857138

y[1]:0.0

y[2]:0.0

{51: [322.1324325395395, 284.23449929959355, 18.94896661997292, 49.26731321192955]}

➤ w = [-5744.07618424],theta =-5744.076184243039

➤ e = [-1482.94612768]

➤ E = [[11.68677943 12.37423704 5.49966091]]

scenario52

123.03906490310668

w[1]:289.0495232236233

w[2]:280.3923578459378

w[3]:6000.0

w[4]:2517.3214285714275

y[1]:0.0

y[2]:0.0

{52: [299.1772753967719, 263.9799488795045, 17.598663258633653, 45.75652447244744]}

➤ w = [-8832.40717713],theta =-8832.407177134657

➤ e = [-2285.36241077]

➤ E = [[17.95623862 19.01248795 8.44999464]]

scenario53

123.03906490310668

w[1]:290.24915410642853

w[2]:281.66887221514287

w[3]:6000.0

w[4]:2538.2142857142844

y[1]:0.0

y[2]:0.0

{53: [284.93344531513156, 251.4118635133513, 16.760790900890107, 43.57805634231422]}

➤ w = [-8036.52695857],theta =-8036.526958565695

➤ e = [-2075.65634517]

➤ E = [[16.34856931 17.31024985 7.69344438]]

scenario54

123.03906490310668

w[1]:292.0024607812979

w[2]:283.5345470624427

w[3]:6000.0

w[4]:2568.75

y[1]:0.0

y[2]:0.0

{54: [277.90983588811355, 245.2145610777472, 16.347637405183168, 42.50385725347617]}

➤ w = [-4966.5899647],theta =-4966.589964695185

➤ e = [-1279.36610333]

➤ E = [[10.1127568 10.70762485 4.75894438]]

scenario55

123.03906490310668

w[1]:286.65026145801266

w[2]:277.83932910752753

w[3]:6000.0

w[4]:2475.5357142857138

y[1]:0.0

y[2]:0.0

{55: [278.5379121172207, 245.7687459857829, 16.384583065718882, 42.599915970869034]}

➤ w = [-1573.52493043],theta =-1573.5249304314566

➤ e = [-408.63150166]

➤ E = [[3.19489252 3.38282738 1.50347883]]

scenario56

123.03906490310668

w[1]:286.6195016917869

w[2]:277.8065979698557

w[3]:6000.0

w[4]:2475.0

y[1]:0.0

y[2]:0.0

{56: [285.22839276627633, 251.67211126436143, 16.77814075095745, 43.6231659524893]}

➤ w = [-277.80096571],theta =-277.8009657082814

➤ e = [-72.14600523]

➤ E = [[0.56403915 0.59721792 0.26543019]]

scenario57

123.03906490310668

w[1]:344.2633035988923

w[2]:339.14474996704337

w[3]:6000.0

w[4]:3478.9285714285706

y[1]:0.0

y[2]:0.0

{57: [237.95749123312137, 209.9624922645188, 13.997499484301272, 36.39349865918325]}

➤ w = [-50.41699987],theta =-50.41699987349258

➤ e = [-12.03784956]

➤ E = [[0.1052605 0.11145229 0.04953435]]

scenario58

123.03906490310668

w[1]:374.0079975392184

w[2]:370.795760095795

w[3]:6000.0

w[4]:3996.9642857142862

y[1]:0.0

y[2]:0.0

{58: [202.5184820637726, 178.692778291564, 11.912851886104283, 30.973414903871088]}

➤ w = [-178.7739603],theta =-178.7739603044875

- $e = [-40.98021049]$
- $E = [[0.37791974 \ 0.40015031 \ 0.17784458]]$

scenario59

123.03906490310668

w[1]:367.8252845278373

w[2]:364.21680142373805

w[3]:6000.0

w[4]:3889.2857142857138

y[1]:0.0

y[2]:0.0

{59: [177.8679324269702, 156.94229331791482, 10.46281955452767, 27.2033308417719]}

- $w = [-622.8402239], \theta = -622.8402238976905$
- $e = [-143.96839707]$
- $E = [[1.31337681 \ 1.39063427 \ 0.61805968]]$

scenario60

123.03906490310668

w[1]:364.53398954167926

w[2]:360.71456969284213

w[3]:6000.0

w[4]:3831.9642857142862

y[1]:0.0

y[2]:0.0

{60: [160.88154488019455, 141.95430430605396, 9.463620287070277, 24.605412746382683]}

➤ $w = [-1016.53766801], \theta = -1016.5376680061356$

➤ $e = [-236.02268996]$

➤ $E = [[2.14067777 \ 2.26659999 \ 1.00737777]]$

scenario61

123.03906490310668

$w[1]: 362.19624730852024$

$w[2]: 358.22700322977585$

$w[3]: 6000.0$

$w[4]: 3791.25$

$y[1]: 0.0$

$y[2]: 0.0$

{61: [149.37047034401664, 131.7974738329558, 8.786498255530399, 22.844895464379]}

➤ $w = [-875.92053168], \theta = -875.9205316777014$

➤ $e = [-204.02248949]$

➤ $E = [[1.84277976 \ 1.95117857 \ 0.86719047]]$

scenario62

123.03906490310668

$w[1]: 364.7493079052597$

$w[2]: 360.94368765654553$

$w[3]: 6000.0$

$w[4]: 3835.7142857142844$

y[1]:0.0

y[2]:0.0

{62: [142.3067408014481, 125.56477129539532, 8.370984753026367, 21.76456035786852]}

➤ w = [-496.24011724],theta =-496.2401172427858

➤ e = [-115.1847502]

➤ E = [[1.0451007 1.10657722 0.4918121]]

scenario63

123.03906490310668

w[1]:359.3971085819745

w[2]:355.2484697016306

w[3]:6000.0

w[4]:3742.4999999999998

y[1]:0.0

y[2]:0.0

{63: [139.14895422306395, 122.7784890203505, 8.185232601356711, 21.28160476352742]}

➤ w = [-180.63651561],theta =-180.63651560796896

➤ e = [-42.23580022]

➤ E = [[0.37958443 0.40191293 0.17862797]]

scenario64

123.03906490310668

w[1]:346.5087665333741

w[2]:341.53412301709386

w[3]:6000.0

w[4]:3518.0357142857138

y[1]:0.0

y[2]:0.0

{64: [139.67493727002713, 123.24259170884743, 8.21617278058984, 21.36204922953355]}

➤ w = [-29.6864262],theta =-29.686426204429605

➤ e = [-7.06590859]

➤ E = [[0.06204012 0.06568953 0.02919535]]

● Termination Criteria Met!

✓ Final Master Problem Objective: -528,449.18

✓ Final Decision Variables: x1 = 123.04, x2 = 109.10,x3 = 267.86

=====

داده تولید شده

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